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(54) **VALIDATION OF A DCI FORMAT**

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H04W 76/11 (2018.01)

H04W 76/20 (2018.01)

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(52) **U.S. Cl.**

CPC **H04L 5/0098** (2013.01); **H04L 1/1812**

(2013.01); **H04L 5/001** (2013.01); **H04W**

72/1268 (2013.01); **H04W 72/23** (2023.01);

H04W 76/11 (2018.02); **H04W 76/20**

(2018.02); **H04W 76/30** (2018.02)

Related U.S. Application Data

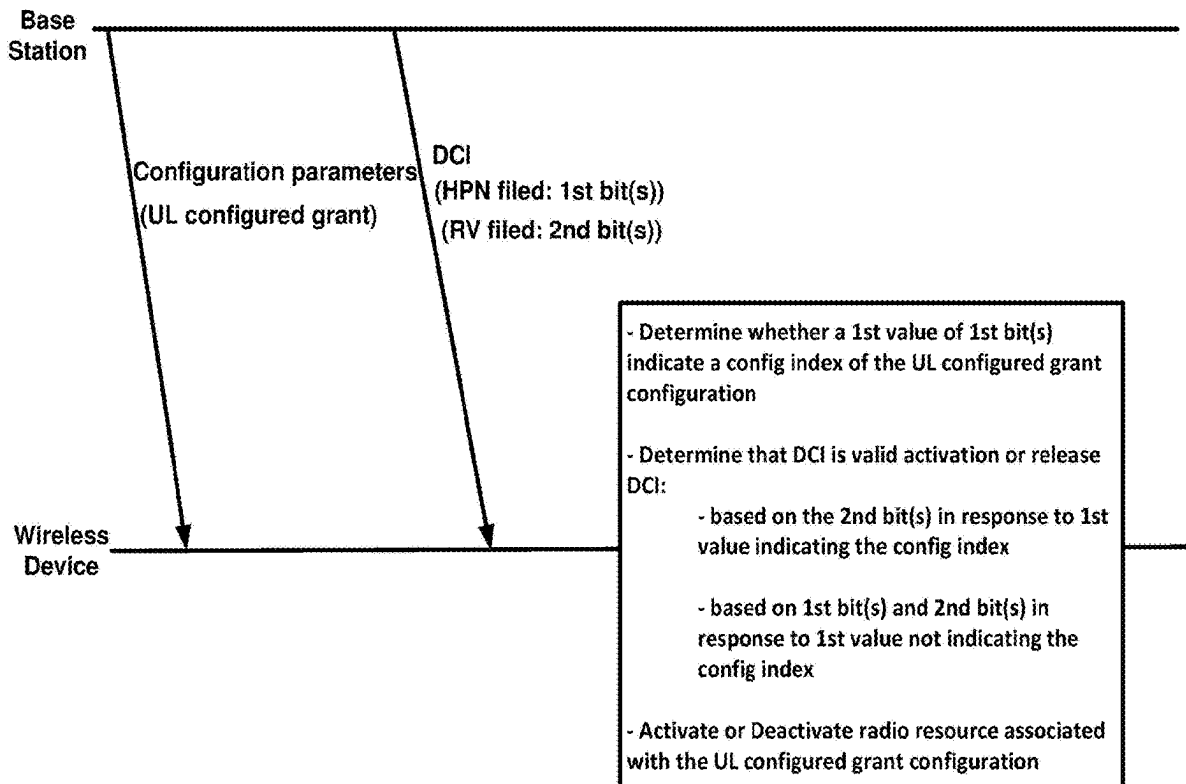
(63) Continuation of application No. 18/395,810, filed on Dec. 26, 2023, now Pat. No. 12,341,728, which is a continuation of application No. 17/884,046, filed on Aug. 9, 2022, now Pat. No. 11,902,216, which is a continuation of application No. 17/467,182, filed on Sep. 4, 2021, now Pat. No. 11,569,970, which is a continuation of application No. 17/180,746, filed on Feb. 20, 2021, now Pat. No. 11,139,944, which is a continuation of application No. 17/084,933, filed on Oct. 30, 2020, now Pat. No. 10,965,434.

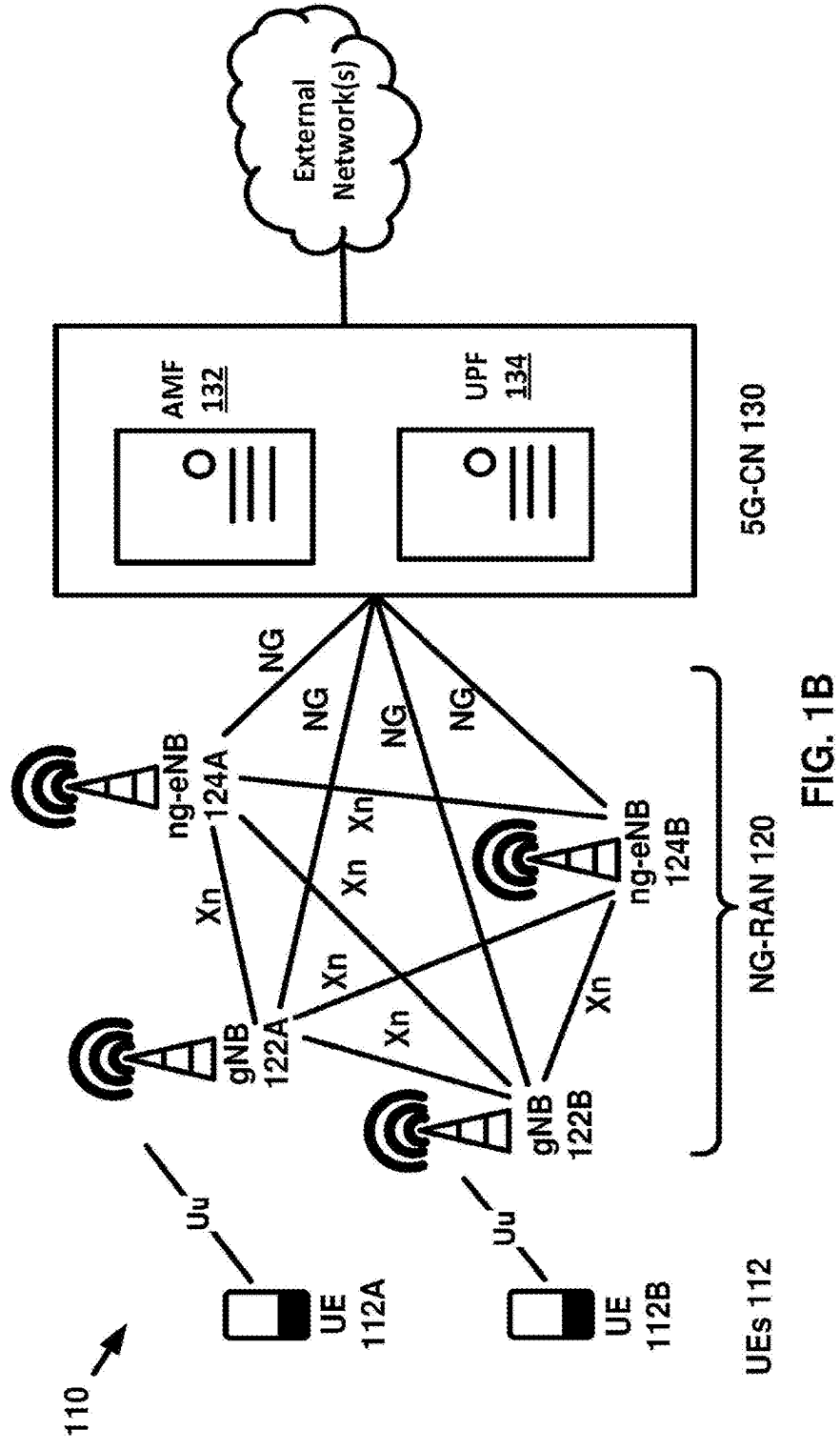
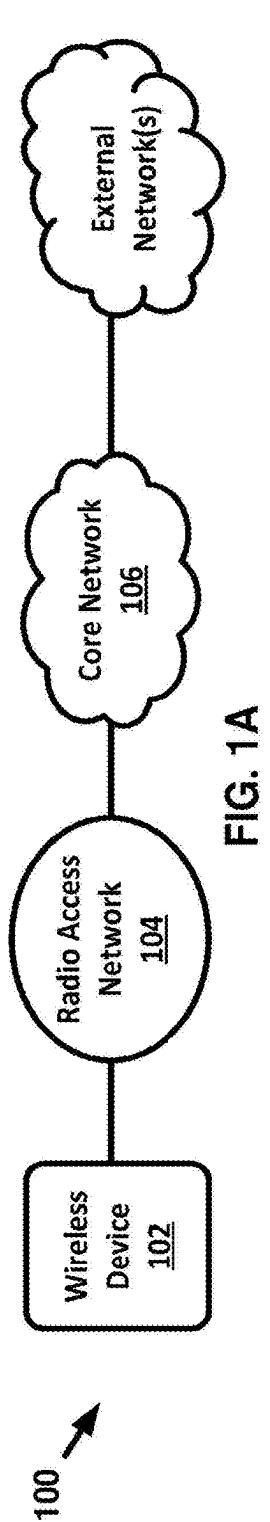
(60) Provisional application No. 62/931,099, filed on Nov. 5, 2019.

(57)

ABSTRACT

A wireless device validates a DCI. The validating may be based on second bit(s) in an RV field of the DCI. The validating may be based on comparing first bit(s) in a HARQ process number field of the DCI with a first sequence of bit(s) in response to the first bit(s) not indicating a configuration index and the DCI having a first format. The first sequence may be determined using a first configuration parameter. The validating may not be based on comparing the first bit(s) with the first sequence in response to the DCI having a second format. The wireless device may transmit a confirmation in response to the validation.





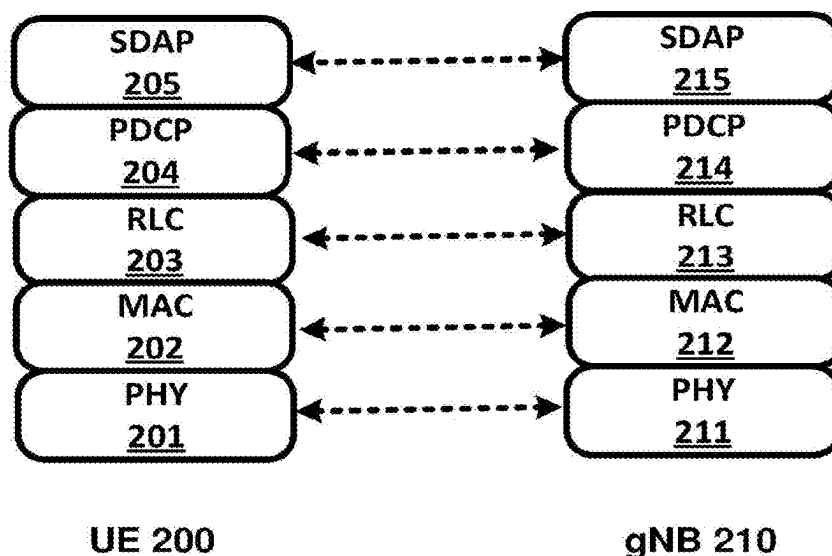


FIG. 2A

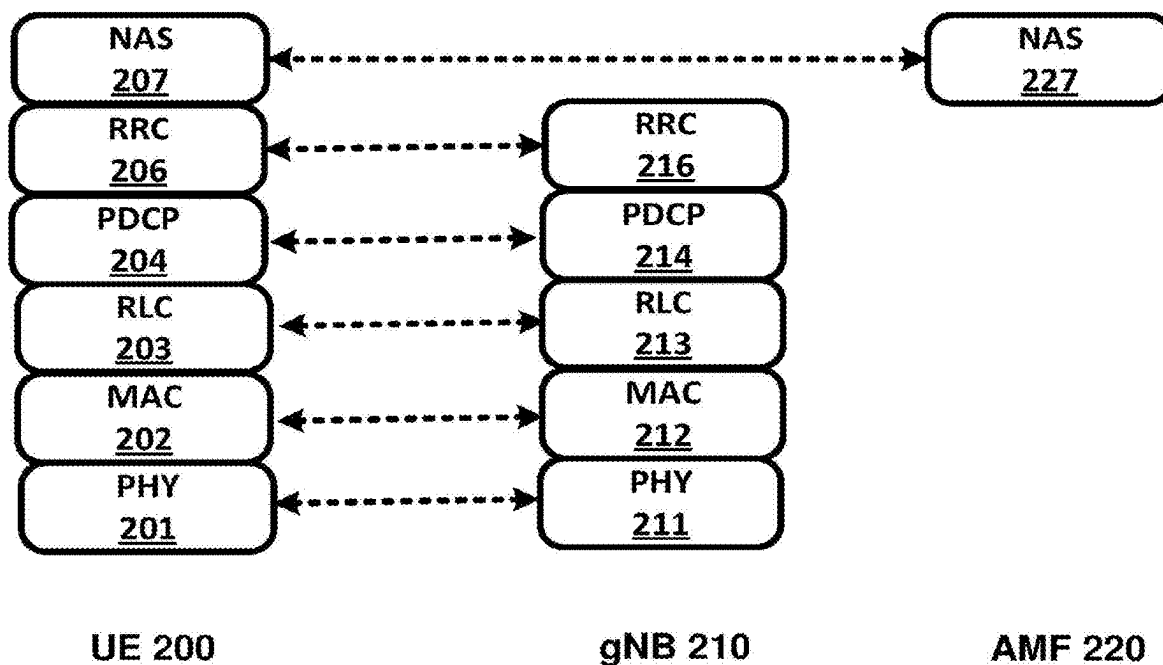


FIG. 2B

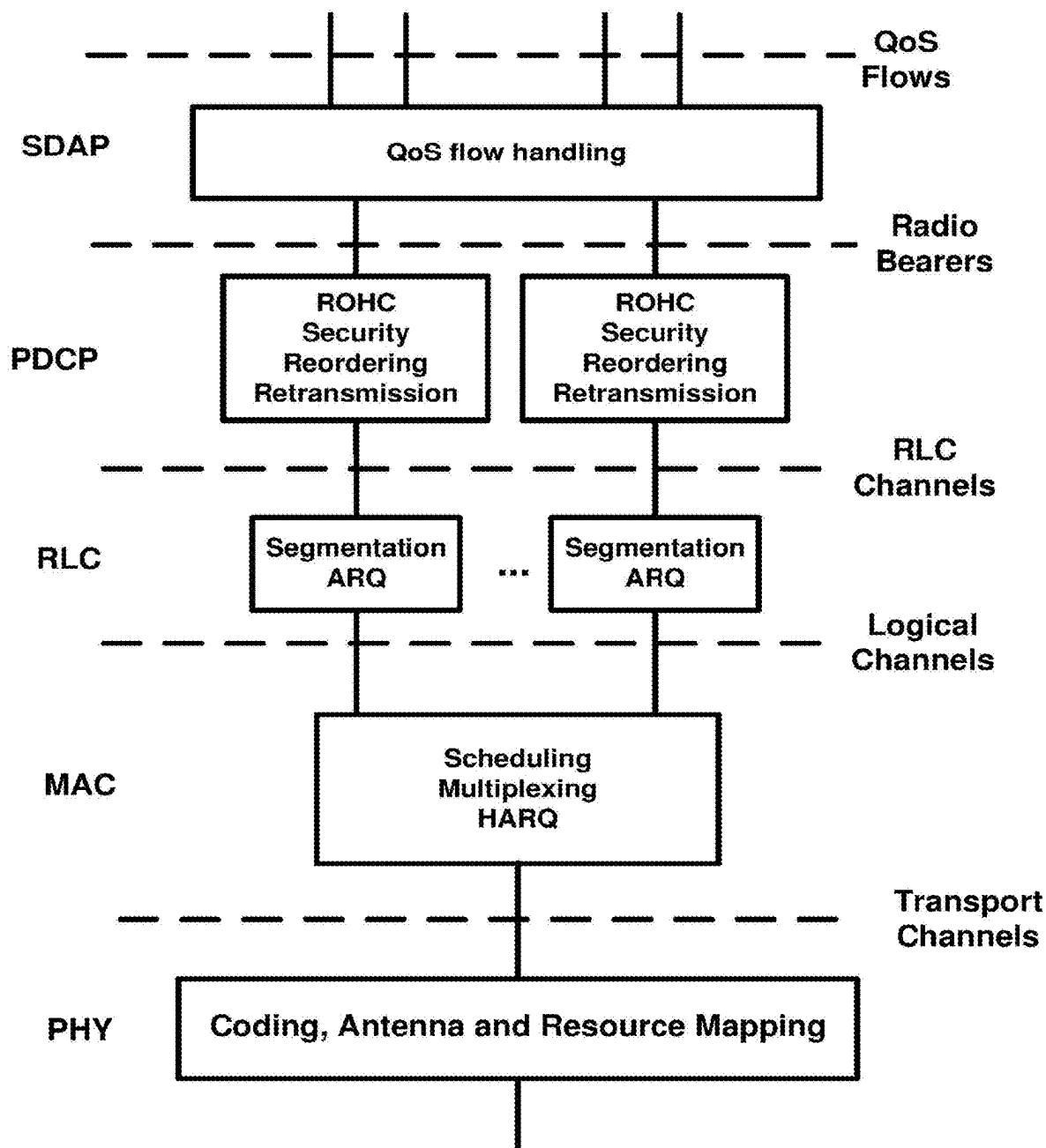


FIG. 3

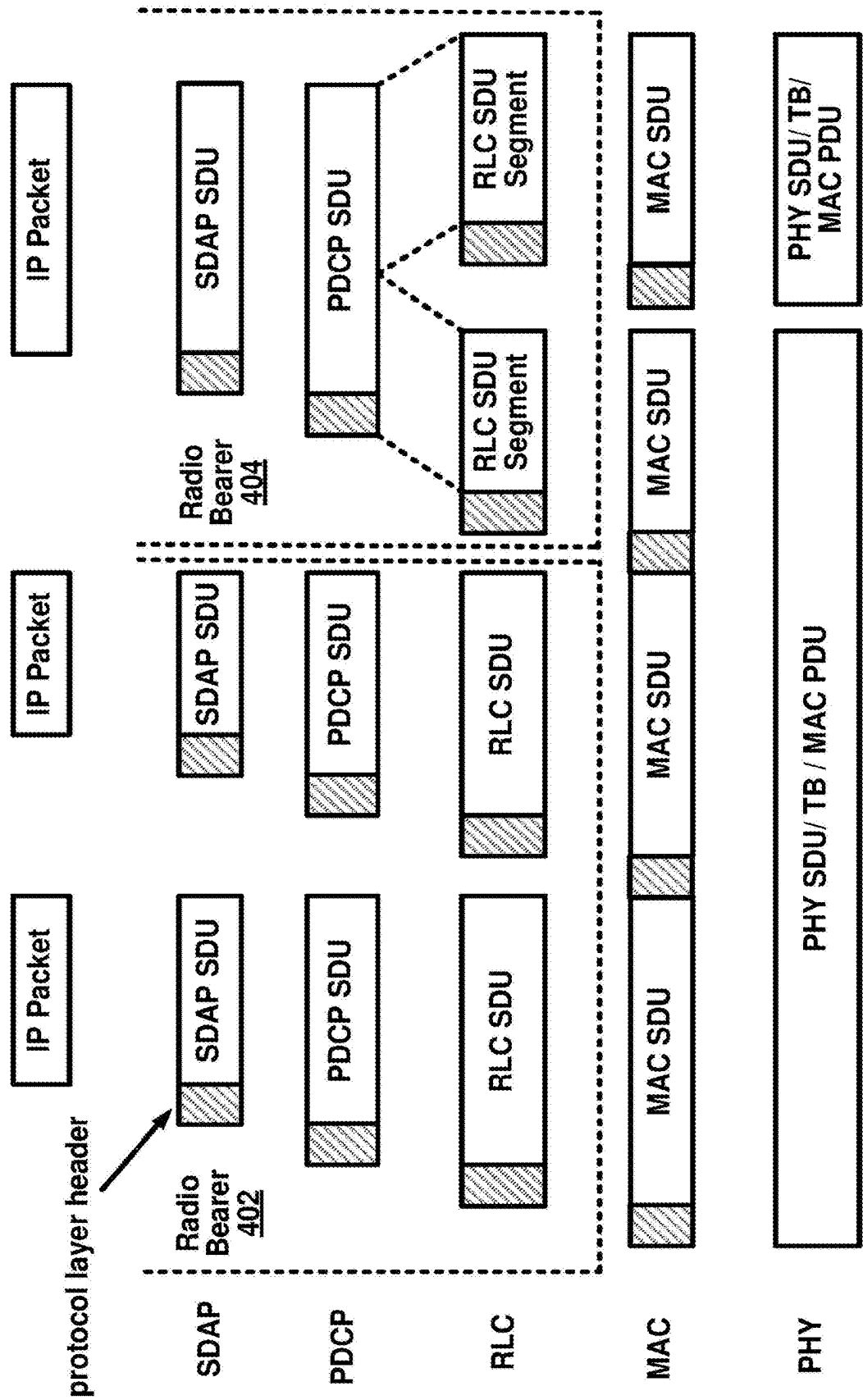


FIG. 4

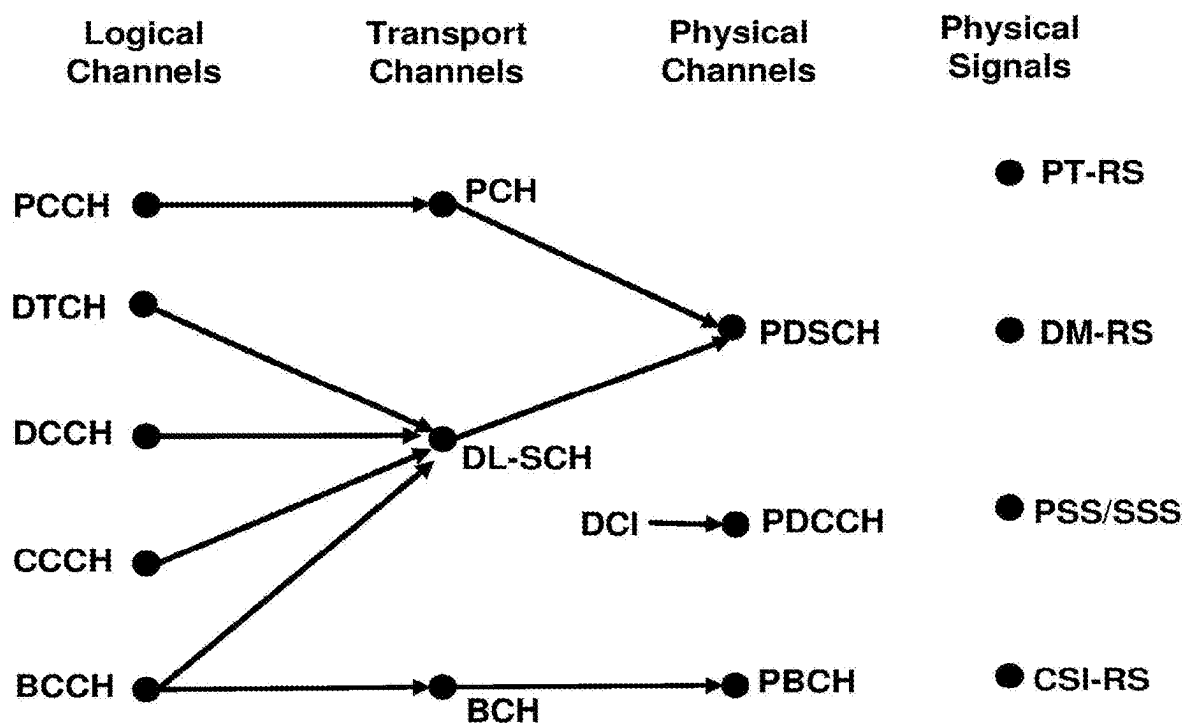


FIG. 5A

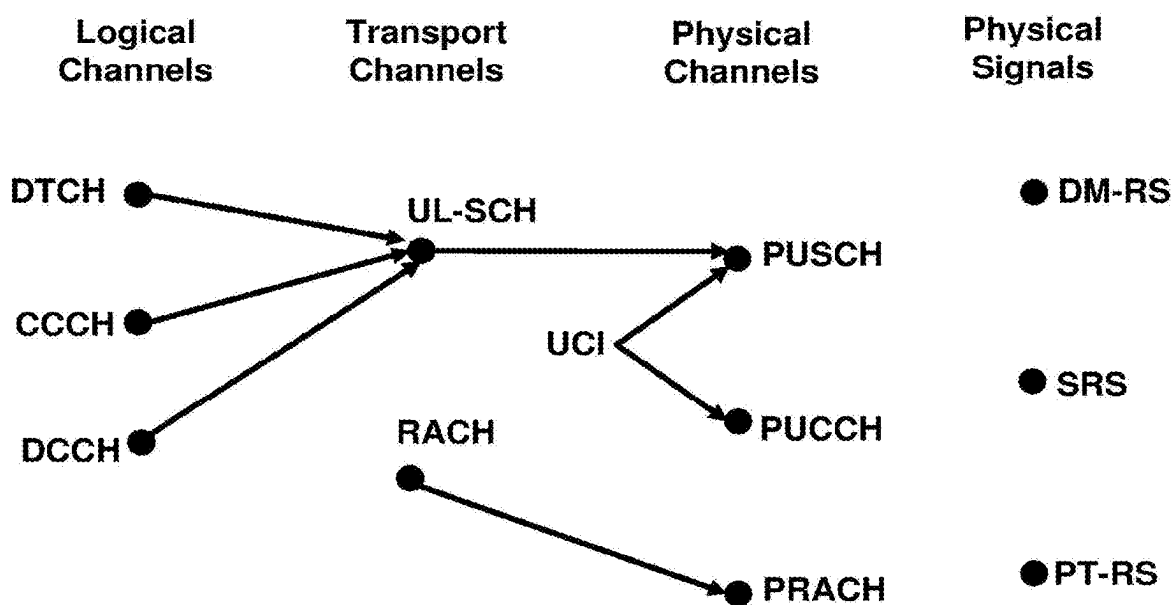


FIG. 5B

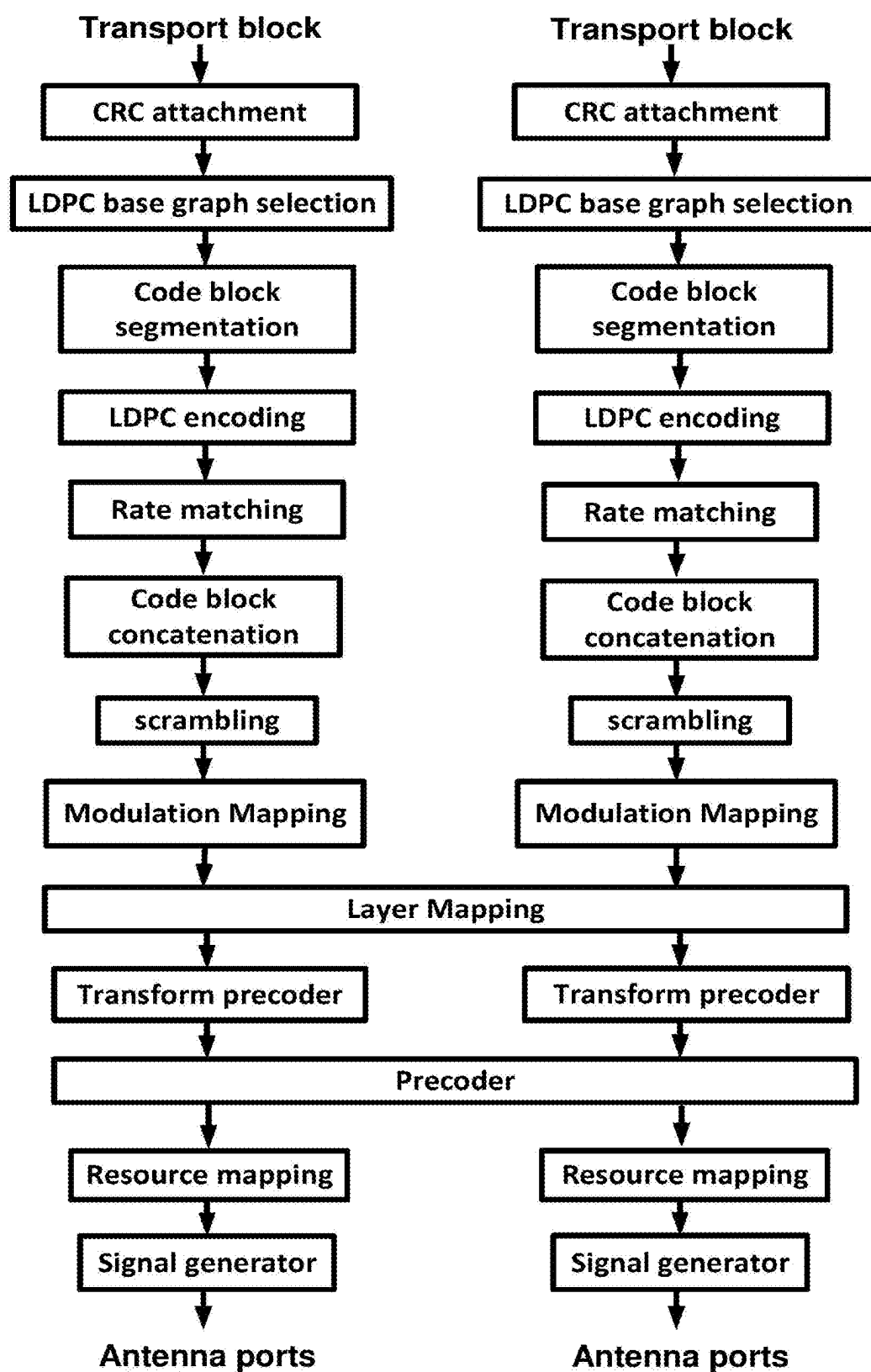


FIG. 6

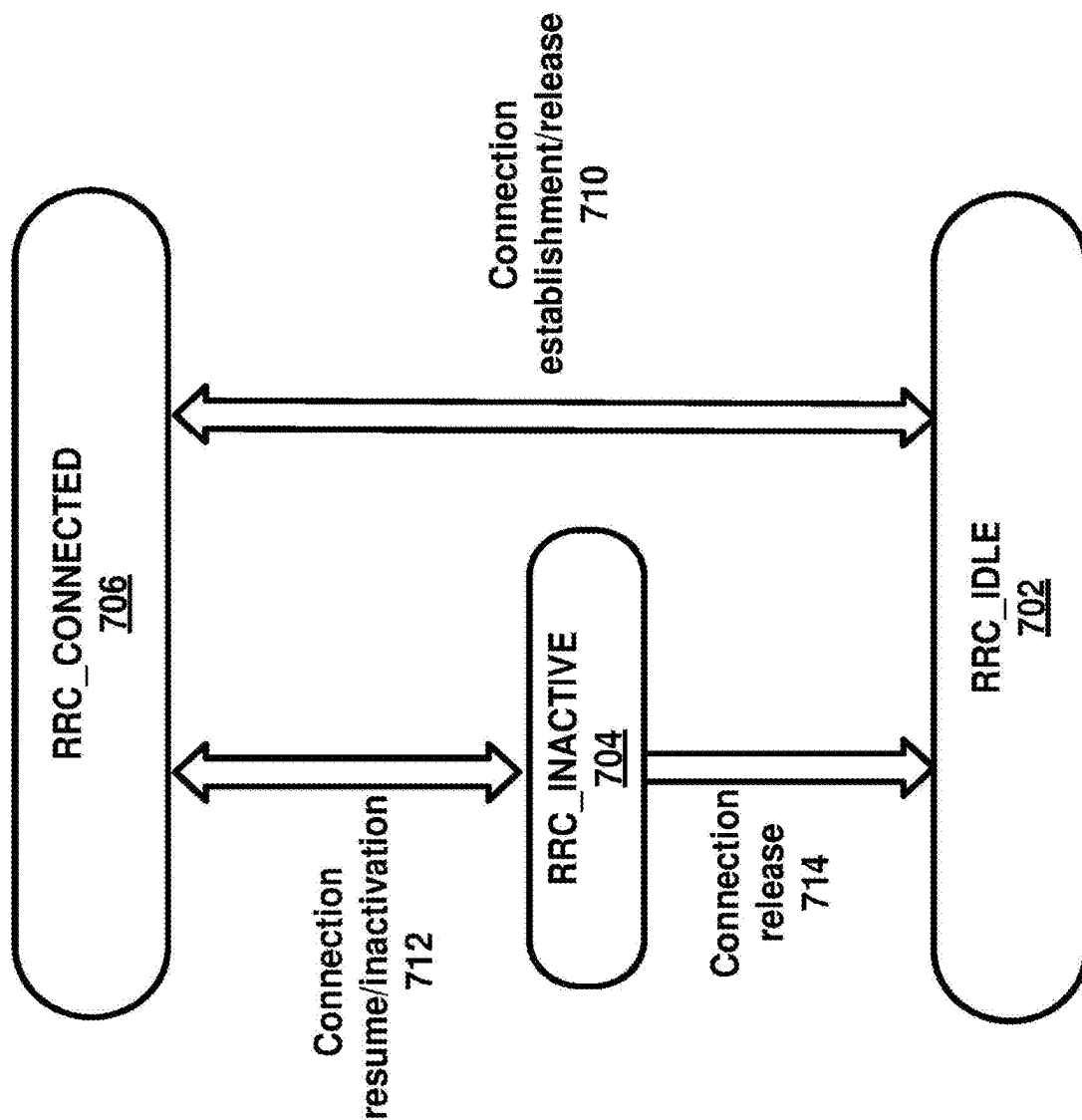


FIG. 7

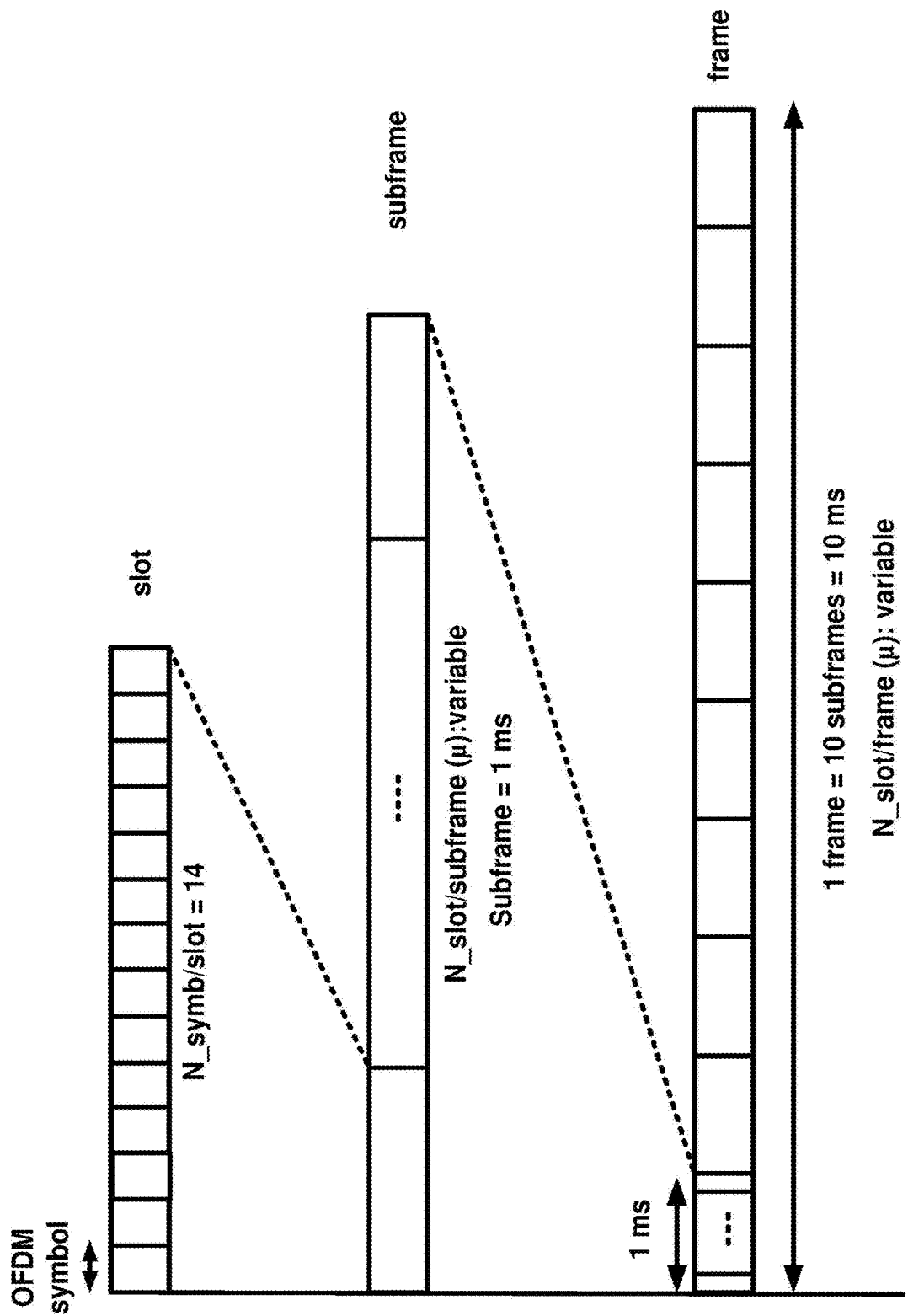


FIG. 8

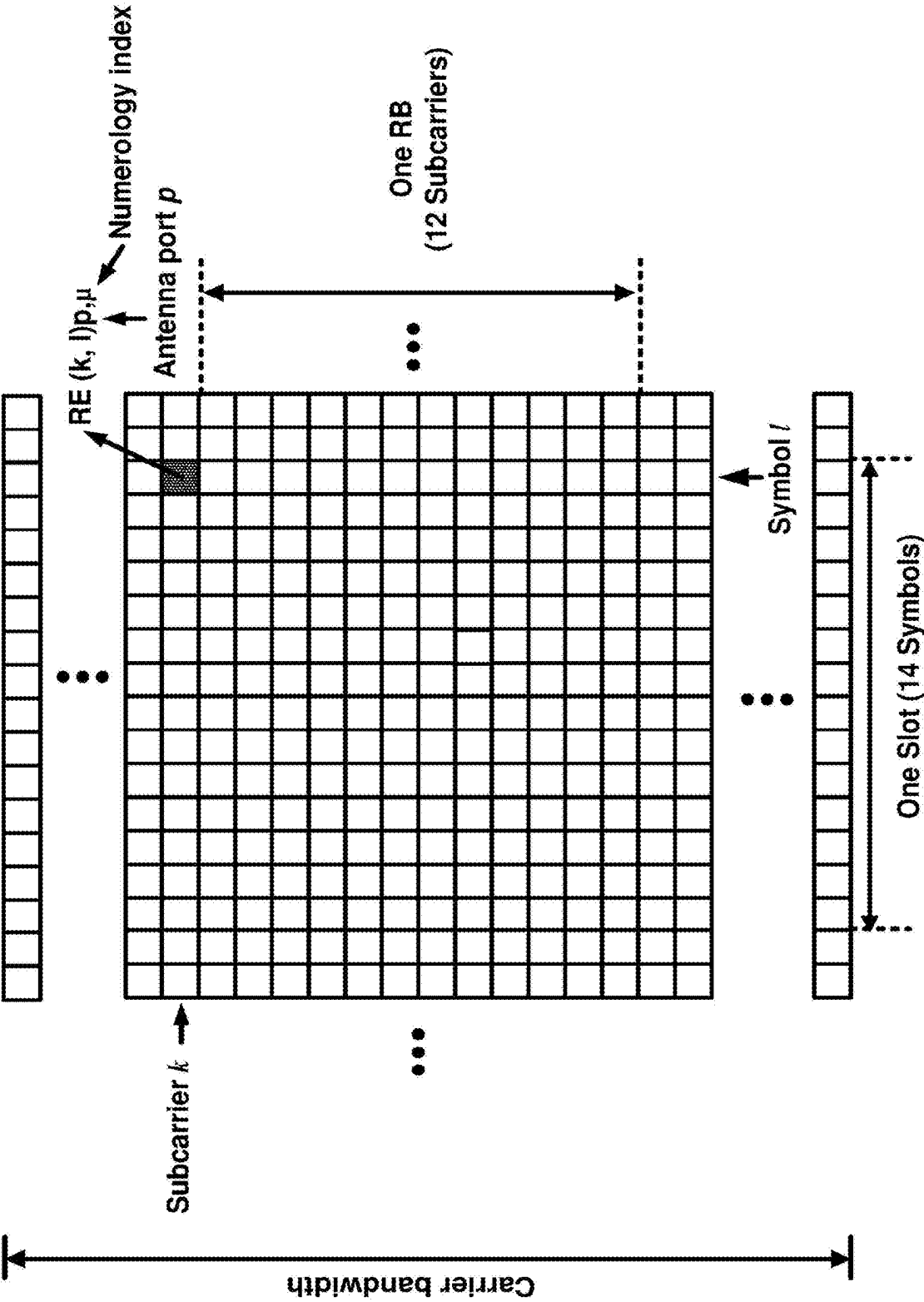


FIG. 9

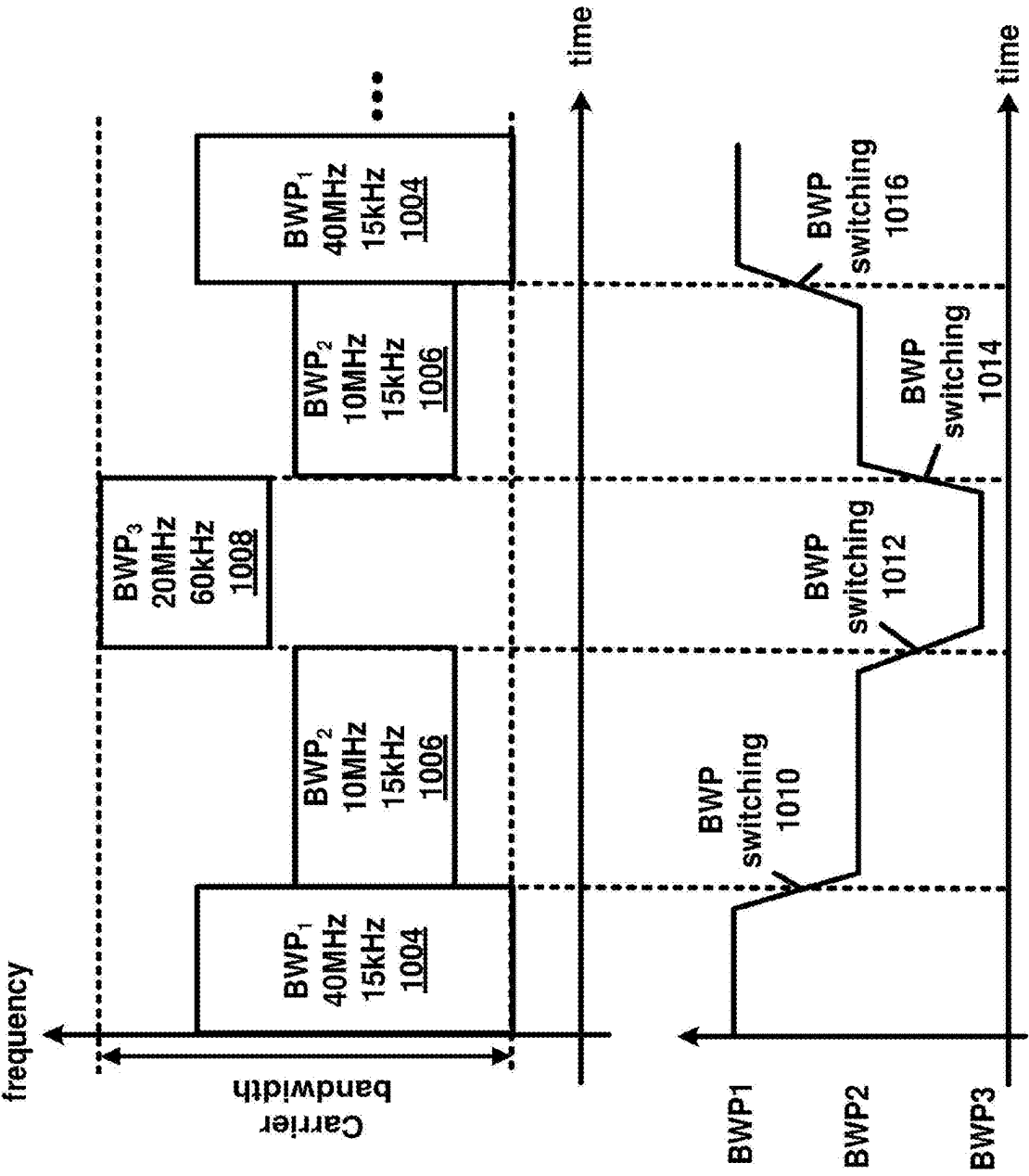


FIG. 10

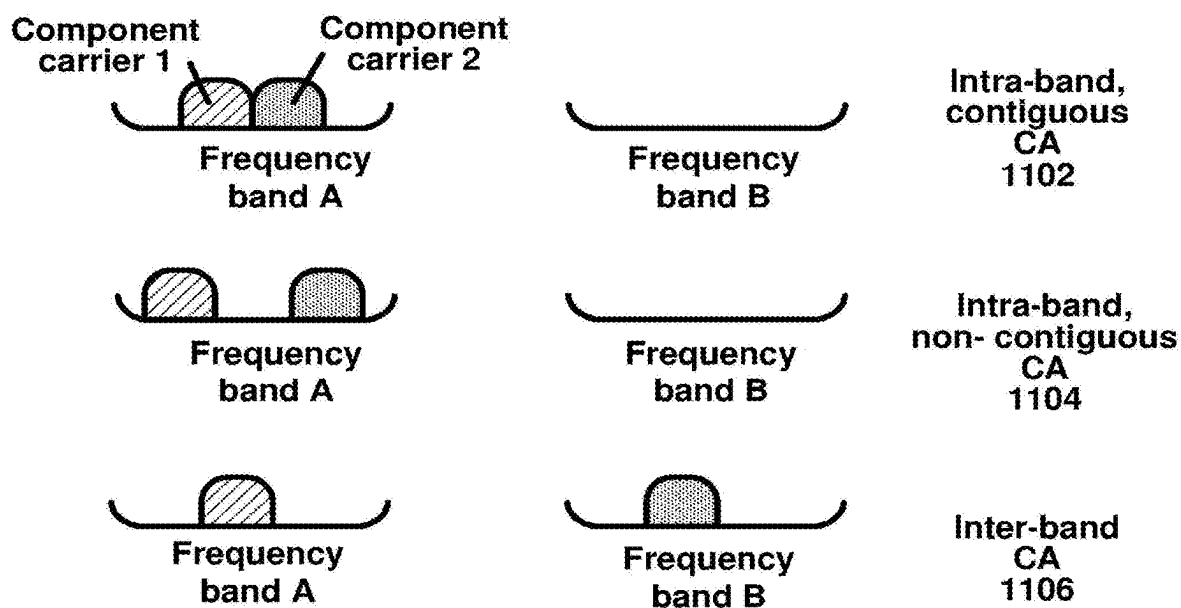


FIG. 11A

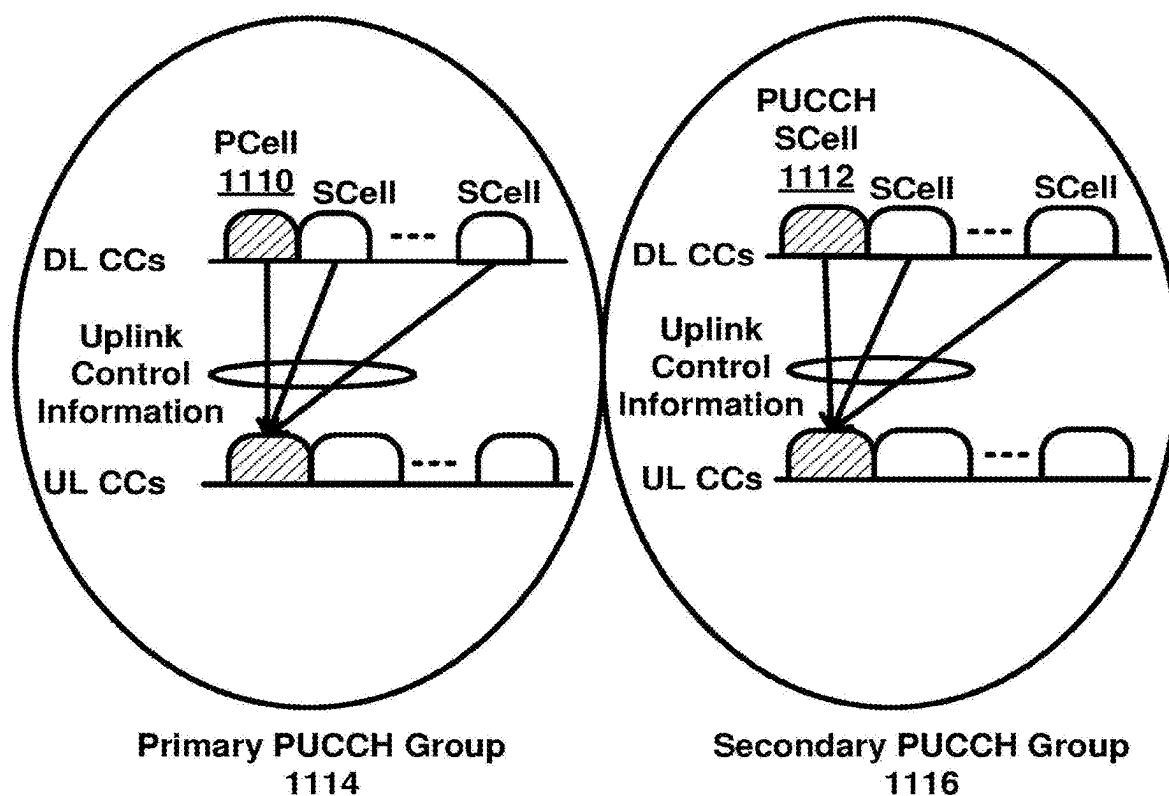


FIG. 11B

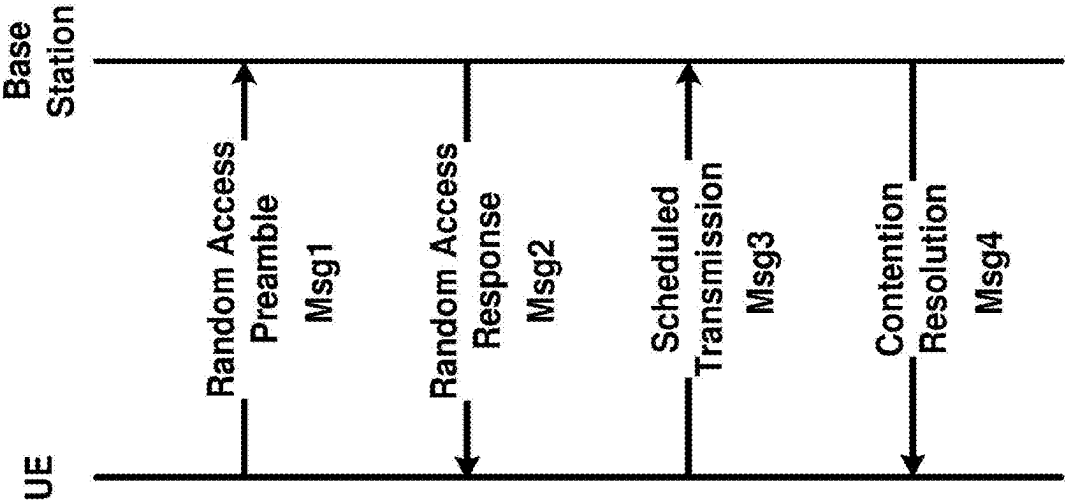


FIG. 12A

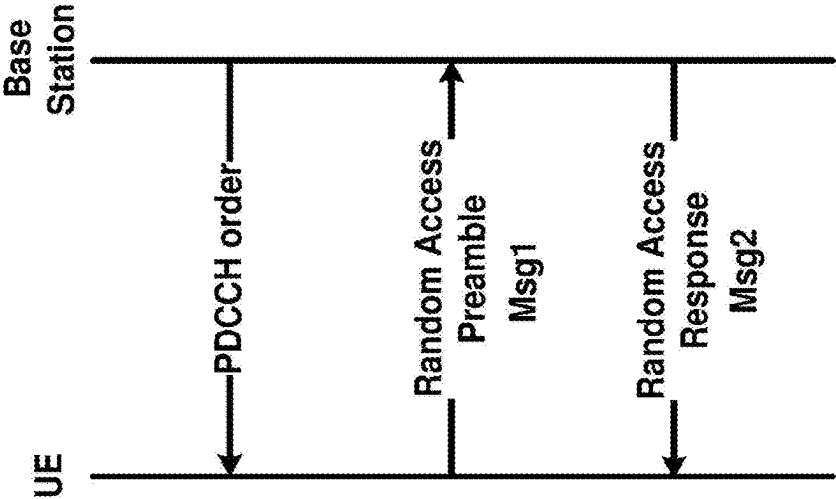


FIG. 12B

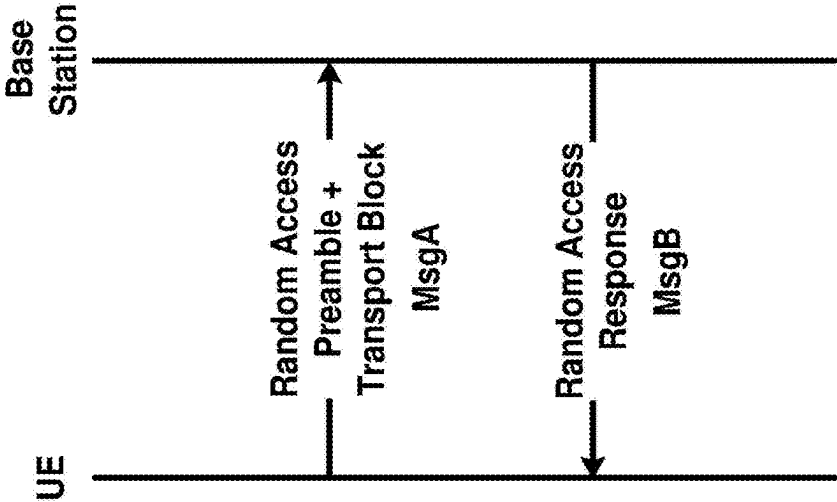


FIG. 12C

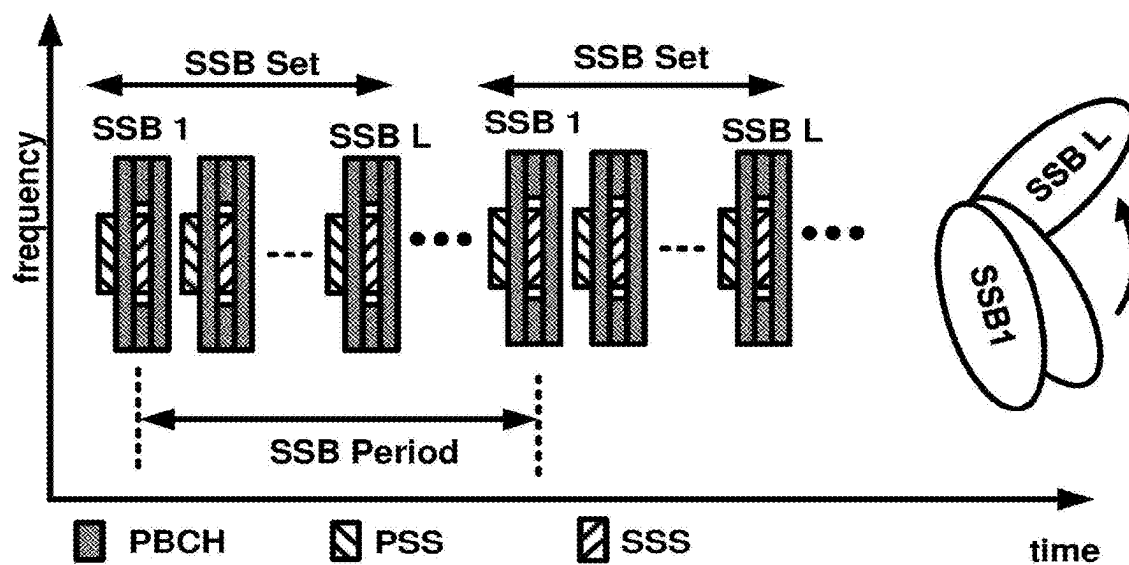


FIG. 13A

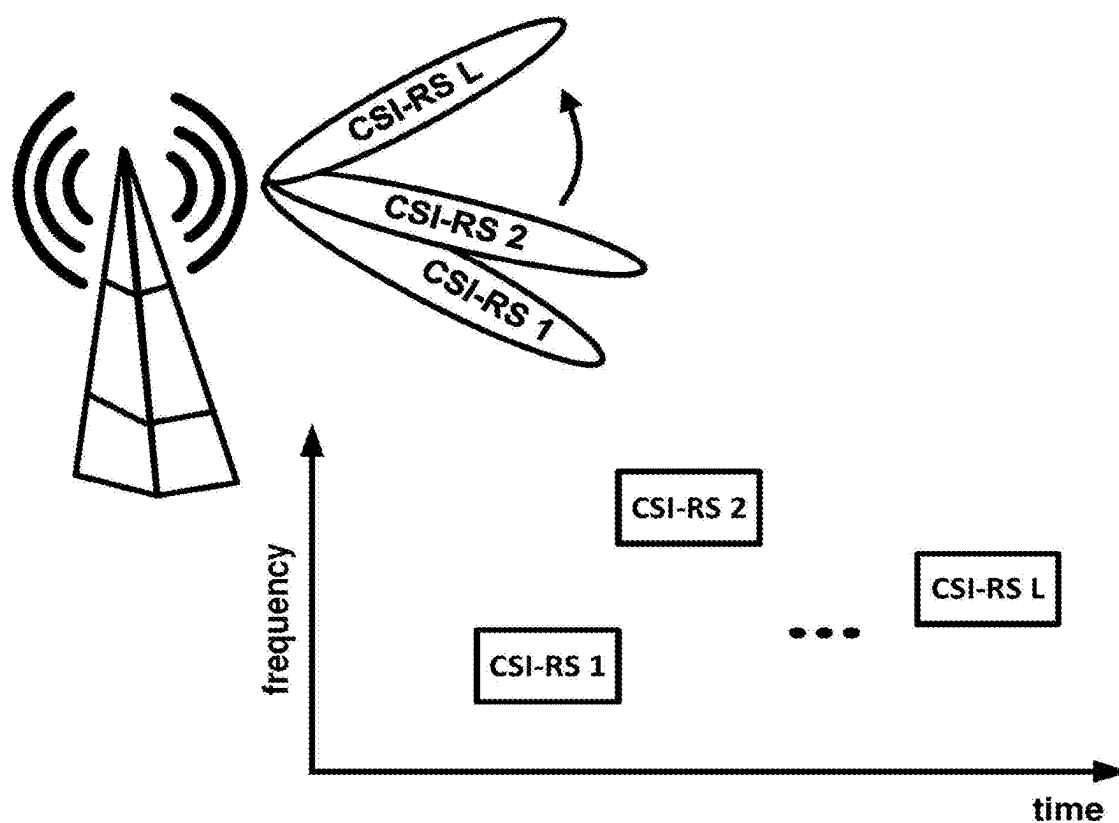


FIG. 13B

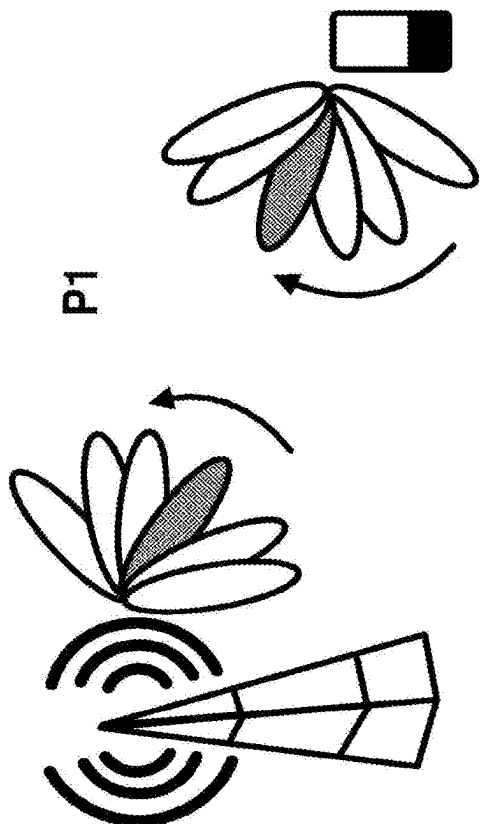


FIG. 14A

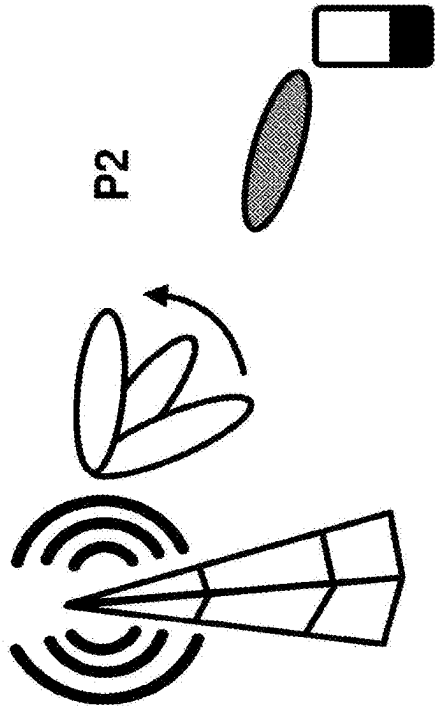


FIG. 14B

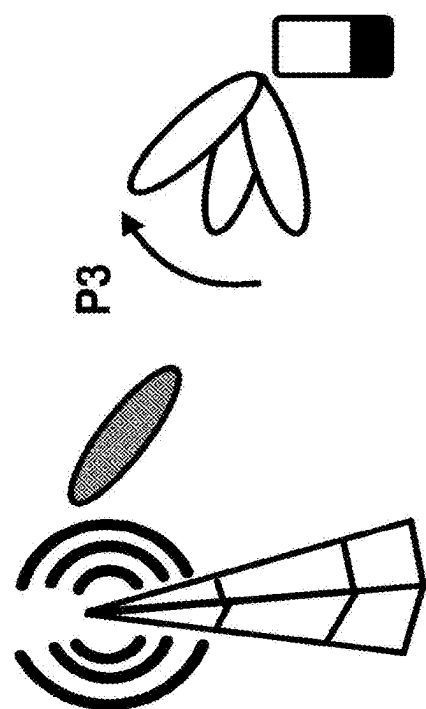


FIG. 14C

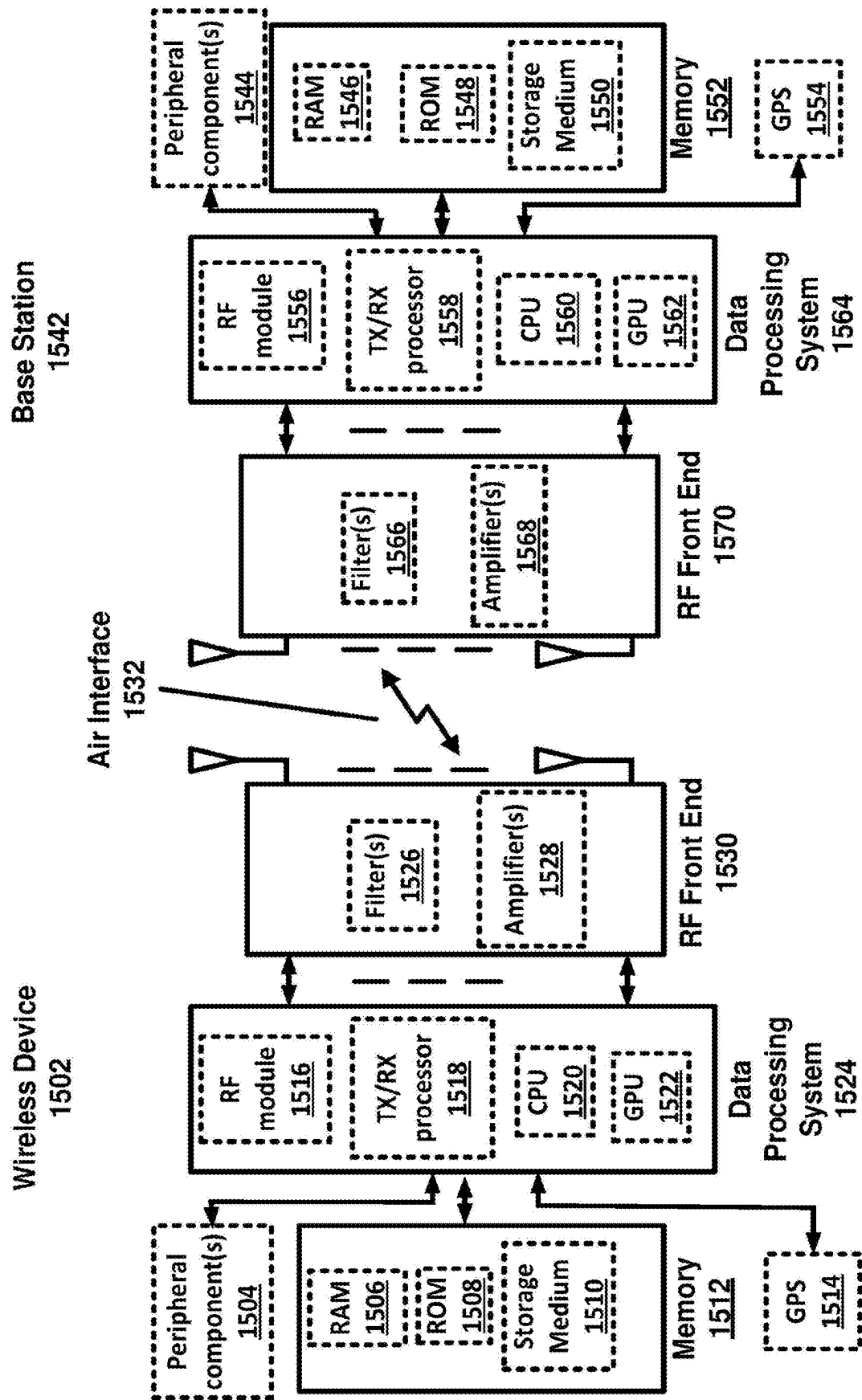


FIG. 15

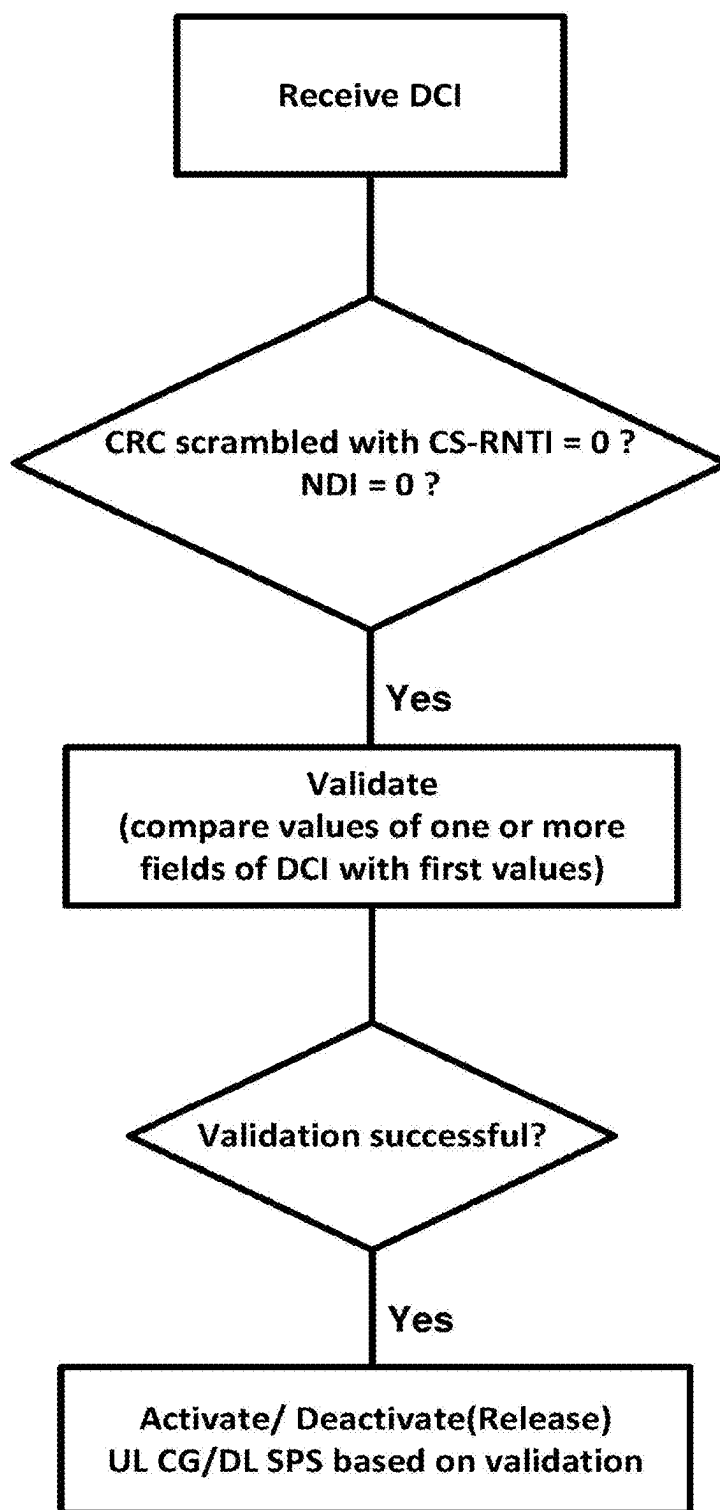


FIG. 16

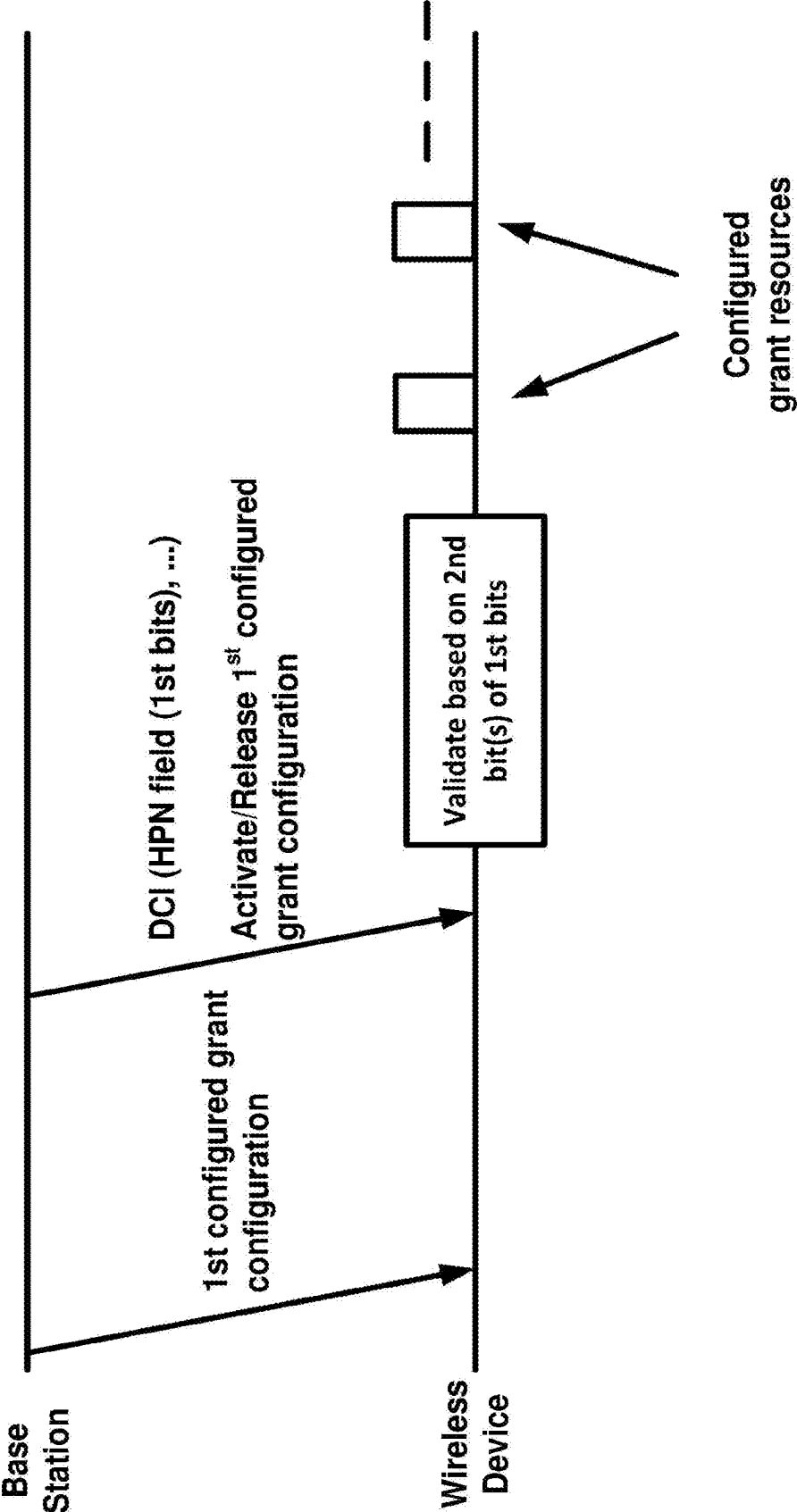


FIG. 17

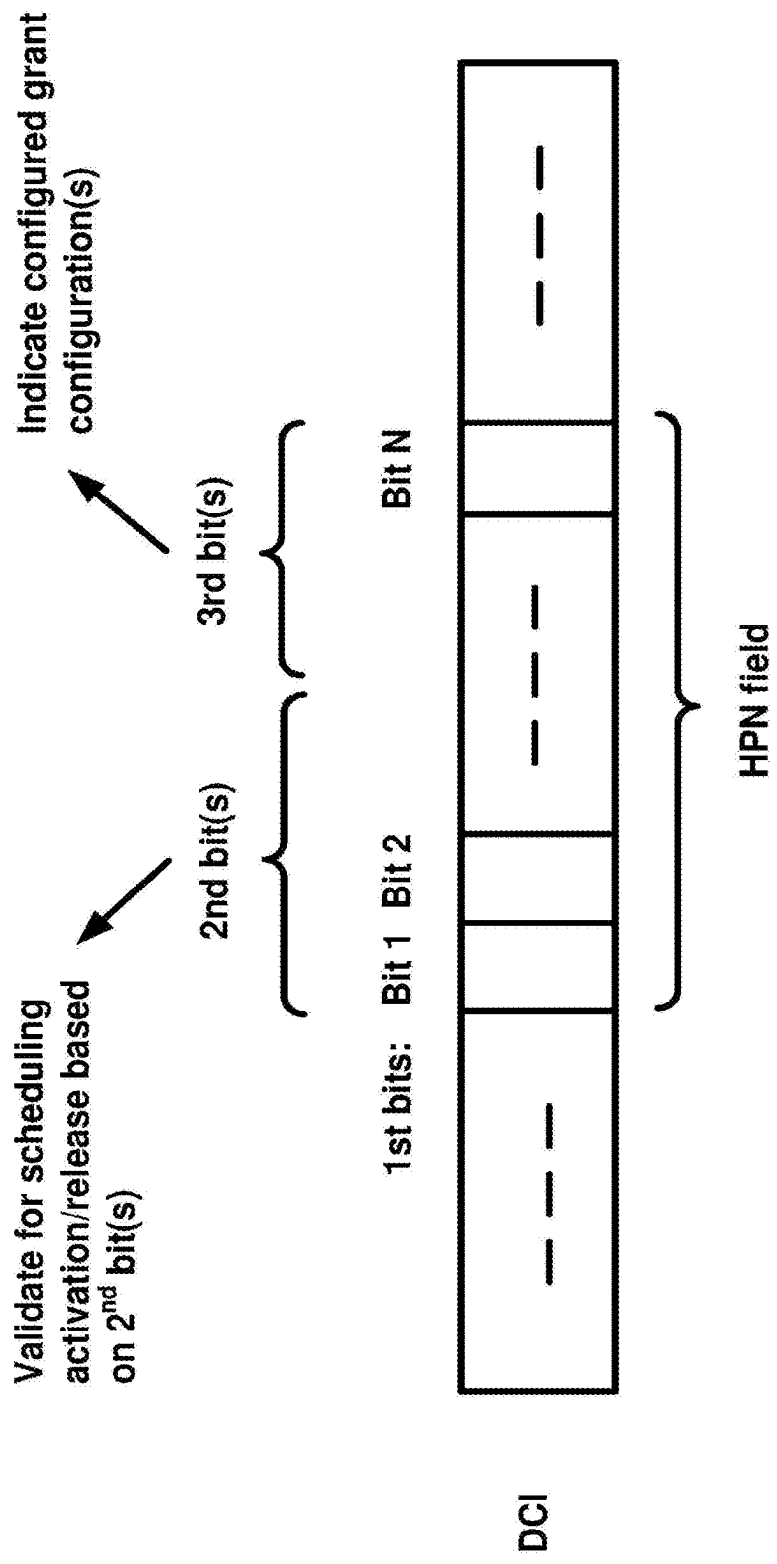


FIG. 18

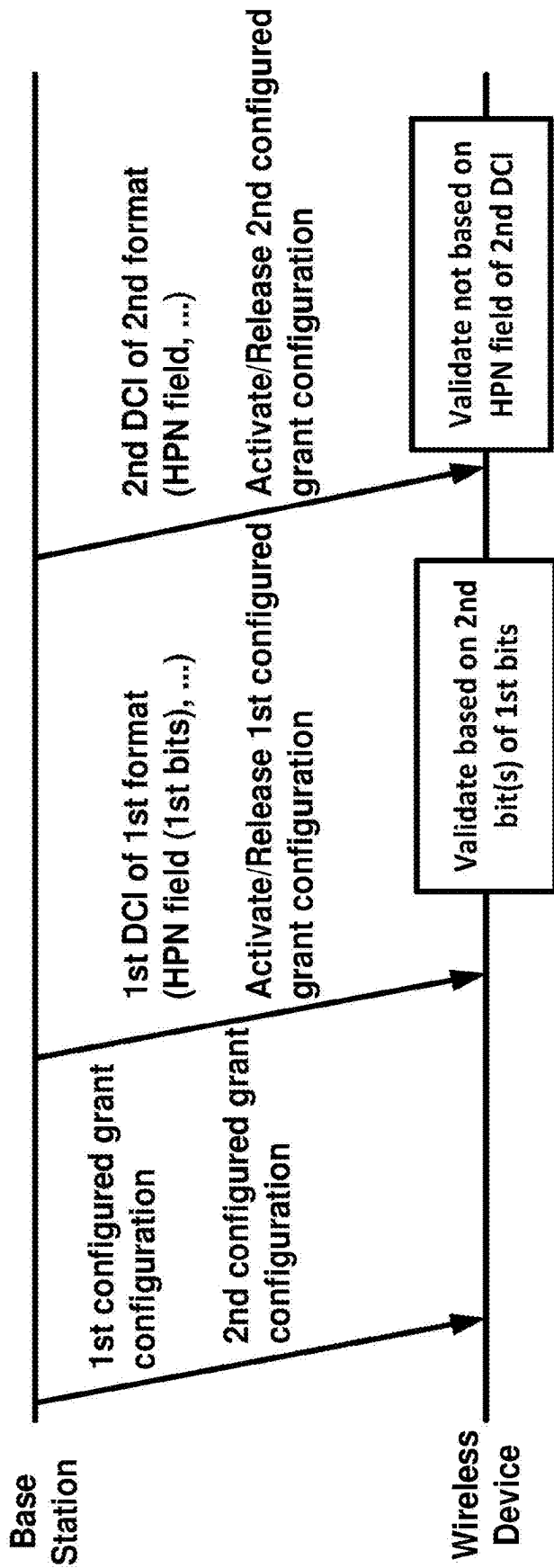


FIG. 19

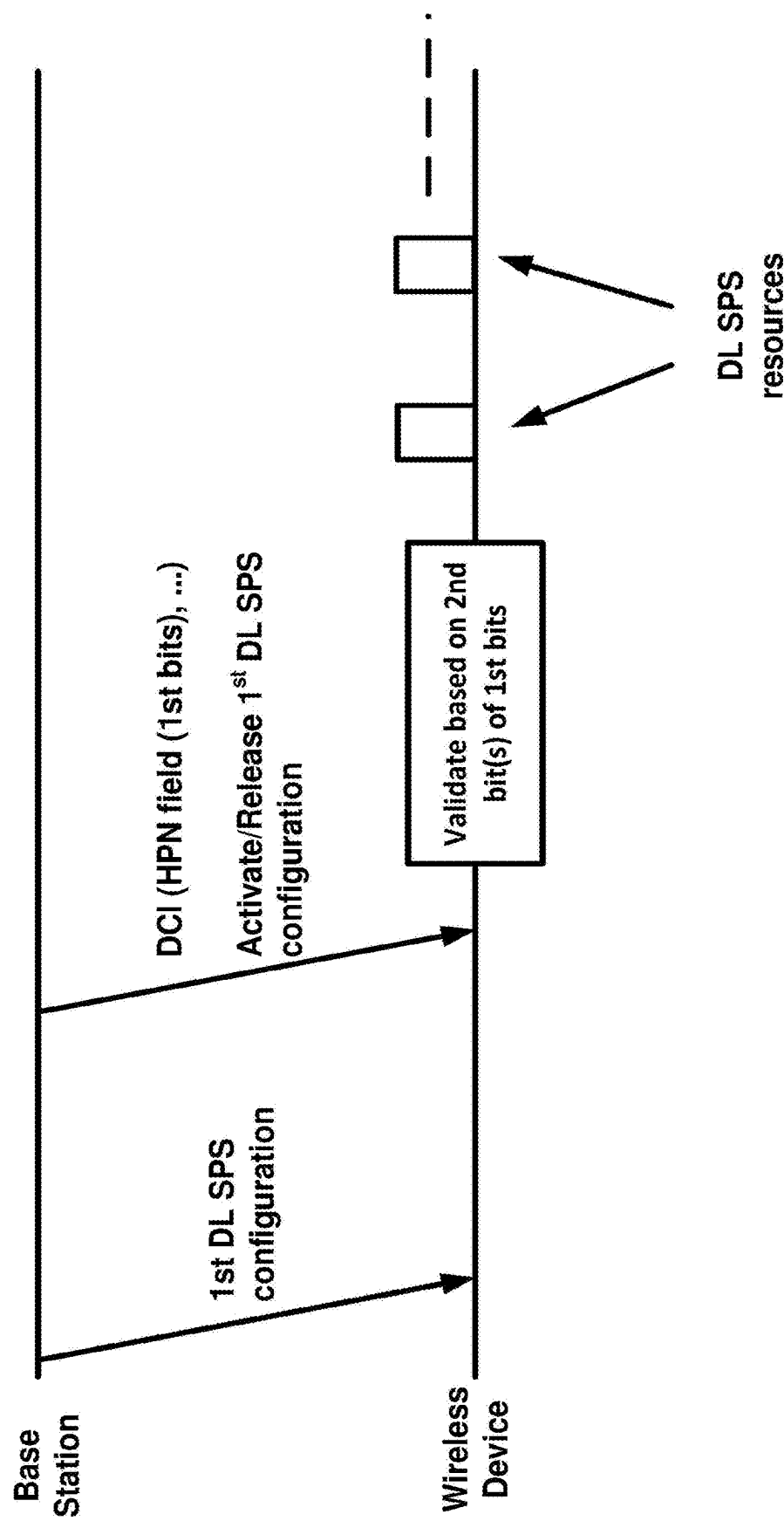


FIG. 20

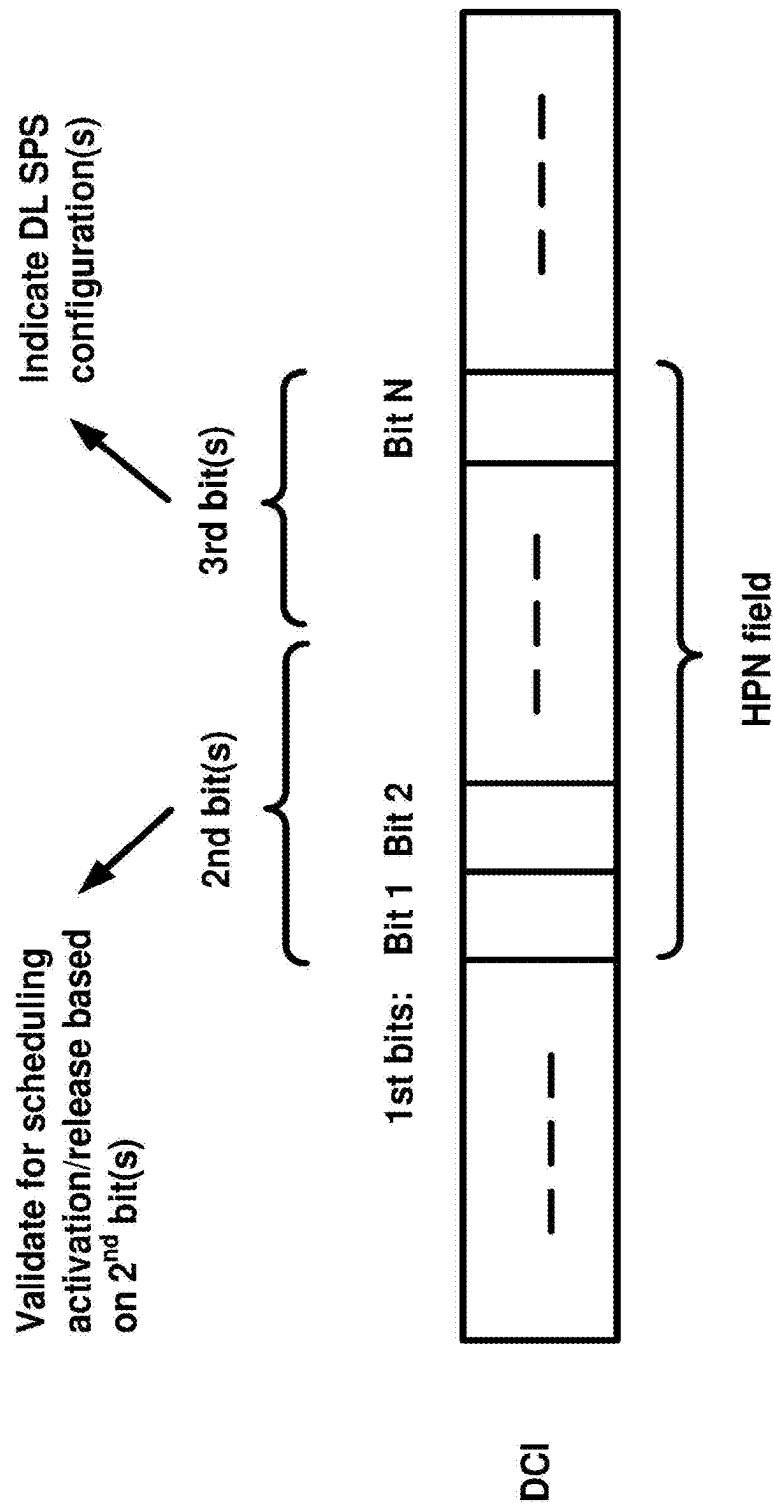


FIG. 21

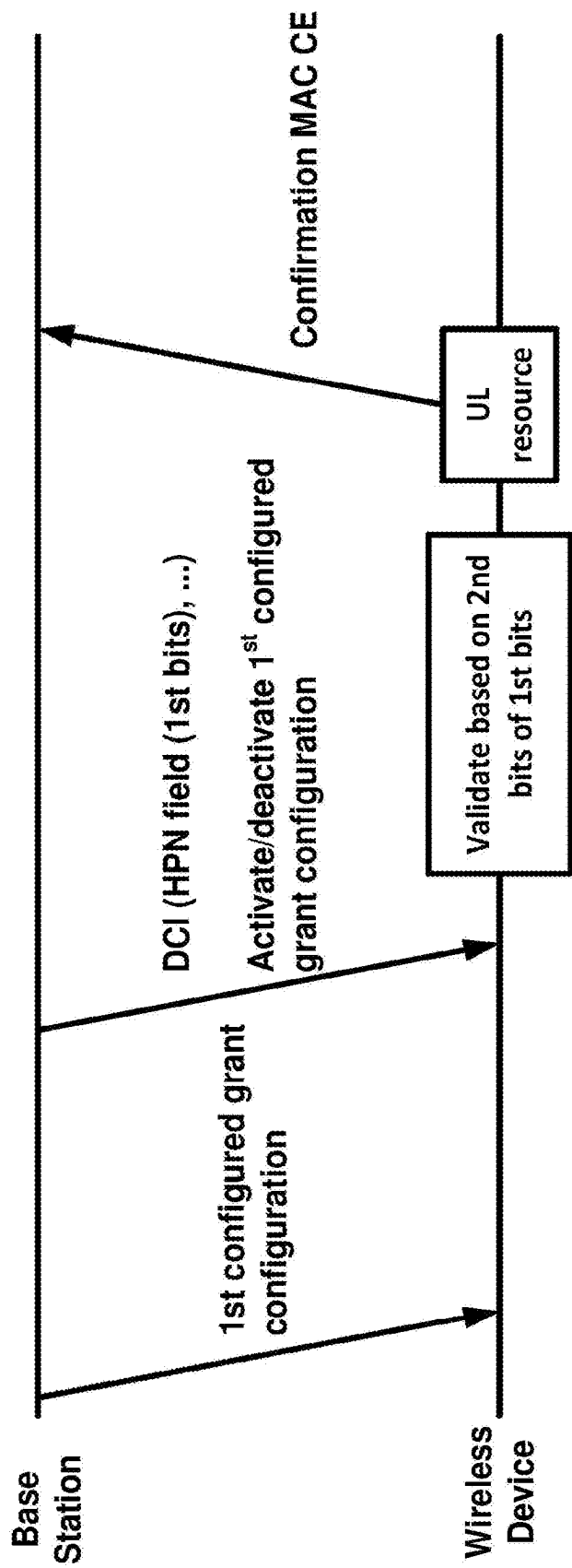


FIG. 22

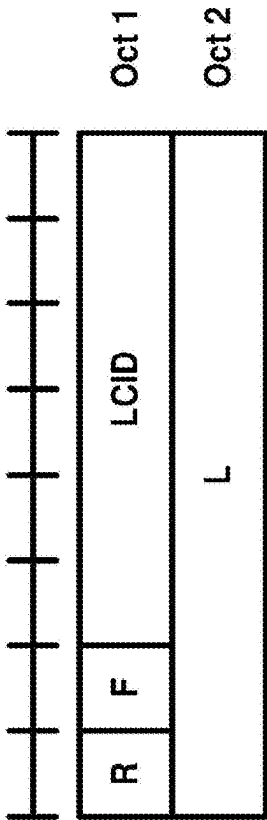


FIG. 23C

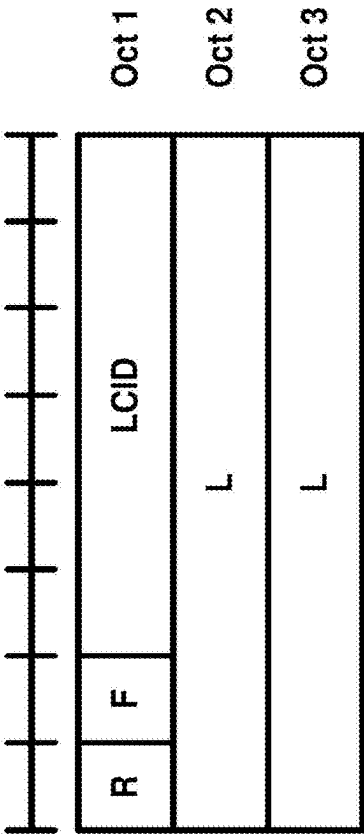


FIG. 23B

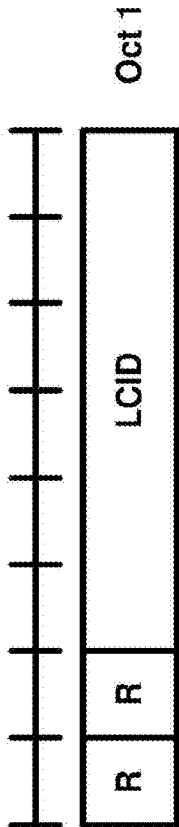


FIG. 23A

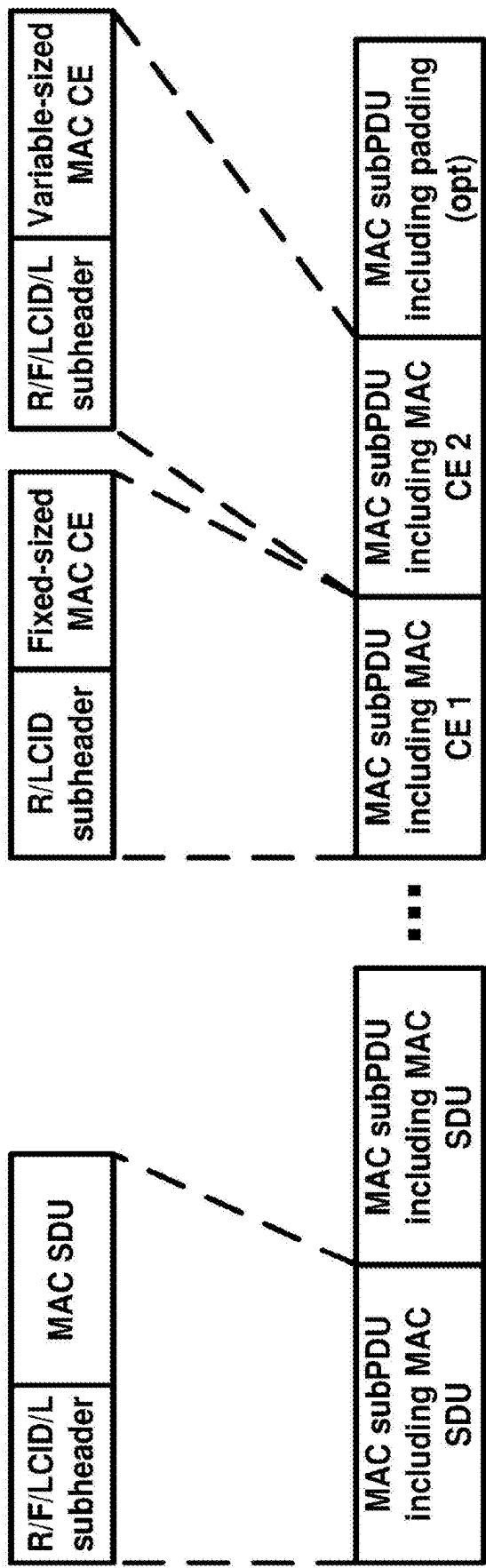


FIG. 24

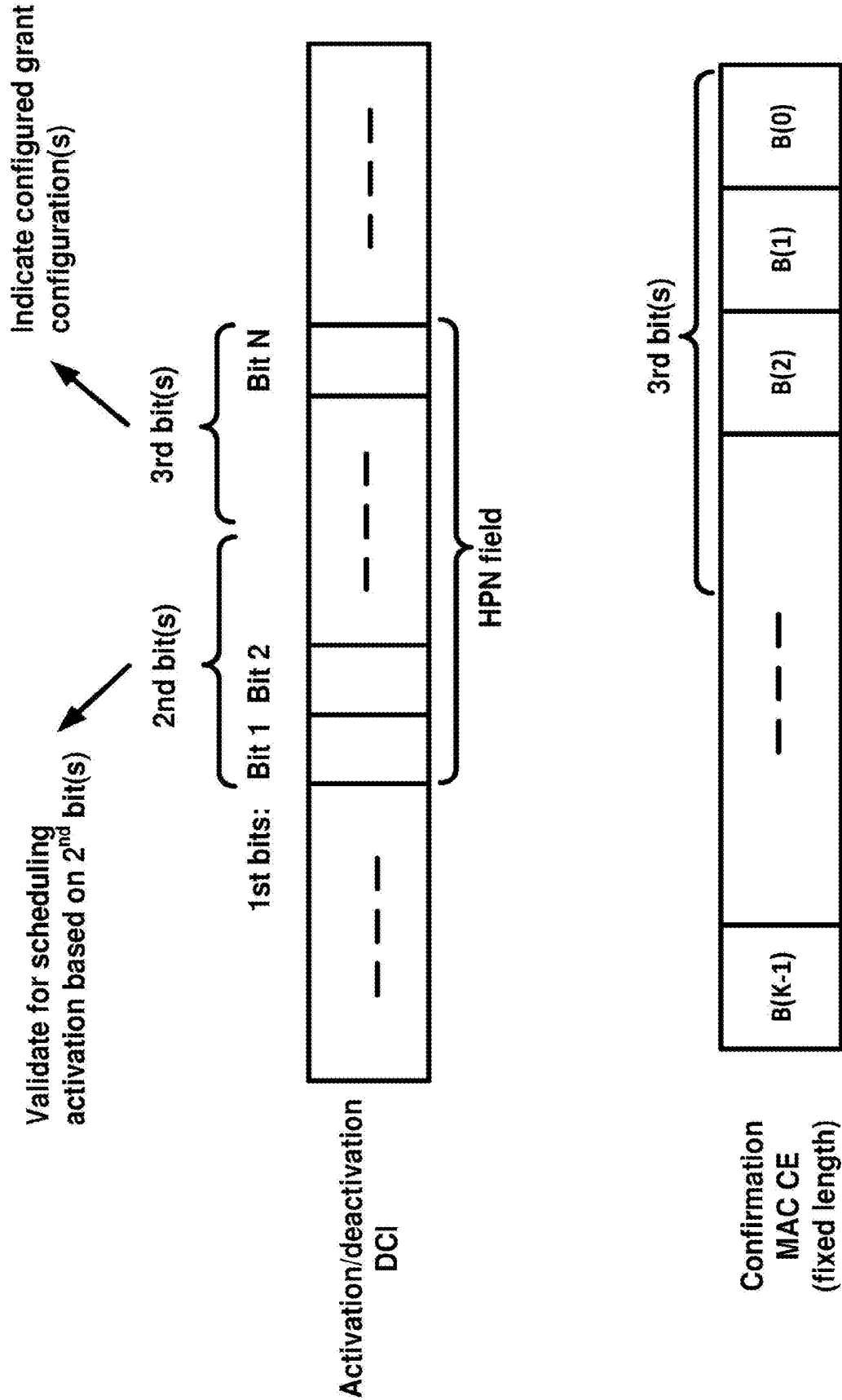


FIG. 25

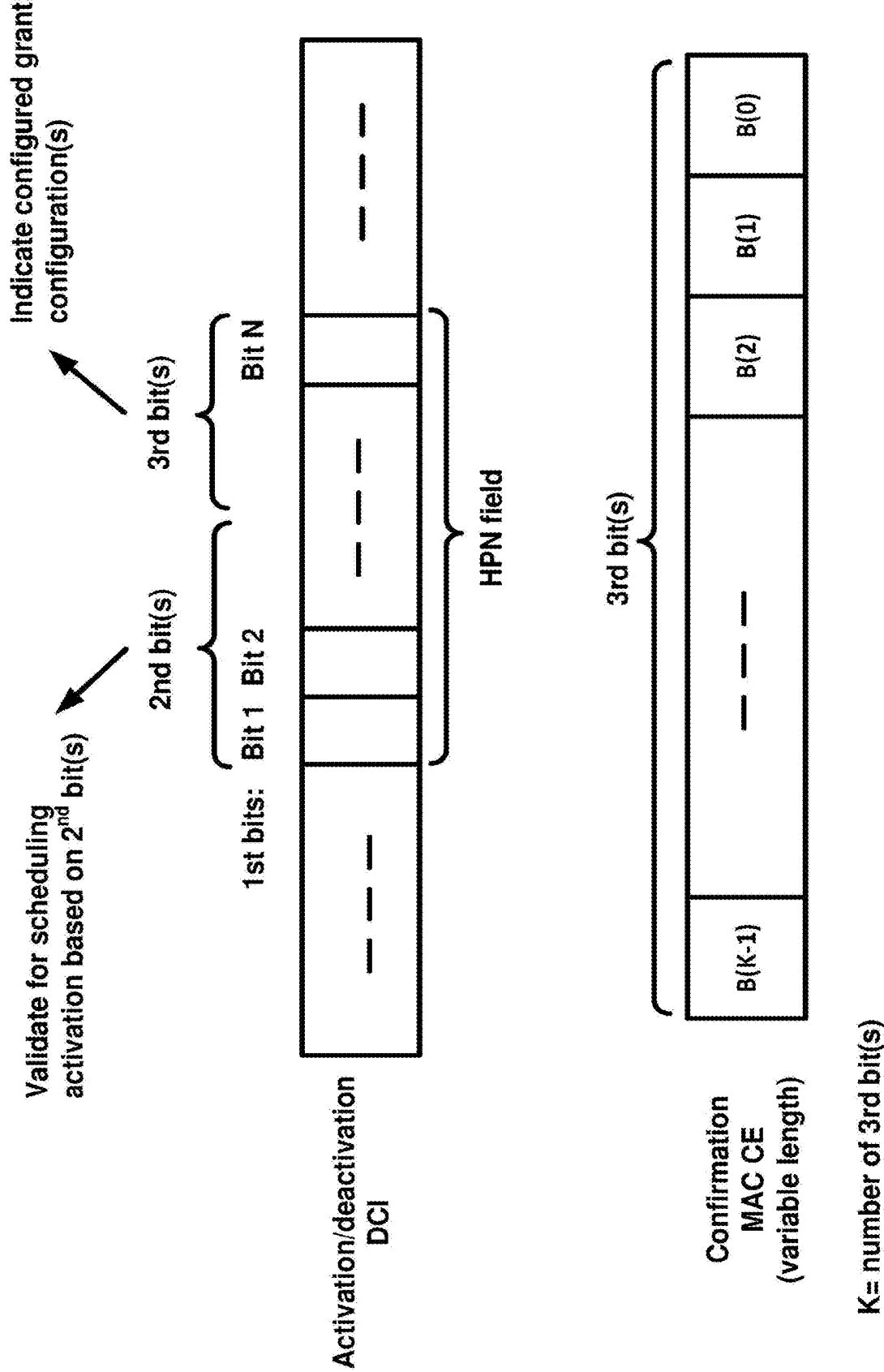


FIG. 26

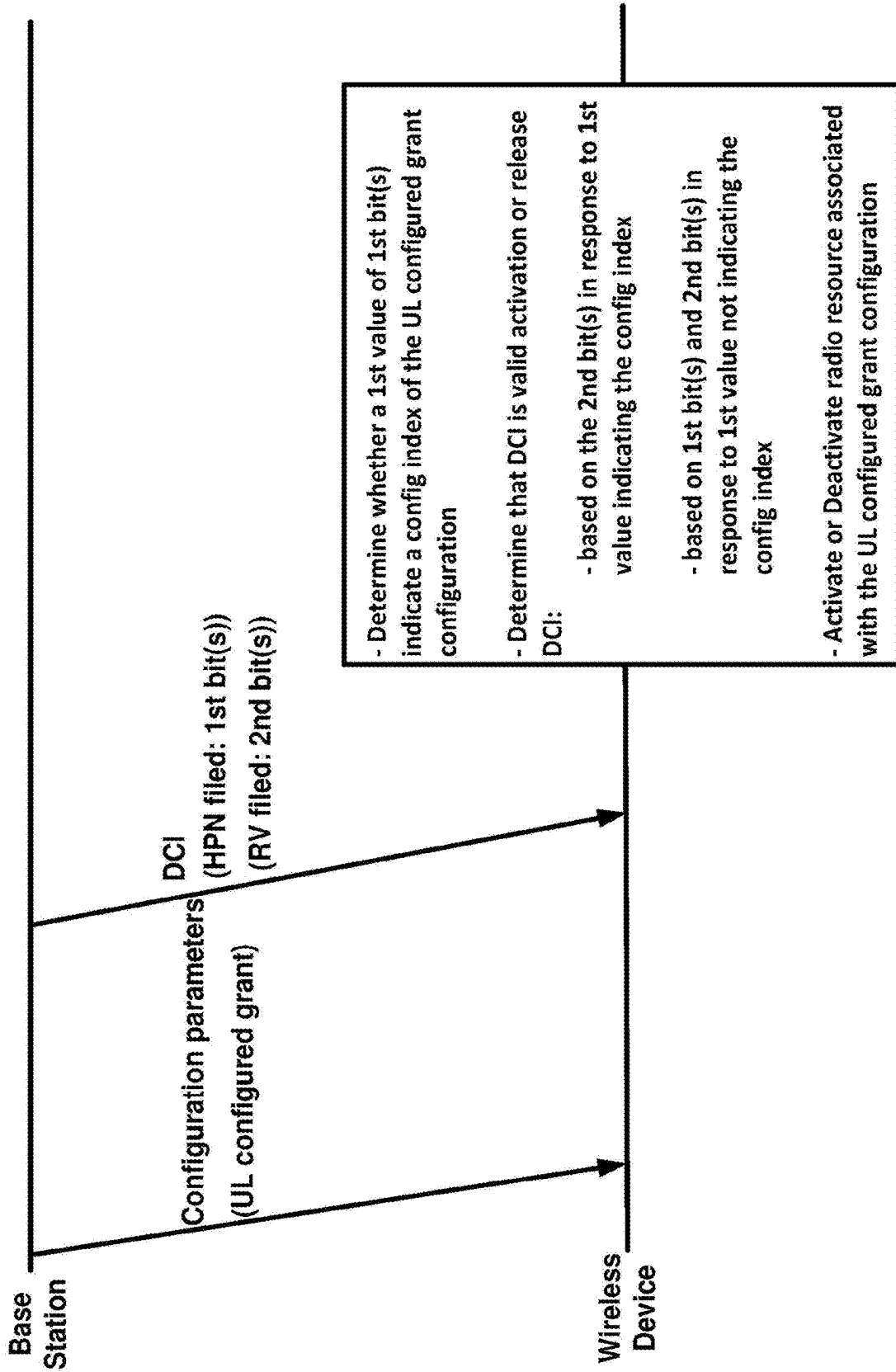


FIG. 27

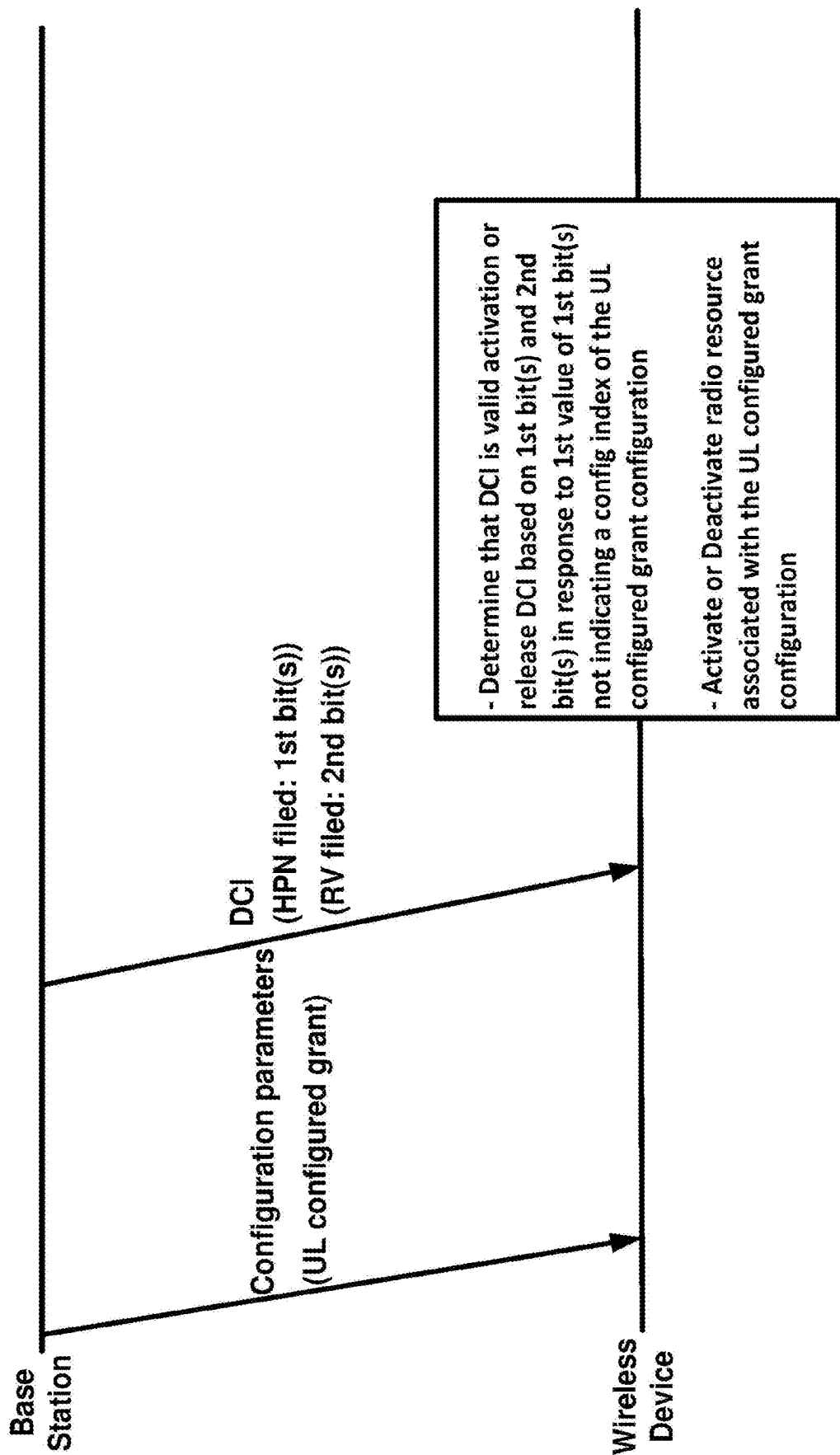


FIG. 28

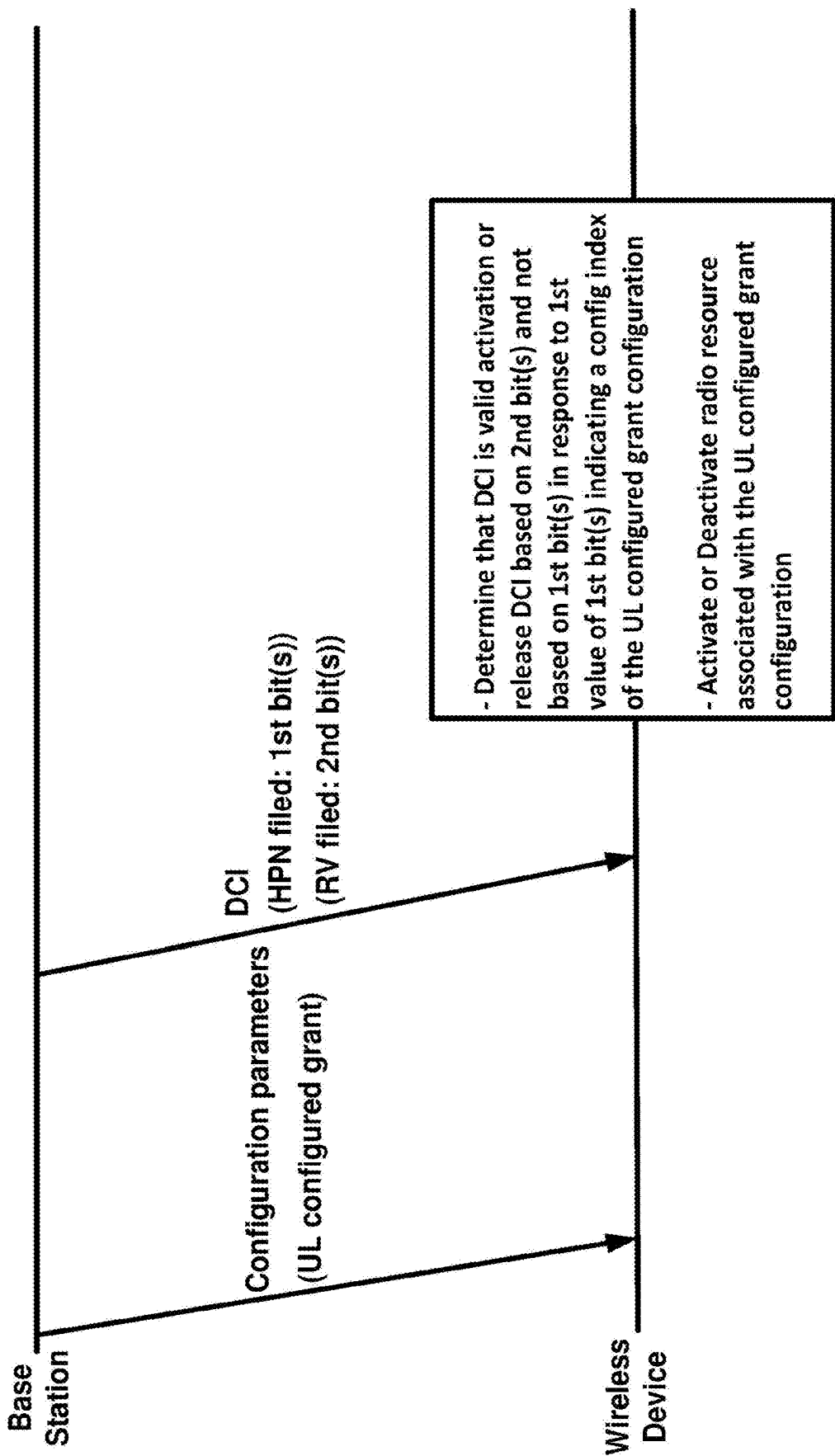


FIG. 29

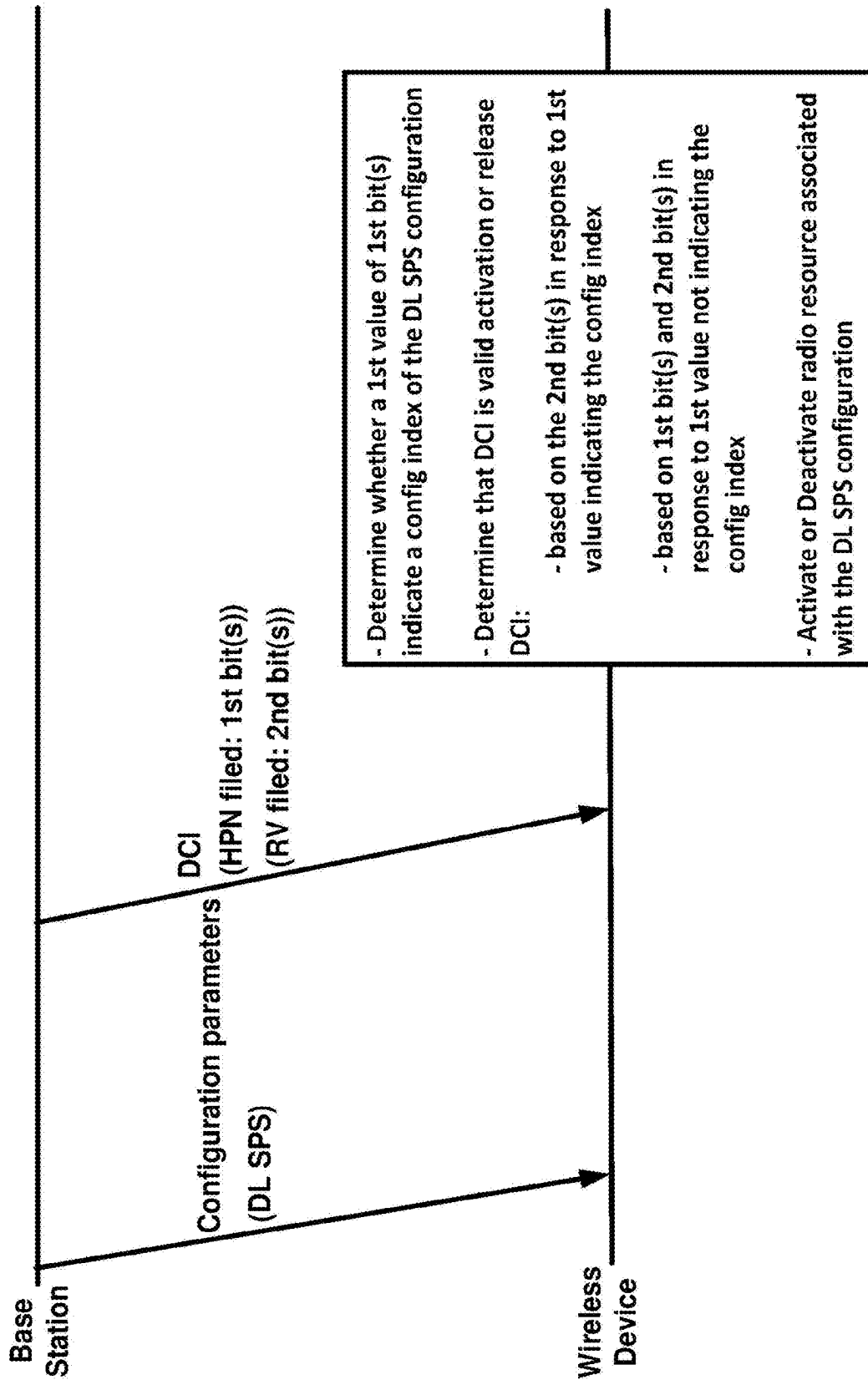


FIG. 30

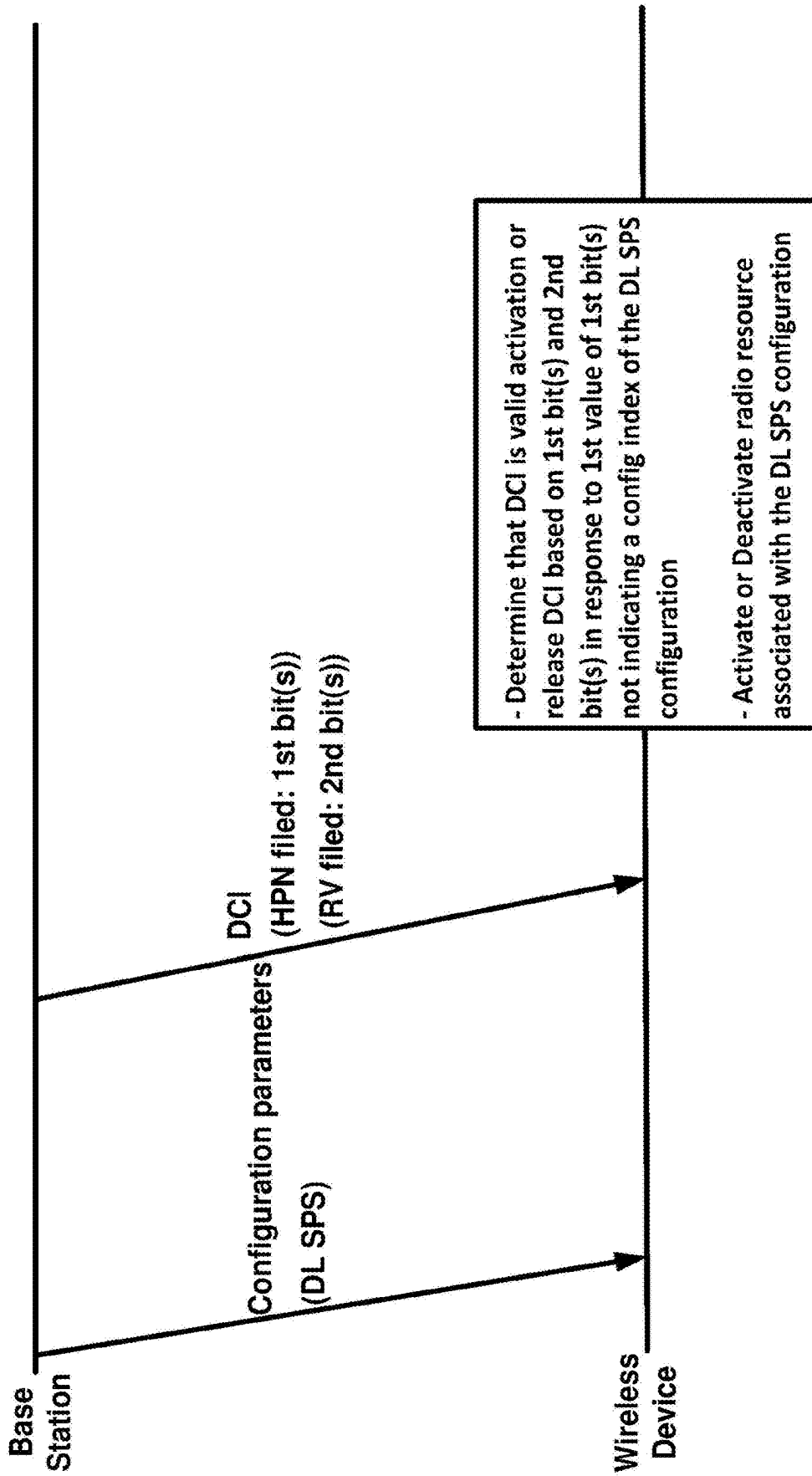


FIG. 31

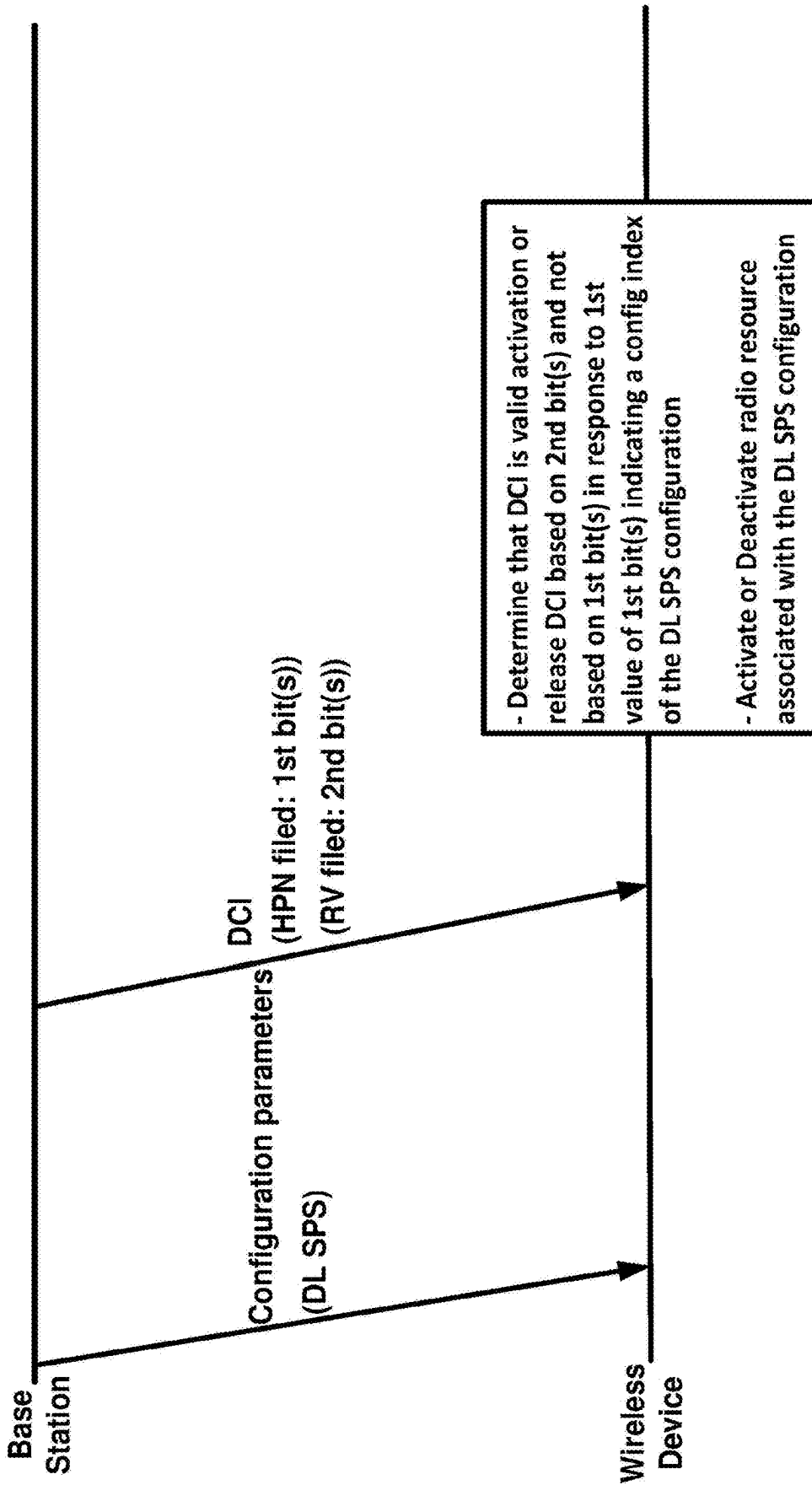


FIG. 32

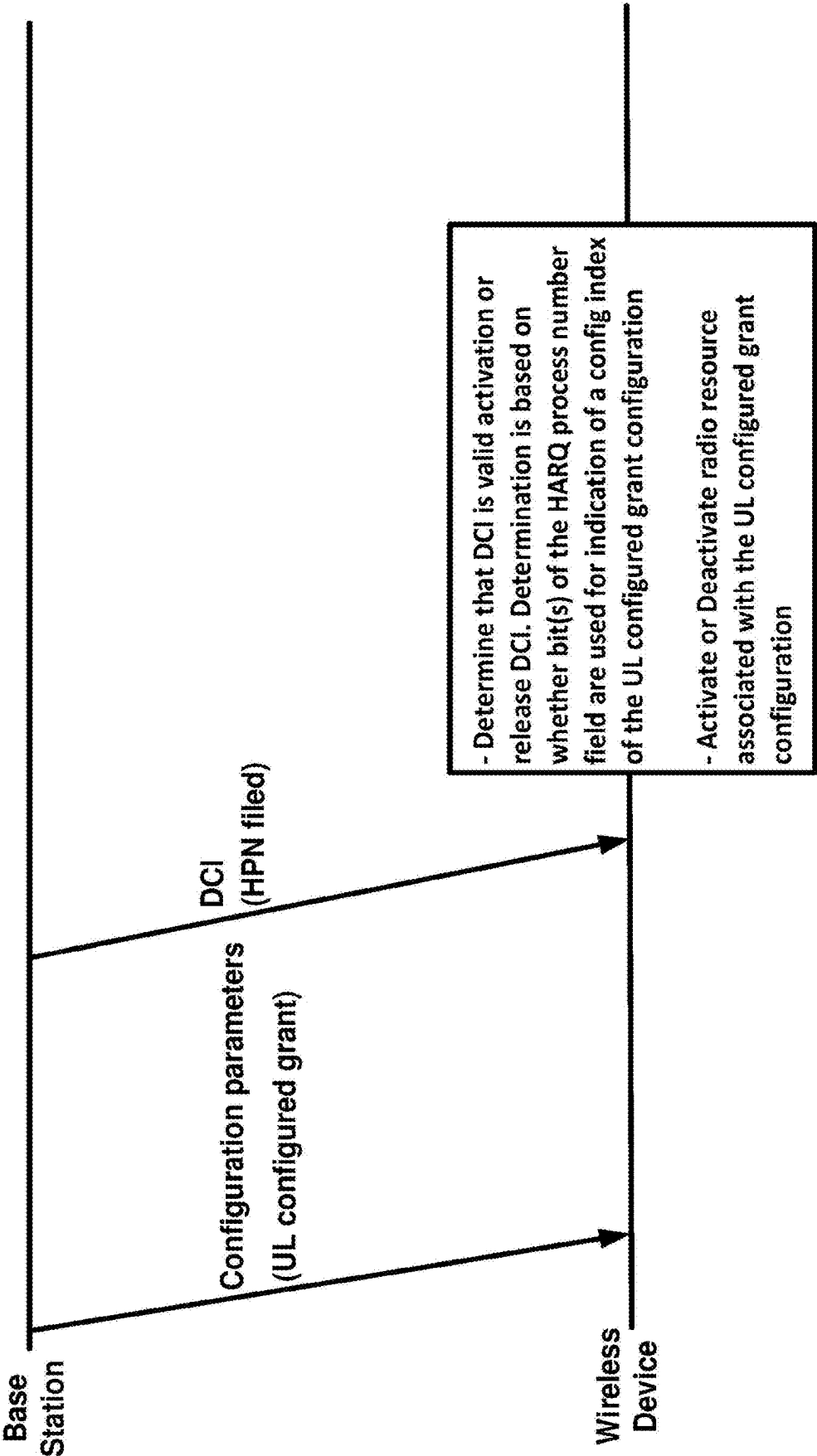


FIG. 33

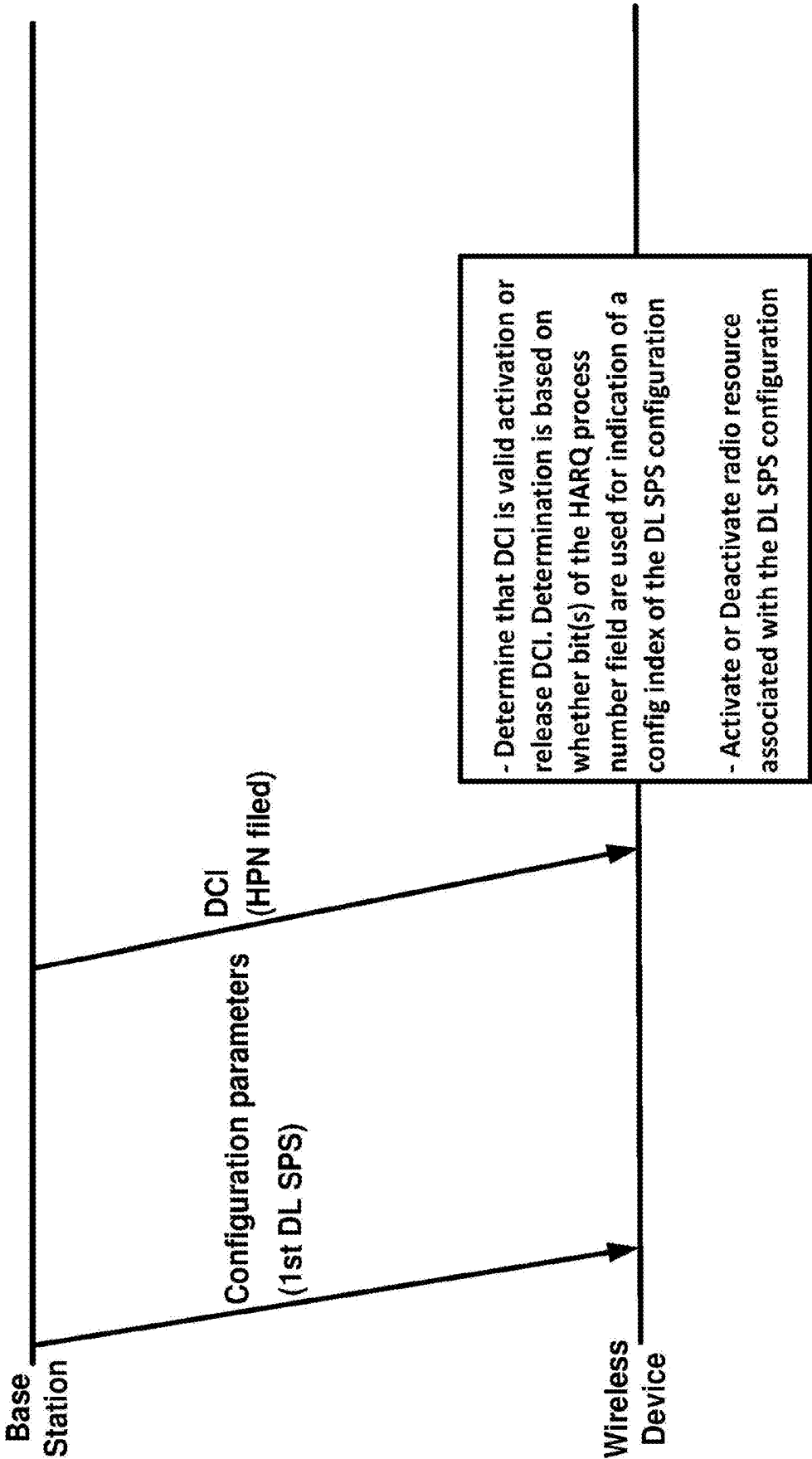


FIG. 34

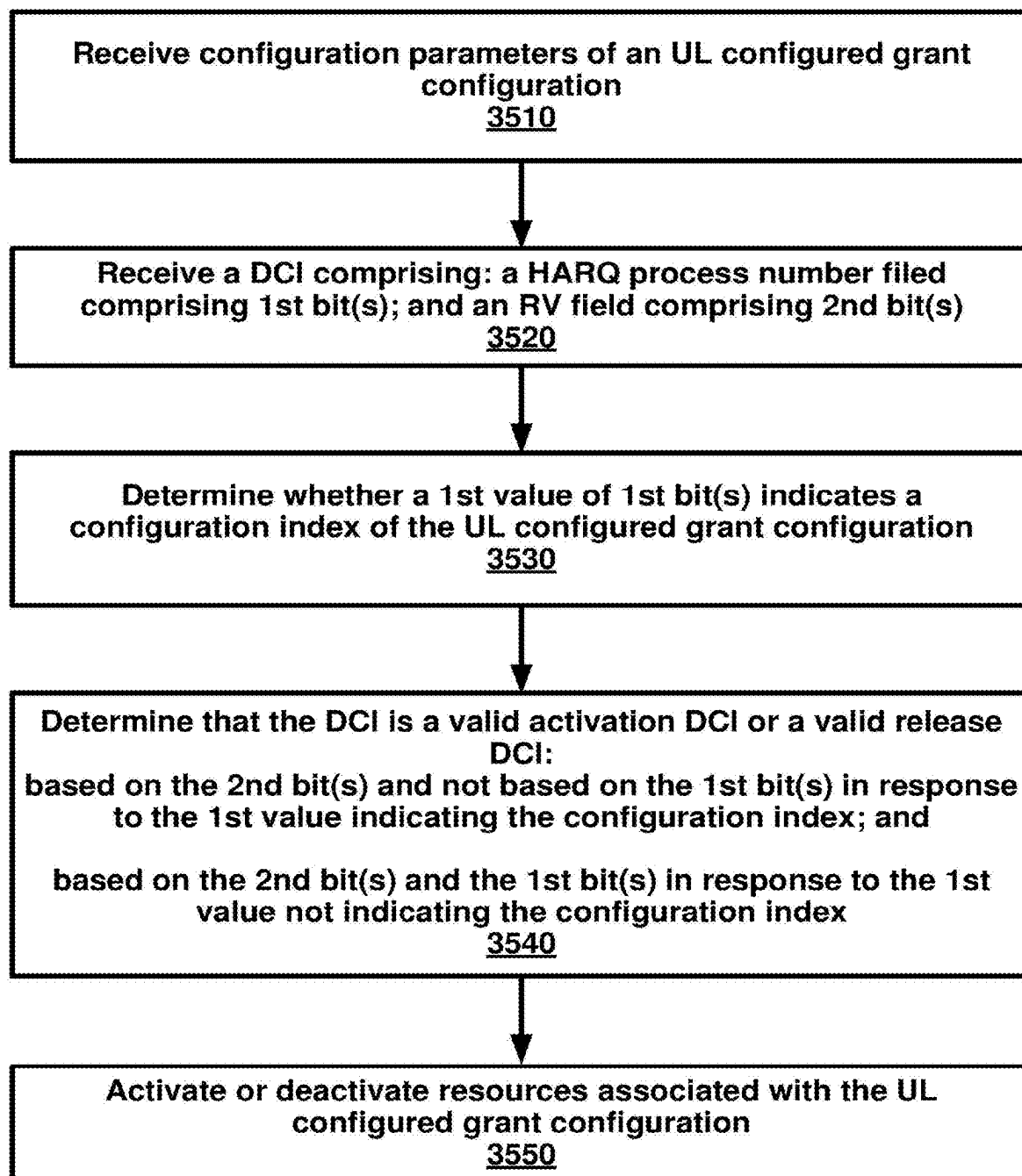


FIG. 35

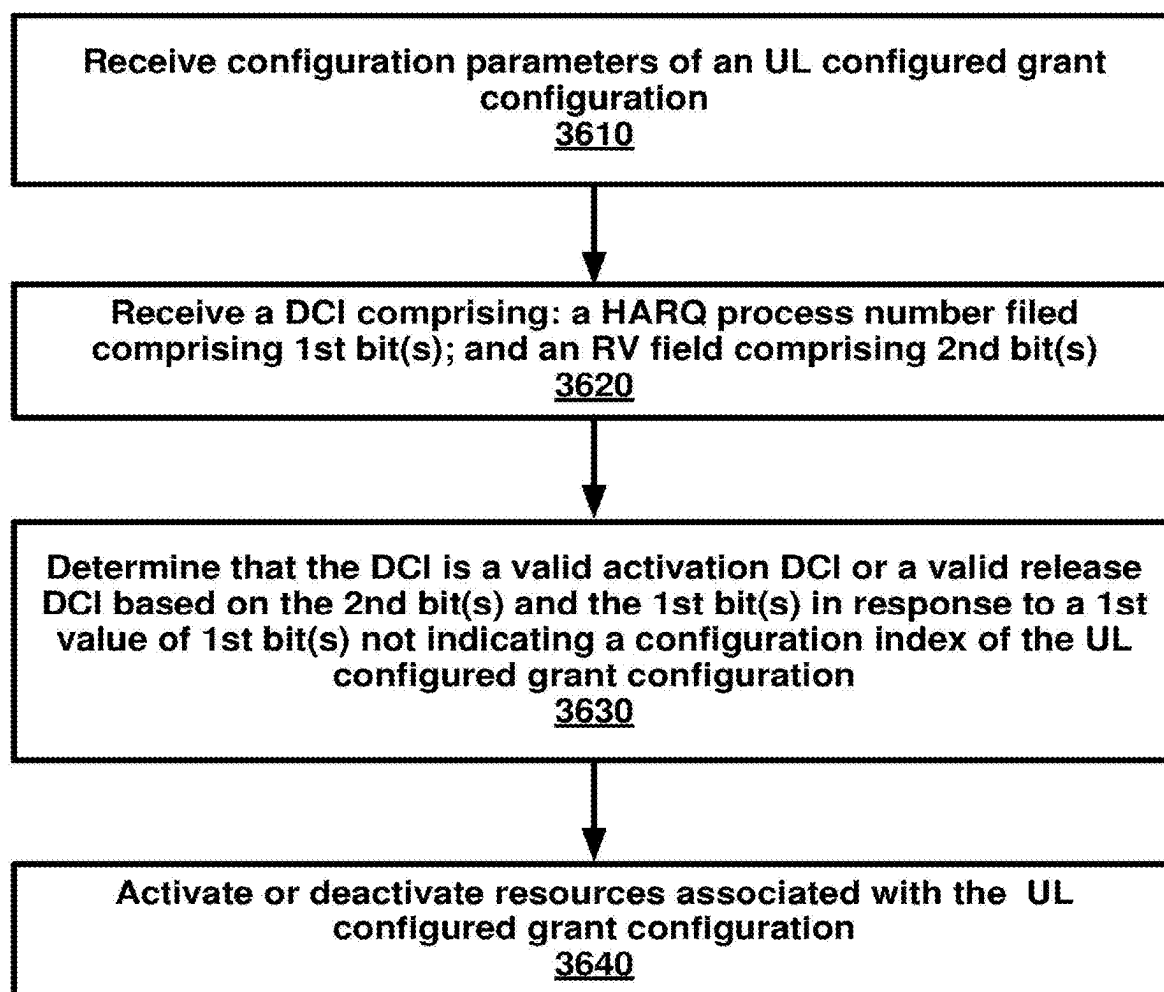


FIG. 36

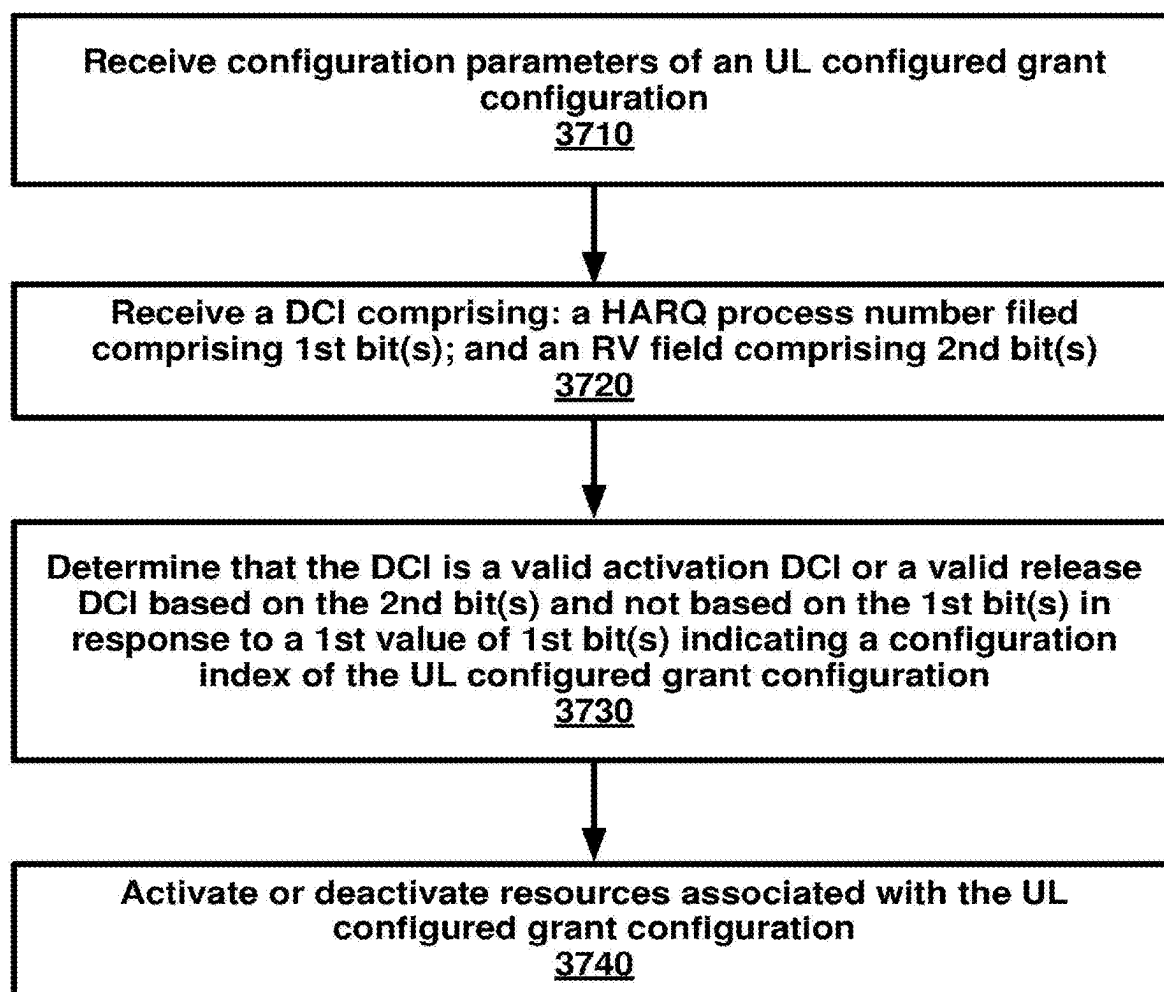


FIG. 37

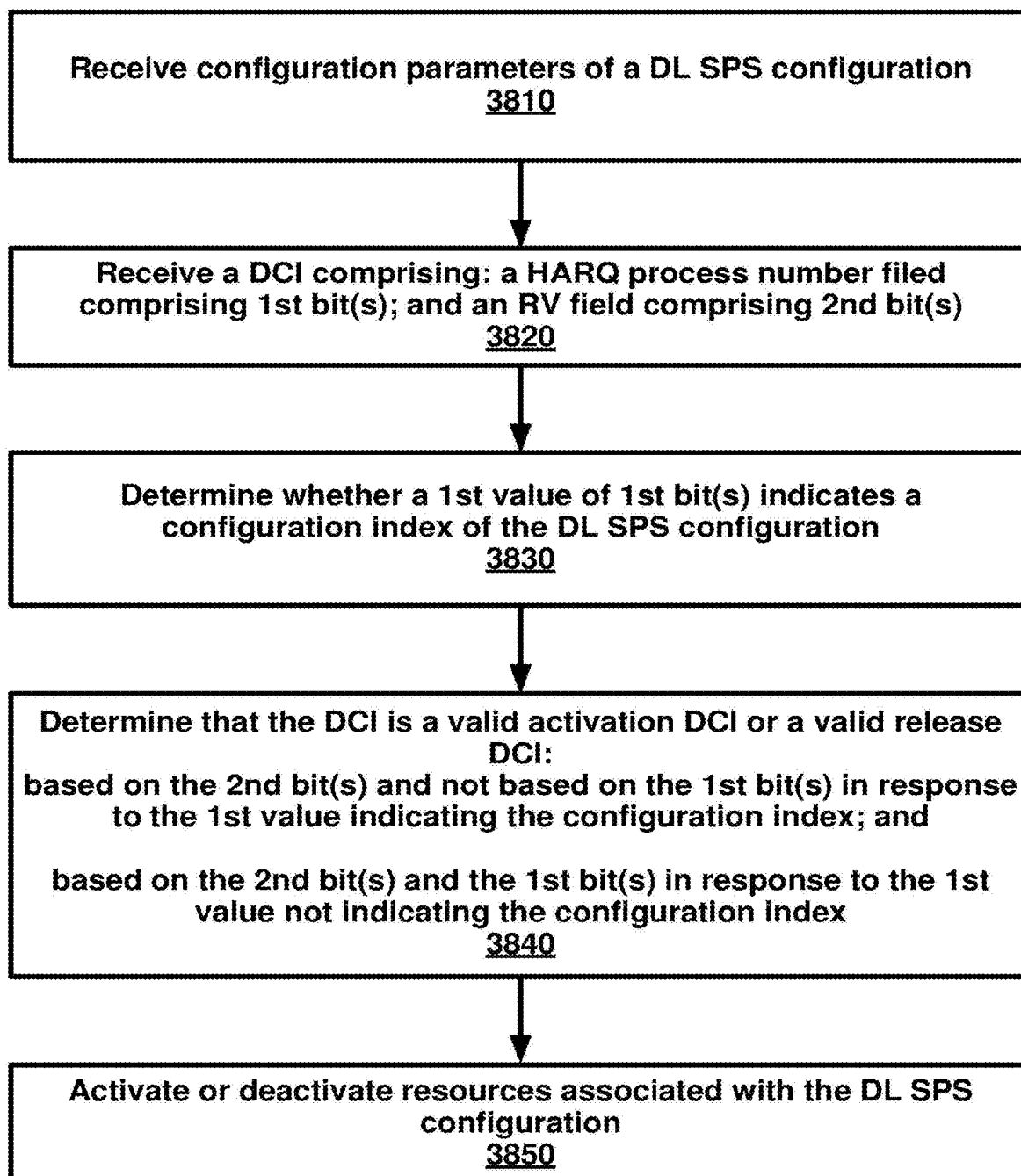


FIG. 38

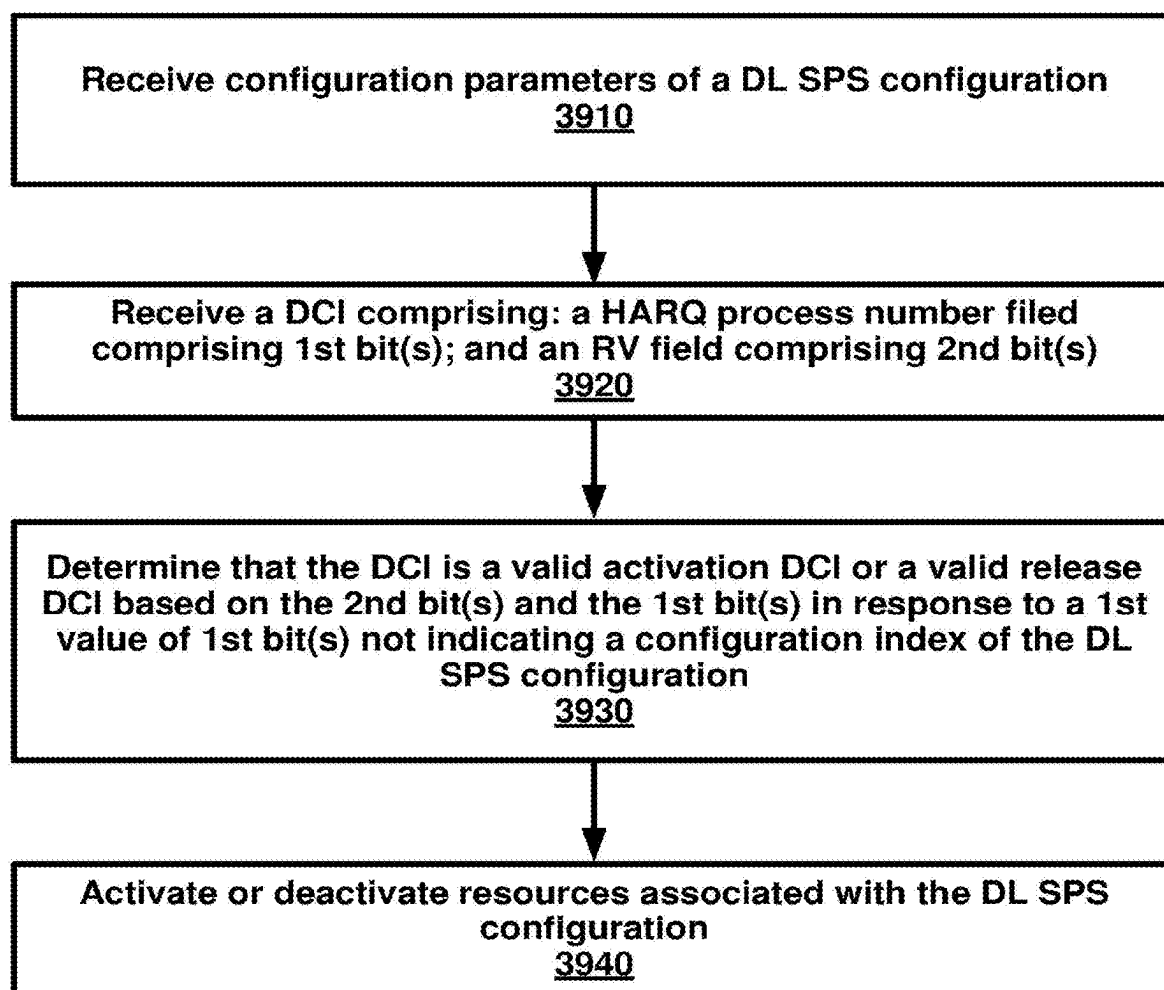


FIG. 39

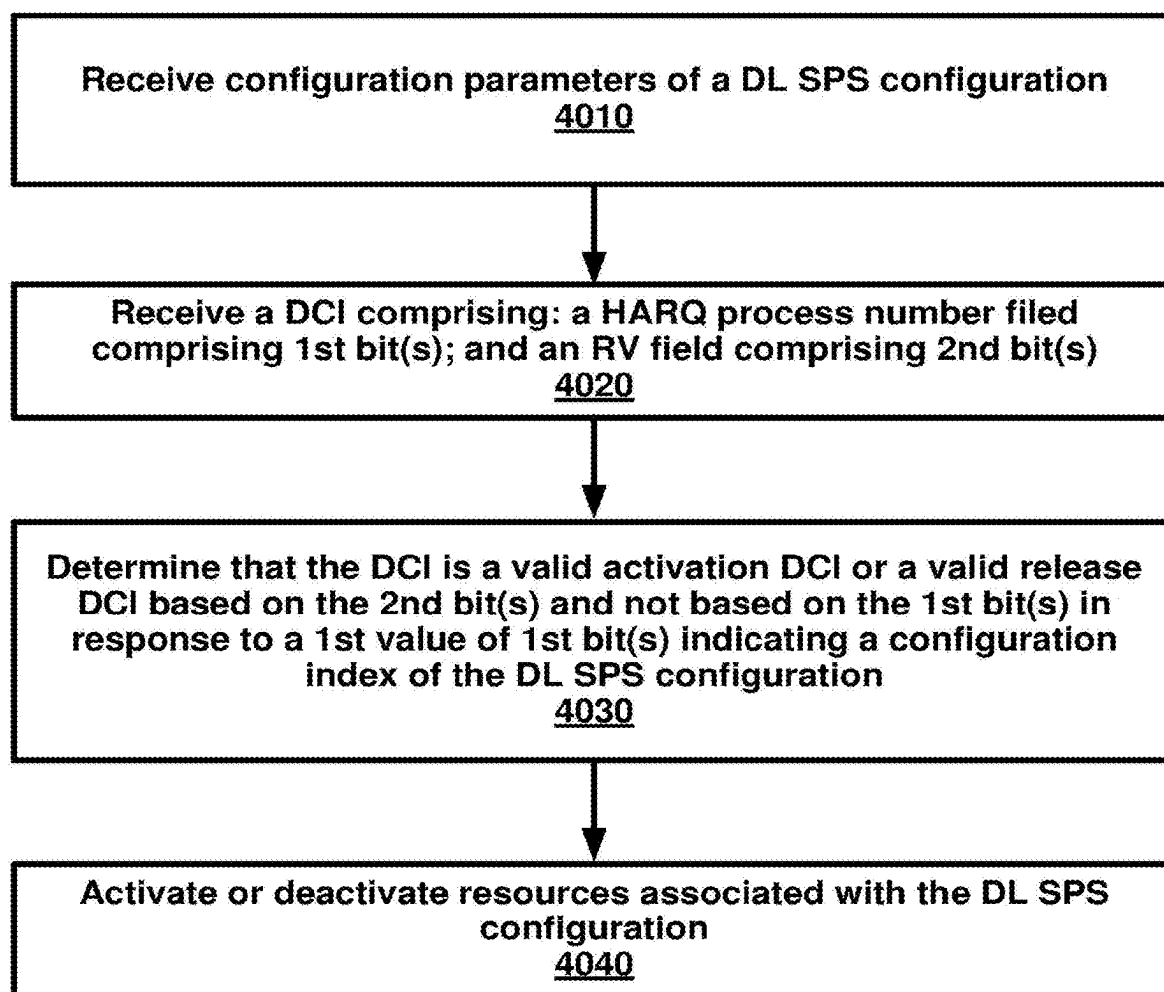


FIG. 40

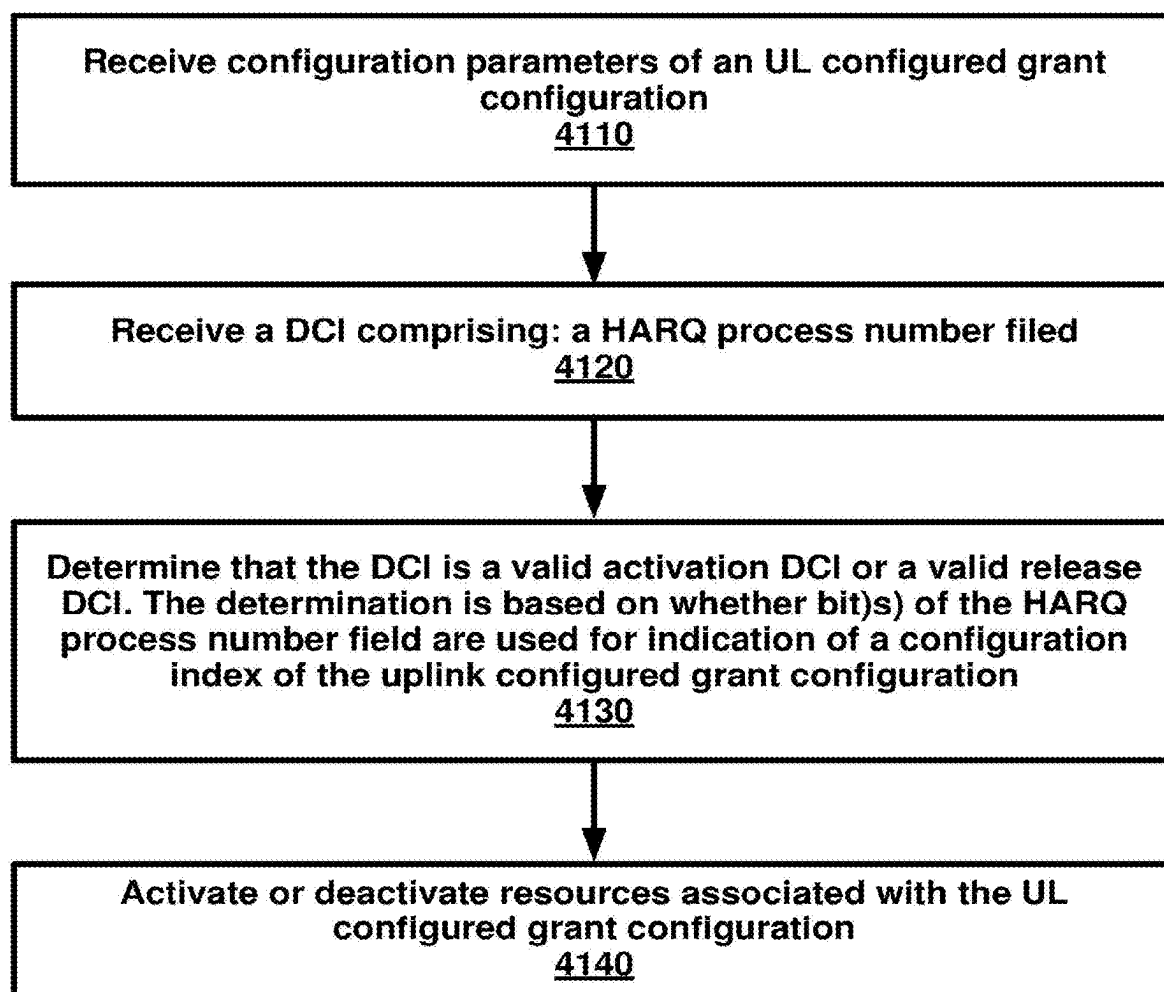


FIG. 41

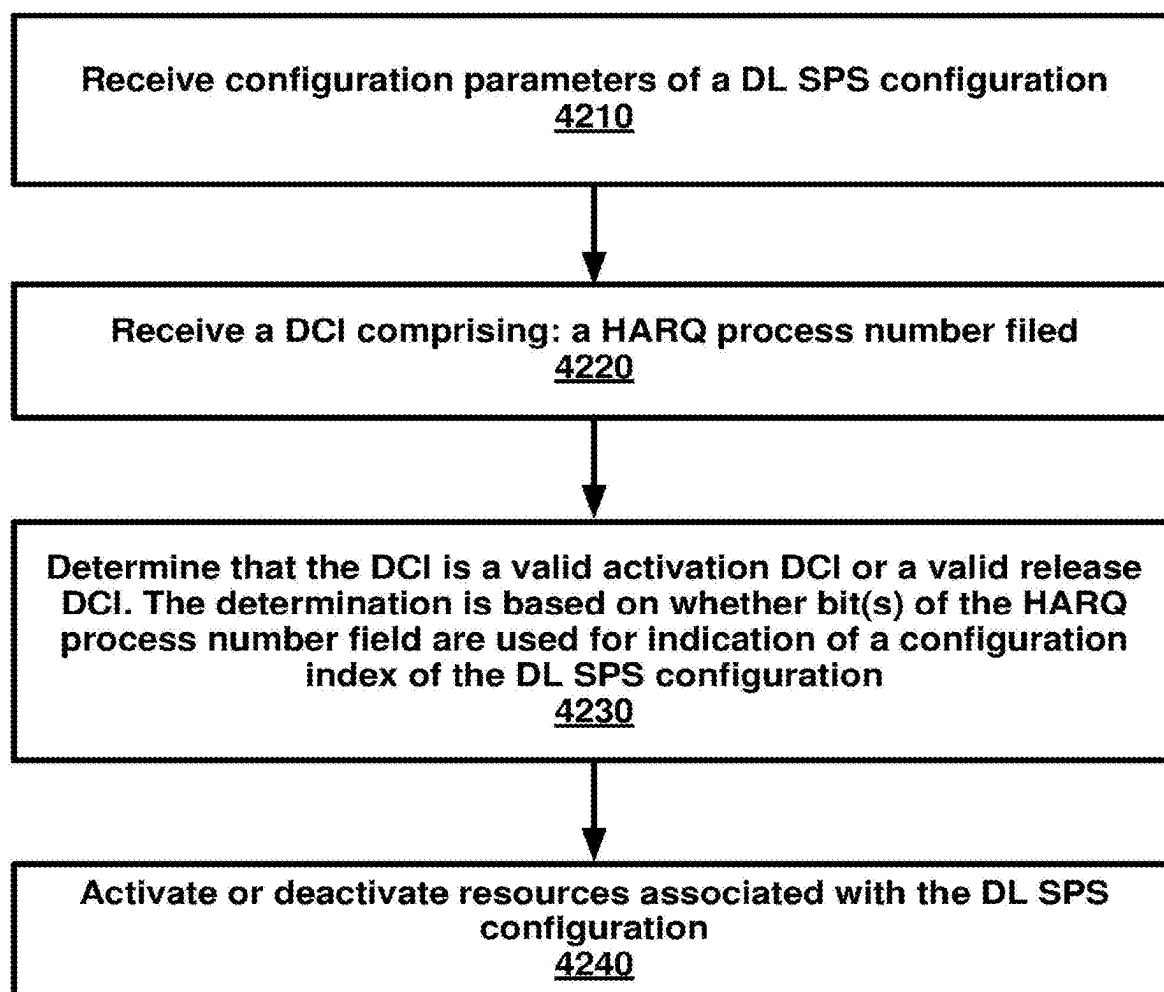


FIG. 42

VALIDATION OF A DCI FORMAT**CROSS-REFERENCE TO RELATED APPLICATIONS**

[0001] This application is a continuation of U.S. application Ser. No. 18/395,810, filed Dec. 26, 2023, which is a continuation of U.S. application Ser. No. 17/884,046, filed Aug. 9, 2022, which is a continuation of U.S. application Ser. No. 17/467,182, filed Sep. 4, 2021, which is a continuation of U.S. application Ser. No. 17/180,746, filed Feb. 20, 2021, which is a continuation of U.S. patent application Ser. No. 17/084,933, filed Oct. 30, 2020, which claims the benefit of U.S. Provisional Application No. 62/931,099, filed Nov. 5, 2019, which is hereby incorporated by reference in its entirety.

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] FIG. 1A and FIG. 1B show examples of mobile communications systems in accordance with several of various embodiments of the present disclosure.

[0003] FIG. 2A and FIG. 2B show examples of user plane and control plane protocol layers in accordance with several of various embodiments of the present disclosure.

[0004] FIG. 3 shows example functions and services offered by protocol layers in a user plane protocol stack in accordance with several of various embodiments of the present disclosure.

[0005] FIG. 4 shows example flow of packets through the protocol layers in accordance with several of various embodiments of the present disclosure.

[0006] FIG. 5A shows example mapping of channels between layers of the protocol stack and different physical signals in downlink in accordance with several of various embodiments of the present disclosure.

[0007] FIG. 5B shows example mapping of channels between layers of the protocol stack and different physical signals in uplink in accordance with several of various embodiments of the present disclosure.

[0008] FIG. 6 shows example physical layer processes for signal transmission in accordance with several of various embodiments of the present disclosure.

[0009] FIG. 7 shows examples of RRC states and RRC state transitions in accordance with several of various embodiments of the present disclosure.

[0010] FIG. 8 shows an example time domain transmission structure in NR by grouping OFDM symbols into slots, subframes and frames in accordance with several of various embodiments of the present disclosure.

[0011] FIG. 9 shows an example of time-frequency resource grid in accordance with several of various embodiments of the present disclosure.

[0012] FIG. 10 shows example adaptation and switching of bandwidth parts in accordance with several of various embodiments of the present disclosure.

[0013] FIG. 11A shows example arrangements of carriers in carrier aggregation in accordance with several of various embodiments of the present disclosure.

[0014] FIG. 11B shows examples of uplink control channel groups in accordance with several of various embodiments of the present disclosure.

[0015] FIG. 12A, FIG. 12B and FIG. 12C show example random access processes in accordance with several of various embodiments of the present disclosure.

[0016] FIG. 13A shows example time and frequency structure of SSBs and their associations with beams in accordance with several of various embodiments of the present disclosure.

[0017] FIG. 13B shows example time and frequency structure of CSI-RSs and their association with beams in accordance with several of various embodiments of the present disclosure.

[0018] FIG. 14A, FIG. 14B and FIG. 14C show example beam management processes in accordance with several of various embodiments of the present disclosure.

[0019] FIG. 15 shows example components of a wireless device and a base station that are in communication via an air interface in accordance with several of various embodiments of the present disclosure.

[0020] FIG. 16 is an example flow diagram for PDCCH validation in accordance with several of various embodiments of the present disclosure.

[0021] FIG. 17 shows an example process in accordance with several of various embodiments of the present disclosure.

[0022] FIG. 18 shows example values of a HARQ process number field of a downlink control information in accordance with several of various embodiments of the present disclosure.

[0023] FIG. 19 shows an example process in accordance with several of various embodiments of the present disclosure.

[0024] FIG. 20 shows an example process in accordance with several of various embodiments of the present disclosure.

[0025] FIG. 21 shows example values of a HARQ process number field of a downlink control information in accordance with several of various embodiments of the present disclosure.

[0026] FIG. 22 shows an example process in accordance with several of various embodiments of the present disclosure.

[0027] FIG. 23A, FIG. 23B and FIG. 23C show example MAC subheaders in accordance with several of various embodiments of the present disclosure.

[0028] FIG. 24 shows an example MAC protocol data unit (MAC PDU) in accordance with several of various embodiments of the present disclosure.

[0029] FIG. 25 shows an example confirmation MAC control element (MAC CE) in accordance with several of various embodiments of the present disclosure.

[0030] FIG. 26 shows an example confirmation MAC control element (MAC CE) in accordance with several of various embodiments of the present disclosure.

[0031] FIG. 27 shows an example scheduling activation or release process in accordance with several of various embodiments of the present disclosure.

[0032] FIG. 28 shows an example scheduling activation or release process in accordance with several of various embodiments of the present disclosure.

[0033] FIG. 29 shows an example scheduling activation or release process in accordance with several of various embodiments of the present disclosure.

[0034] FIG. 30 shows an example scheduling activation or release process in accordance with several of various embodiments of the present disclosure.

[0035] FIG. 31 shows an example scheduling activation or release process in accordance with several of various embodiments of the present disclosure.

[0036] FIG. 32 shows an example scheduling activation or release process in accordance with several of various embodiments of the present disclosure.

[0037] FIG. 33 shows an example scheduling activation or release process in accordance with several of various embodiments of the present disclosure.

[0038] FIG. 34 shows an example scheduling activation or release process in accordance with several of various embodiments of the present disclosure.

[0039] FIG. 35 shows an example flow diagram in accordance with several of various embodiments of the present disclosure.

[0040] FIG. 36 shows an example flow diagram in accordance with several of various embodiments of the present disclosure.

[0041] FIG. 37 shows an example flow diagram in accordance with several of various embodiments of the present disclosure.

[0042] FIG. 38 shows an example flow diagram in accordance with several of various embodiments of the present disclosure.

[0043] FIG. 39 shows an example flow diagram in accordance with several of various embodiments of the present disclosure.

[0044] FIG. 40 shows an example flow diagram in accordance with several of various embodiments of the present disclosure.

[0045] FIG. 41 shows an example flow diagram in accordance with several of various embodiments of the present disclosure.

[0046] FIG. 42 shows an example flow diagram in accordance with several of various embodiments of the present disclosure.

DETAILED DESCRIPTION

[0047] The exemplary embodiments of the disclosed technology enable semi-persistent scheduling and/or configured grants operation in a wireless device and/or one or more base stations. The exemplary disclosed embodiments may be implemented in the technical field of wireless communication systems. More particularly, the embodiment of the disclosed technology may relate to scheduling activation or scheduling release of semi-persistent or configured grants.

[0048] The devices and/or nodes of the mobile communications system disclosed herein may be implemented based on various technologies and/or various releases/versions/amendments of a technology. The various technologies include various releases of long-term evolution (LTE) technologies, various releases of 5G new radio (NR) technologies, various wireless local area networks technologies and/or a combination thereof and/or alike. For example, a base station may support a given technology and may communicate with wireless devices with different characteristics. The wireless devices may have different categories that define their capabilities in terms of supporting various features. The wireless device with the same category may have different capabilities. The wireless devices may support various technologies such as various releases of LTE technologies, various releases of 5G NR technologies and/or a combination thereof and/or alike. At least some of the wireless devices in the mobile communications system of

the present disclosure may be stationary or almost stationary. In this disclosure, the terms “mobile communications system” and “wireless communications system” may be used interchangeably.

[0049] FIG. 1A shows an example of a mobile communications system 100 in accordance with several of various embodiments of the present disclosure. The mobile communications system 100 may be, for example, run by a mobile network operator (MNO) or a mobile virtual network operator (MVNO). The mobile communications system 100 may be a public land mobile network (PLMN) run by a network operator providing a variety of service including voice, data, short messaging service (SMS), multimedia messaging service (MMS), emergency calls, etc. The mobile communications system 100 includes a core network (CN) 106, a radio access network (RAN) 104 and at least one wireless device 102.

[0050] The CN 106 connects the RAN 104 to one or more external networks (e.g., one or more data networks such as the Internet) and is responsible for functions such as authentication, charging and end-to-end connection establishment. Several radio access technologies (RATs) may be served by the same CN 106.

[0051] The RAN 104 may implement a RAT and may operate between the at least one wireless device 102 and the CN 106. The RAN 104 may handle radio related functionalities such as scheduling, radio resource control, modulation and coding, multi-antenna transmissions and retransmission protocols. The wireless device and the RAN may share a portion of the radio spectrum by separating transmissions from the wireless device to the RAN and the transmissions from the RAN to the wireless device. The direction of the transmissions from the wireless device to the RAN is known as the uplink and the direction of the transmissions from the RAN to the wireless device is known as the downlink. The separation of uplink and downlink transmissions may be achieved by employing a duplexing technique. Example duplexing techniques include frequency division duplexing (FDD), time division duplexing (TDD) or a combination of FDD and TDD.

[0052] In this disclosure, the term wireless device may refer to a device that communicates with a network entity or another device using wireless communication techniques. The wireless device may be a mobile device or a non-mobile (e.g., fixed) device. Examples of the wireless device include cellular phone, smart phone, tablet, laptop computer, wearable device (e.g., smart watch, smart shoe, fitness trackers, smart clothing, etc.), wireless sensor, wireless meter, extended reality (XR) devices including augmented reality (AR) and virtual reality (VR) devices, Internet of Things (IoT) device, vehicle to vehicle communications device, road-side units (RSU), automobile, relay node or any combination thereof. In some examples, the wireless device (e.g., a smart phone, tablet, etc.) may have an interface (e.g., a graphical user interface (GUI)) for configuration by an end user. In some examples, the wireless device (e.g., a wireless sensor device, etc.) may not have an interface for configuration by an end user. The wireless device may be referred to as a user equipment (UE), a mobile station (MS), a subscriber unit, a handset, an access terminal, a user terminal, a wireless transmit and receive unit (WTRU) and/or other terminology.

[0053] The at least one wireless device may communicate with at least one base station in the RAN 104. In this

disclosure, the term base station may encompass terminologies associated with various RATs. For example, a base station may be referred to as a Node B in a 3G cellular system such as Universal Mobile Telecommunication Systems (UMTS), an evolved Node B (eNB) in a 4G cellular system such as evolved universal terrestrial radio access (E-UTRA), a next generation eNB (ng-eNB), a Next Generation Node B (gNB) in NR and/or a 5G system, an access point (AP) in Wi-Fi and/or other wireless local area networks. A base station may be referred to as a remote radio head (RRH), a baseband unit (BBU) in connection with one or more RRHs, a repeater or relay for coverage extension and/or any combination thereof. In some examples, all protocol layers of a base station may be implemented in one unit. In some example, some of the protocol layers (e.g., upper layers) of the base station may be implemented in a first unit (e.g., a central unit (CU)) and some other protocol layer (e.g., lower layers) may be implemented in one or more second units (e.g., distributed units (DUs)).

[0054] A base station in the RAN **104** includes one or more antennas to communicate with the at least one wireless device. The base station may communicate with the at least one wireless device using radio frequency (RF) transmissions and receptions via RF transceivers. The base station antennas may control one or more cells (or sectors). The size and/or radio coverage area of a cell may depend on the range that transmissions by a wireless device can be successfully received by the base station when the wireless device transmits using the RF frequency of the cell. The base station may be associated with cells of various sizes. At a given location, the wireless device may be in coverage area of a first cell of the base station and may not be in coverage area of a second cell of the base station depending on the sizes of the first cell and the second cell.

[0055] A base station in the RAN **104** may have various implementations. For example, a base station may be implemented by connecting a BBU (or a BBU pool) coupled to one or more RRHs and/or one or more relay nodes to extend the cell coverage. The BBU pool may be located at a centralized site like a cloud or data center. The BBU pool may be connected to a plurality of RRHs that control a plurality of cells. The combination of BBU with the one or more RRHs may be referred to as a centralized or cloud RAN (C-RAN) architecture. In some implementations, the BBU functions may be implemented on virtual machines (VMs) on servers at a centralized location. This architecture may be referred to as virtual RAN (vRAN). All, most or a portion of the protocol layer functions (e.g., all or portions of physical layer, medium access control (MAC) layer and/or higher layers) may be implemented at the BBU pool and the processed data may be transmitted to the RRHs for further processing and/or RF transmission. The links between the BBU pool and the RRHs may be referred to as fronthaul.

[0056] In some deployment scenarios, the RAN **104** may include macrocell base stations with high transmission power levels and large coverage areas. In other deployment scenarios, the RAN **104** may include base stations that employ different transmission power levels and/or have cells with different coverage areas. For example, some base station may be macrocell base stations with high transmission powers and/or large coverage areas and other base station may be small cell base stations with comparatively smaller transmission powers and/or coverage areas. In some

deployment scenarios, a small cell base station may have coverage that is within or has overlap with coverage area of a macrocell base station. A wireless device may communicate with the macrocell base station while within the coverage area of the macrocell base station. For additional capacity, the wireless device may communicate with both the macrocell base station and the small cell base station while in the overlapped coverage area of the macrocell base station and the small cell base station. Depending on their coverage areas, a small cell base station may be referred to as a microcell base station, a picocell base station, a femtocell base station or a home base station.

[0057] Different standard development organizations (SDOs) have specified, or may specify in future, mobile communications systems that have similar characteristics as the mobile communications system **100** of FIG. 1A. For example, the Third-Generation Partnership Project (3GPP) is a group of SDOs that provides specifications that define 3GPP technologies for mobile communications systems that are akin to the mobile communications system **100**. The 3GPP has developed specifications for third generation (3G) mobile networks, fourth generation (4G) mobile networks and fifth generation (5G) mobile networks. The 3G, 4G and 5G networks are also known as Universal Mobile Telecommunications System (UMTS), Long Term Evolution (LTE) and 5G system (5GS), respectively. In this disclosure, embodiments are described with respect to the RAN implemented in a 3GPP 5G mobile network that is also referred to as next generation RAN (NG-RAN). The embodiments may also be implemented in other mobile communications systems such as 3G or 4G mobile networks or mobile networks that may be standardized in future such as sixth generation (6G) mobile networks or mobile networks that are implemented by standards bodies other than 3GPP. The NG-RAN may be based on a new RAT known as new radio (NR) and/or other radio access technologies such as LTE and/or non-3GPP RATs.

[0058] FIG. 1B shows an example of a mobile communications system **110** in accordance with several of various embodiments of the present disclosure. The mobile communications system **110** of FIG. 1B is an example of a 5G mobile network and includes a 5G CN (5G-CN) **130**, an NG-RAN **120** and UEs (collectively **112** and individually UE **112A** and UE **112B**). The 5G-CN **130**, the NG-RAN **120** and the UEs **112** of FIG. 1B operate substantially alike the CN **106**, the RAN **104** and the at least one wireless device **102**, respectively, as described for FIG. 1A.

[0059] The 5G-CN **130** of FIG. 1B connects the NG-RAN **120** to one or more external networks (e.g., one or more data networks such as the Internet) and is responsible for functions such as authentication, charging and end-to-end connection establishment. The 5G-CN has new enhancements compared to previous generations of CNs (e.g., evolved packet core (EPC) in the 4G networks) including service-based architecture, support for network slicing and control plane/user plane split. The service-based architecture of the 5G-CN provides a modular framework based on service and functionalities provided by the core network wherein a set of network functions are connected via service-based interfaces. The network slicing enables multiplexing of independent logical networks (e.g., network slices) on the same physical network infrastructure. For example, a network slice may be for mobile broadband applications with full mobility support and a different network slice may be for

non-mobile latency-critical applications such as industry automation. The control plane/user plane split enables independent scaling of the control plane and the user plane. For example, the control plane capacity may be increased without affecting the user plane of the network.

[0060] The 5G-CN **130** of FIG. 1B includes an access and mobility management function (AMF) **132** and a user plane function (UPF) **134**. The AMF **132** may support termination of non-access stratum (NAS) signaling, NAS signaling security such as ciphering and integrity protection, inter-3GPP access network mobility, registration management, connection management, mobility management, access authentication and authorization and security context management. The NAS is a functional layer between a UE and the CN and the access stratum (AS) is a functional layer between the UE and the RAN. The UPF **134** may serve as an interconnect point between the NG-RAN and an external data network. The UPF may support packet routing and forwarding, packet inspection and Quality of Service (QoS) handling and packet filtering. The UPF may further act as a Protocol Data Unit (PDU) session anchor point for mobility within and between RATs.

[0061] The 5G-CN **130** may include additional network functions (not shown in FIG. 1B) such as one or more Session Management Functions (SMFs), a Policy Control Function (PCF), a Network Exposure Function (NEF), a Unified Data Management (UDM), an Application Function (AF), and/or an Authentication Server Function (AUSF). These network functions along with the AMF **132** and UPF **134** enable a service-based architecture for the 5G-CN.

[0062] The NG-RAN **120** may operate between the UEs **112** and the 5G-CN **130** and may implement one or more RATs. The NG-RAN **120** may include one or more gNBs (e.g., gNB **122A** or gNB **122B** or collectively gNBs **122**) and/or one or more ng-eNBs (e.g., ng-eNB **124A** or ng-eNB **124B** or collectively ng-eNBs **124**). The general terminology for gNBs **122** and/or an ng-eNBs **124** is a base station and may be used interchangeably in this disclosure. The gNBs **122** and the ng-eNBs **124** may include one or more antennas to communicate with the UEs **112**. The one or more antennas of the gNBs **122** or ng-eNBs **124** may control one or more cells (or sectors) that provide radio coverage for the UEs **112**.

[0063] A gNB and/or an ng-eNB of FIG. 1B may be connected to the 5G-CN **130** using an NG interface. A gNB and/or an ng-eNB may be connected with other gNBs and/or ng-eNBs using an Xn interface. The NG or the Xn interfaces are logical connections that may be established using an underlying transport network. The interface between a UE and a gNB or between a UE and an ng-eNBs may be referred to as the Uu interface. An interface (e.g., Uu, NG or Xn) may be established by using a protocol stack that enables data and control signaling exchange between entities in the mobile communications system of FIG. 1B. When a protocol stack is used for transmission of user data, the protocol stack may be referred to as user plane protocol stack. When a protocol stack is used for transmission of control signaling, the protocol stack may be referred to as control plane protocol stack. Some protocol layer may be used in both of the user plane protocol stack and the control plane protocol stack while other protocol layers may be specific to the user plane or control plane.

[0064] The NG interface of FIG. 1B may include an NG-User plane (NG-U) interface between a gNB and the

UPF **134** (or an ng-eNB and the UPF **134**) and an NG-Control plane (NG-C) interface between a gNB and the AMF **132** (or an ng-eNB and the AMF **132**). The NG-U interface may provide non-guaranteed delivery of user plane PDUs between a gNB and the UPF or an ng-eNB and the UPF. The NG-C interface may provide services such as NG interface management, UE context management, UE mobility management, transport of NAS messages, paging, PDU session management, configuration transfer and/or warning message transmission.

[0065] The UEs **112** and a gNB may be connected using the Uu interface and using the NR user plane and control plane protocol stack. The UEs **112** and an ng-eNB may be connected using the Uu interface using the LTE user plane and control plane protocol stack.

[0066] In the example mobile communications system of FIG. 1B, a 5G-CN is connected to a RAN comprised of 4G LTE and/or 5G NR RATs. In other example mobile communications systems, a RAN based on the 5G NR RAT may be connected to a 4G CN (e.g., EPC). For example, earlier releases of 5G standards may support a non-standalone mode of operation where a NR based RAN is connected to the 4G EPC. In an example non-standalone mode, a UE may be connected to both a 5G NR gNB and a 4G LTE eNB (e.g., a ng-eNB) and the control plane functionalities (such as initial access, paging and mobility) may be provided through the 4G LTE eNB. In a standalone of operation, the 5G NR gNB is connected to a 5G-CN and the user plane and the control plane functionalities are provided by the 5G NR gNB.

[0067] FIG. 2A shows an example of the protocol stack for the user plan of an NR Uu interface in accordance with several of various embodiments of the present disclosure. The user plane protocol stack comprises five protocol layers that terminate at the UE **200** and the gNB **210**. The five protocol layers, as shown in FIG. 2A, include physical (PHY) layer referred to as PHY **201** at the UE **200** and PHY **211** at the gNB **210**, medium access control (MAC) layer referred to as MAC **202** at the UE **200** and MAC **212** at the gNB **210**, radio link control (RLC) layer referred to as RLC **203** at the UE **200** and RLC **213** at the gNB **210**, packet data convergence protocol (PDCP) layer referred to as PDCP **204** at the UE **200** and PDCP **214** at the gNB **210**, and service data application protocol (SDAP) layer referred to as SDAP **205** at the UE **200** and SDAP **215** at the gNB **210**. The PHY layer, also known as layer 1 (L1), offers transport services to higher layers. The other four layers of the protocol stack (MAC, RLC, PDCP and SDAP) are collectively known as layer 2 (L2).

[0068] FIG. 2B shows an example of the protocol stack for the control plan of an NR Uu interface in accordance with several of various embodiments of the present disclosure. Some of the protocol layers (PHY, MAC, RLC and PDCP) are common between the user plane protocol stack shown in FIG. 2A and the control plan protocol stack. The control plane protocol stack also includes the RRC layer, referred to RRC **206** at the UE **200** and RRC **216** at the gNB **210**, that also terminates at the UE **200** and the gNB **210**. In addition, the control plane protocol stack includes the NAS layer that terminates at the UE **200** and the AMF **220**. In FIG. 2B, the NAS layer is referred to as NAS **207** at the UE **200** and NAS **227** at the AMF **220**.

[0069] FIG. 3 shows example functions and services offered to other layers by a layer in the NR user plane

protocol stack of FIG. 2A in accordance with several of various embodiments of the present disclosure. For example, the SDAP layer of FIG. 3 (shown in FIG. 2A as SDAP 205 at the UE side and SDAP 215 at the gNB side) may perform mapping and de-mapping of QoS flows to data radio bearers. The mapping and de-mapping may be based on QoS (e.g., delay, throughput, jitter, error rate, etc.) associated with a QoS flow. A QoS flow may be a QoS differentiation granularity for a PDU session which is a logical connection between a UE 200 and a data network. A PDU session may contain one or more QoS flows. The functions and services of the SDAP layer include mapping and de-mapping between one or more QoS flows and one or more data radio bearers. The SDAP layer may also mark the uplink and/or downlink packets with a QoS flow ID (QFI).

[0070] The PDCP layer of FIG. 3 (shown in FIG. 2A as PDCP 204 at the UE side and PDCP 214 at the gNB side) may perform header compression and decompression (e.g., using Robust Header Compression (ROHC) protocol) to reduce the protocol header overhead, ciphering and deciphering and integrity protection and verification to enhance the security over the air interface, reordering and in-order delivery of packets and discarding of duplicate packets. A UE may be configured with one PDCP entity per bearer.

[0071] In an example scenario not shown in FIG. 3, a UE may be configured with dual connectivity and may connect to two different cell groups provided by two different base stations. For example, a base station of the two base stations may be referred to as a master base station and a cell group provided by the master base station may be referred to as a master cell group (MCG). The other base station of the two base stations may be referred to as a secondary base station and the cell group provided by the secondary base station may be referred to as a secondary cell group (SCG). A bearer may be configured for the UE as a split bearer that may be handled by the two different cell groups. The PDCP layer may perform routing of packets corresponding to a split bearer to and/or from RLC channels associated with the cell groups.

[0072] In an example scenario not shown in FIG. 3, a bearer of the UE may be configured (e.g., with control signaling) with PDCP packet duplication. A bearer configured with PDCP duplication may be mapped to a plurality of RLC channels each corresponding to different one or more cells. The PDCP layer may duplicate packets of the bearer configured with PDCP duplication and the duplicated packets may be mapped to the different RLC channels. With PDCP packet duplication, the likelihood of correct reception of packets increases thereby enabling higher reliability.

[0073] The RLC layer of FIG. 3 (shown in FIG. 2A as RLC 203 at the UE side and RLC 213 at the gNB side) provides service to upper layers in the form of RLC channels. The RLC layer may include three transmission modes: transparent mode (TM), Unacknowledged mode (UM) and Acknowledged mode (AM). The RLC layer may perform error correction through automatic repeat request (ARQ) for the AM transmission mode, segmentation of RLC service data units (SDUs) for the AM and UM transmission modes and re-segmentation of RLC SDUs for AM transmission mode, duplicate detection for the AM transmission mode, RLC SDU discard for the AM and UM transmission modes, etc. The UE may be configured with one RLC entity per RLC channel.

[0074] The MAC layer of FIG. 3 (shown in FIG. 2A as MAC 202 at the UE side and MAC 212 at the gNB side) provides services to the RLC layer in form of logical channels. The MAC layer may perform mapping between logical channels and transport channels, multiplexing/de-multiplexing of MAC SDUs belonging to one or more logical channels into/from transport blocks (TBs) delivered to/from the physical layer on transport channels, reporting of scheduling information, error correction through hybrid automatic repeat request (HARQ), priority handling between UEs by means of dynamic scheduling, priority handling between logical channels of one UE by means of logical channel prioritization and/or padding. In case of carrier aggregation, a MAC entity may comprise one HARQ entity per cell. A MAC entity may support multiple numerologies, transmission timings and cells. The control signaling may configure logical channels with mapping restrictions. The mapping restrictions in logical channel prioritization may control the numerology (ies), cell(s), and/or transmission timing(s)/duration(s) that a logical channel may use.

[0075] The PHY layer of FIG. 3 (shown in FIG. 2A as PHY 201 at the UE side and PHY 211 at the gNB side) provides transport services to the MAC layer in form of transport channels. The physical layer may handle coding/decoding, HARQ soft combining, rate matching of a coded transport channel to physical channels, mapping of coded transport channels to physical channels, modulation and demodulation of physical channels, frequency and time synchronization, radio characteristics measurements and indication to higher layers, RF processing, and mapping to antennas and radio resources.

[0076] FIG. 4 shows example processing of packets at different protocol layers in accordance with several of various embodiments of the present disclosure. In this example, three Internet Protocol (IP) packets that are processed by the different layers of the NR protocol stack. The term SDU shown in FIG. 4 is the data unit that is entered from/to a higher layer. In contrast, a protocol data unit (PDU) is the data unit that is entered to/from a lower layer. The flow of packets in FIG. 4 is for downlink. An uplink data flow through layers of the NR protocol stack is similar to FIG. 4. In this example, the two leftmost IP packets are mapped by the SDAP layer (shown as SDAP 205 and SDAP 215 in FIG. 2A) to radio bearer 402 and the rightmost packet is mapped by the SDAP layer to the radio bearer 404. The SDAP layer adds SDAP headers to the IP packets which are entered into the PDCP layer as PDCP SDUs. The PDCP layer is shown as PDCP 204 and PDCP 214 in FIG. 2A. The PDCP layer adds the PDCP headers to the PDCP SDUs which are entered into the RLC layer as RLC SDUs. The RLC layer is shown as RLC 203 and RLC 213 in FIG. 2A. An RLC SDU may be segmented at the RLC layer. The RLC layer adds RLC headers to the RLC SDUs after segmentation (if segmented) which are entered into the MAC layer as MAC SDUs. The MAC layer adds the MAC headers to the MAC SDUs and multiplexes one or more MAC SDUs to form a PHY SDU (also referred to as a transport block (TB) or a MAC PDU).

[0077] In FIG. 4, the MAC SDUs are multiplexed to form a transport block. The MAC layer may multiplex one or more MAC control elements (MAC CEs) with zero or more MAC SDUs to form a transport block. The MAC CEs may also be referred to as MAC commands or MAC layer control

signaling and may be used for in-band control signaling. The MAC CEs may be transmitted by a base station to a UE (e.g., downlink MAC CEs) or by a UE to a base station (e.g., uplink MAC CEs). The MAC CEs may be used for transmission of information useful by a gNB for scheduling (e.g., buffer status report (BSR) or power headroom report (PHR)), activation/deactivation of one or more cells, activation/deactivation of configured radio resources for one or more processes, activation/deactivation of one or more processes, indication of parameters used in one or more processes, etc.

[0078] FIG. 5A and FIG. 5B show example mapping between logical channels, transport channels and physical channels for downlink and uplink, respectively in accordance with several of various embodiments of the present disclosure. As discussed before, the MAC layer provides services to higher layer in the form of logical channels. A logical channel may be classified as a control channel, if used for transmission of control and/or configuration information, or a traffic channel if used for transmission of user data. Example logical channels in NR include Broadcast Control Channel (BCCH) used for transmission of broadcast system control information, Paging Control Channel (PCCH) used for carrying paging messages for wireless devices with unknown locations, Common Control Channel (CCCH) used for transmission of control information between UEs and network and for UEs that have no RRC connection with the network, Dedicated Control Channel (DCCH) which is a point-to-point bi-directional channel for transmission of dedicated control information between a UE that has an RRC connection and the network and Dedicated Traffic Channel (DTCH) which is point-to-point channel, dedicated to one UE, for the transfer of user information and may exist in both uplink and downlink.

[0079] As discussed before, the PHY layer provides services to the MAC layer and higher layers in the form of transport channels. Example transport channels in NR include Broadcast Channel (BCH) used for transmission of part of the BCCH referred to as master information block (MIB), Downlink Shared Channel (DL-SCH) used for transmission of data (e.g., from DTCH in downlink) and various control information (e.g., from DCCH and CCCH in downlink and part of the BCCH that is not mapped to the BCH), Uplink Shared Channel (UL-SCH) used for transmission of uplink data (e.g., from DTCH in uplink) and control information (e.g., from CCCH and DCCH in uplink) and Paging Channel (PCH) used for transmission of paging information from the PCCH. In addition, Random Access Channel (RACH) is a transport channel used for transmission of random access preambles. The RACH does not carry a transport block. Data on a transport channel (except RACH) may be organized in transport blocks, wherein One or more transport blocks may be transmitted in a transmission time interval (TTI).

[0080] The PHY layer may map the transport channels to physical channels. A physical channel may correspond to time-frequency resources that are used for transmission of information from one or more transport channels. In addition to mapping transport channels to physical channels, the physical layer may generate control information (e.g., downlink control information (DCI) or uplink control information (UCI)) that may be carried by the physical channels. Example DCI include scheduling information (e.g., downlink assignments and uplink grants), request for channel

state information report, power control command, etc. Example UCI include HARQ feedback indicating correct or incorrect reception of downlink transport blocks, channel state information report, scheduling request, etc. Example physical channels in NR include a Physical Broadcast Channel (PBCH) for carrying information from the BCH, a Physical Downlink Shared Channel (PDSCH) for carrying information from the PCH and the DL-SCH, a Physical Downlink Control Channel (PDCCH) for carrying DCI, a Physical Uplink Shared Channel (PUSCH) for carrying information from the UL-SCH and/or UCI, a Physical Uplink Control Channel (PUCCH) for carrying UCI and Physical Random Access Channel (PRACH) for transmission of RACH (e.g., random access preamble).

[0081] The PHY layer may also generate physical signals that are not originated from higher layers. As shown in FIG. 5A, example downlink physical signals include Demodulation Reference Signal (DM-RS), Phase Tracking Reference Signal (PT-RS), Channel State Information Reference Signal (CSI-RS), Primary Synchronization Signal (PSS) and Secondary Synchronization Signal (SSS). As shown in FIG. 5B, example uplink physical signals include DM-RS, PT-RS and sounding reference signal (SRS).

[0082] As indicated earlier, some of the protocol layers (PHY, MAC, RLC and PDCP) of the control plane of an NR Uu interface, are common between the user plane protocol stack (as shown in FIG. 2A) and the control plane protocol stack (as shown in FIG. 2B). In addition to PHY, MAC, RLC and PDCP, the control plane protocol stack includes the RRC protocol layer and the NAS protocol layer.

[0083] The NAS layer, as shown in FIG. 2B, terminates at the UE 200 and the AMF 220 entity of the 5G-C 130. The NAS layer is used for core network related functions and signaling including registration, authentication, location update and session management. The NAS layer uses services from the AS of the Uu interface to transmit the NAS messages.

[0084] The RRC layer, as shown in FIG. 2B, operates between the UE 200 and the gNB 210 (more generally NG-RAN 120) and may provide services and functions such as broadcast of system information (SI) related to AS and NAS as well as paging initiated by the 5G-C 130 or NG-RAN 120. In addition, the RRC layer is responsible for establishment, maintenance and release of an RRC connection between the UE 200 and the NG-RAN 120, carrier aggregation configuration (e.g., addition, modification and release), dual connectivity configuration (e.g., addition, modification and release), security related functions, radio bearer configuration/maintenance and release, mobility management (e.g., maintenance and context transfer), UE cell selection and reselection, inter-RAT mobility, QoS management functions, UE measurement reporting and control, radio link failure (RLF) detection and NAS message transfer. The RRC layer uses services from PHY, MAC, RLC and PDCP layers to transmit RRC messages using signaling radio bearers (SRBs). The SRBs are mapped to CCCH logical channel during connection establishment and to DCCH logical channel after connection establishment.

[0085] FIG. 6 shows example physical layer processes for signal transmission in accordance with several of various embodiments of the present disclosure. Data and/or control streams from MAC layer may be encoded/decoded to offer transport and control services over the radio transmission link. For example, one or more (e.g., two as shown in FIG.

6) transport blocks may be received from the MAC layer for transmission via a physical channel (e.g., a physical downlink shared channel or a physical uplink shared channel). A cyclic redundancy check (CRC) may be calculated and attached to a transport block in the physical layer. The CRC calculation may be based on one or more cyclic generator polynomials. The CRC may be used by the receiver for error detection. Following the transport block CRC attachment, a low-density parity check (LDPC) base graph selection may be performed. In example embodiments, two LDPC base graphs may be used wherein a first LDPC base graph may be optimized for small transport blocks and a second LDPC base graph may be optimized for comparatively larger transport blocks.

[0086] The transport block may be segmented into code blocks and code block CRC may be calculated and attached to a code block. A code block may be LDPC coded and the LDPC coded blocks may be individually rate matched. The code blocks may be concatenated to create one or more codewords. The contents of a codeword may be scrambled and modulated to generate a block of complex-valued modulation symbols. The modulation symbols may be mapped to a plurality of transmission layers (e.g., multiple-input multiple-output (MIMO) layers) and the transmission layers may be subject to transform precoding and/or precoding. The precoded complex-valued symbols may be mapped to radio resources (e.g., resource elements). The signal generator block may create a baseband signal and up-convert the baseband signal to a carrier frequency for transmission via antenna ports. The signal generator block may employ mixers, filters and/or other radio frequency (RF) components prior to transmission via the antennas. The functions and blocks in FIG. 6 are illustrated as examples and other mechanisms may be implemented in various embodiments.

[0087] FIG. 7 shows examples of RRC states and RRC state transitions at a UE in accordance with several of various embodiments of the present disclosure. A UE may be in one of three RRC states: RRC_IDLE 702, RRC_INACTIVE 704 and RRC_CONNECTED 706. In RRC_IDLE 702 state, no RRC context (e.g., parameters needed for communications between the UE and the network) may be established for the UE in the RAN. In RRC_IDLE 702 state, no data transfer between the UE and the network may take place and uplink synchronization is not maintained. The wireless device may sleep most of the time and may wake up periodically to receive paging messages. The uplink transmission of the UE may be based on a random access process and to enable transition to the RRC_CONNECTED 706 state. The mobility in RRC_IDLE 702 state is through a cell reselection procedure where the UE camps on a cell based on one or more criteria including signal strength that is determined based on the UE measurements.

[0088] In RRC_CONNECTED 706 state, the RRC context is established and both the UE and the RAN have necessary parameters to enable communications between the UE and the network. In the RRC_CONNECTED 706 state, the UE is configured with an identity known as a Cell Radio Network Temporary Identifier (C-RNTI) that is used for signaling purposes (e.g., uplink and downlink scheduling, etc.) between the UE and the RAN. The wireless device mobility in the RRC_CONNECTED 706 state is managed by the RAN. The wireless device provides neighboring cells and/or current serving cell measurements to the network and the network may make hand over decisions. Based on the

wireless device measurements, the current serving base station may send a handover request message to a neighboring base station and may send a handover command to the wireless device to handover to a cell of the neighboring base station. The transition of the wireless device from the RRC_IDLE 702 state to the RRC_CONNECTED 706 state or from the RRC_CONNECTED 706 state to the RRC_IDLE 702 state may be based on connection establishment and connection release procedures (shown collectively as connection establishment/release 710 in FIG. 7).

[0089] To enable a faster transition to the RRC_CONNECTED 706 state (e.g., compared to transition from RRC_IDLE 702 state to RRC_CONNECTED 706 state), an RRC_INACTIVE 704 state is used for an NR UE wherein, the RRC context is kept at the UE and the RAN. The transition from the RRC_INACTIVE 704 state to the RRC_CONNECTED 706 state is handled by RAN without CN signaling. Similar to the RRC_IDLE 702 state, the mobility in RRC_INACTIVE 704 state is based on a cell reselection procedure without involvement from the network. The transition of the wireless device from the RRC_INACTIVE 704 state to the RRC_CONNECTED 706 state or from the RRC_CONNECTED 706 state to the RRC_INACTIVE 704 state may be based on connection resume and connection inactivation procedures (shown collectively as connection resume/inactivation 712 in FIG. 7). The transition of the wireless device from the RRC_INACTIVE 704 state to the RRC_IDLE 702 state may be based on a connection release 714 procedure as shown in FIG. 7.

[0090] In NR, Orthogonal Frequency Division Multiplexing (OFDM), also called cyclic prefix OFDM (CP-OFDM), is the baseline transmission scheme in both downlink and uplink of NR and the Discrete Fourier Transform (DFT) spread OFDM (DFT-s-OFDM) is a complementary uplink transmission in addition to the baseline OFDM scheme. OFDM is multi-carrier transmission scheme wherein the transmission bandwidth may be composed of several narrowband subcarriers. The subcarriers are modulated by the complex valued OFDM modulation symbols resulting in an OFDM signal. The complex valued OFDM modulation symbols are obtained by mapping, by a modulation mapper, the input data (e.g., binary digits) to different points of a modulation constellation diagram. The modulation constellation diagram depends on the modulation scheme. NR may use different types of modulation schemes including Binary Phase Shift Keying (BPSK), $\pi/2$ -BPSK, Quadrature Phase Shift Keying (QPSK), 16 Quadrature Amplitude Modulation (16QAM), 64QAM and 256QAM. Different and/or higher order modulation schemes (e.g., M-QAM in general) may be used. An OFDM signal with N subcarriers may be generated by processing N subcarriers in parallel for example by using Inverse Fast Fourier Transform (IFFT) processing. The OFDM receiver may use FFT processing to recover the transmitted OFDM modulation symbols. The subcarrier spacing of subcarriers in an OFDM signal is inversely proportional to an OFDM modulation symbol duration. For example, for a 15 KHz subcarrier spacing, duration of an OFDM signal is nearly 66.7 μ s. To enhance the robustness of OFDM transmission in time dispersive channels, a cyclic prefix (CP) may be inserted at the beginning of an OFDM symbol. For example, the last part of an OFDM symbol may be copied and inserted at the beginning of an OFDM

symbol. The CP insertion enhanced the OFDM transmission scheme by preserving subcarrier orthogonality in time dispersive channels.

[0091] In NR, different numerologies may be used for OFDM transmission. A numerology of OFDM transmission may indicate a subcarrier spacing and a CP duration for the OFDM transmission. For example, a subcarrier spacing in NR may generally be a multiple of 15 KHz and expressed as $\Delta f = 2^\mu \cdot 15$ KHz ($\mu=0, 1, 2, \dots$). Example subcarrier spacings used in NR include 15 KHz ($\mu=0$), 30 KHz ($\mu=1$), 60 KHz ($\mu=2$), 120 KHz ($\mu=3$) and 240 KHz ($\mu=4$). As discussed before, a duration of OFDM symbol is inversely proportional to the subcarrier spacing and therefore OFDM symbol duration may depend on the numerology (e.g. the μ value).

[0092] FIG. 8 shows an example time domain transmission structure in NR wherein OFDM symbols are grouped into slots, subframes and frames in accordance with several of various embodiments of the present disclosure. A slot is a group of $N_{\text{slot}}^{\text{OFDM symbol}}$ OFDM symbols, wherein the $N_{\text{slot}}^{\text{OFDM symbol}}$ may have a constant value (e.g., 14). Since different numerologies results in different OFDM symbol durations, duration of a slot may also depend on the numerology and may be variable. A subframe may have a duration of 1 ms and may be composed of one or more slots, the number of which may depend on the slot duration. The number of slots per subframe is therefore a function of μ and may generally expressed as $N_{\text{slot}}^{\text{subframe}, \mu}$ and the number of symbols per subframe may be expressed as $N_{\text{symbol}}^{\text{subframe}, \mu}$. A frame may have a duration of 10 ms and may consist of 10 subframes. The number of slots per frame may depend on the numerology and therefore may be variable. The number of slots per frame may generally be expressed as $N_{\text{slot}}^{\text{frame}, \mu}$.

[0093] An antenna port may be defined as a logical entity such that channel characteristics over which a symbol on the antenna port is conveyed may be inferred from the channel characteristics over which another symbol on the same antenna port is conveyed. For example, for DM-RS associated with a PDSCH, the channel over which a PDSCH symbol on an antenna port is conveyed may be inferred from the channel over which a DM-RS symbol on the same antenna port is conveyed, for example, if the two symbols are within the same resource as the scheduled PDSCH and/or in the same slot and/or in the same precoding resource block group (PRG). For example, for DM-RS associated with a PDCCH, the channel over which a PDCCH symbol on an antenna port is conveyed may be inferred from the channel over which a DM-RS symbol on the same antenna port is conveyed if, for example, the two symbols are within resources for which the UE may assume the same precoding being used. For example, for DM-RS associated with a PBCH, the channel over which a PBCH symbol on one antenna port is conveyed may be inferred from the channel over which a DM-RS symbol on the same antenna port is conveyed if, for example, the two symbols are within a SS/PBCH block transmitted within the same slot, and with the same block index. The antenna port may be different from a physical antenna. An antenna port may be associated with an antenna port number and different physical channels may correspond to different ranges of antenna port numbers.

[0094] FIG. 9 shows an example of time-frequency resource grid in accordance with several of various embodiments of the present disclosure. The number of subcarriers

in a carrier bandwidth may be based on the numerology of OFDM transmissions in the carrier. A resource element, corresponding to one symbol duration and one subcarrier, may be the smallest physical resource in the time-frequency grid. A resource element (RE) for antenna port p and subcarrier spacing configuration μ may be uniquely identified by $(k, l)_{p, \mu}$ where k is the index of a subcarrier in the frequency domain and l may refer to the symbol position in the time domain relative to some reference point. A resource block may be defined as $N_{\text{SC}}^{\text{RB}} = 12$ subcarriers. Since subcarrier spacing depends on the numerology of OFDM transmission, the frequency domain span of a resource block may be variable and may depend on the numerology. For example, for a subcarrier spacing of 15 KHz (e.g., $\mu=0$), a resource block may be 180 KHz and for a subcarrier spacing of 30 KHz (e.g., $\mu=1$), a resource block may be 360 KHz.

[0095] With large carrier bandwidths defined in NR and due to limited capabilities for some UEs (e.g., due to hardware limitations), a UE may not support an entire carrier bandwidth. Receiving on the full carrier bandwidth may imply high energy consumption. For example, transmitting downlink control channels on the full downlink carrier bandwidth may result in high power consumption for wide carrier bandwidths. NR may use a bandwidth adaptation procedure to dynamically adapt the transmit and receive bandwidths. The transmit and receive bandwidth of a UE on a cell may be smaller than the bandwidth of the cell and may be adjusted. For example, the width of the transmit and/or receive bandwidth may change (e.g. shrink during period of low activity to save power); the location of the transmit and/or receive bandwidth may move in the frequency domain (e.g. to increase scheduling flexibility); and the subcarrier spacing of the transmit or receive bandwidth may change (e.g. to allow different services). A subset of the cell bandwidth may be referred to as a Bandwidth Part (BWP) and bandwidth adaptation may be achieved by configuring the UE with one or more BWPs. The base station may configure a UE with a set of downlink BWPs and a set of uplink BWPs. A BWP may be characterized by a numerology (e.g., subcarrier spacing and cyclic prefix) and a set of consecutive resource blocks in the numerology of the BWP. One or more first BWPs of the one or more BWPs of the cell may be active at a time. An active BWP may be an active downlink BWP or an active uplink BWP.

[0096] FIG. 10 shows an example of bandwidth part adaptation and switching. In this example, three BWPs (BWP₁ 1004, BWP₂ 1006 and BWP₃ 1008) are configured for a UE on a carrier bandwidth. The BWP₁ is configured with a bandwidth of 40 MHz and a numerology with subcarrier spacing of 15 KHz, the BWP₂ is configured with a bandwidth of 10 MHz and a numerology with subcarrier spacing of 15 KHz and the BWP₃ is configured with a bandwidth of 20 MHz and a subcarrier spacing of 60 KHz. The wireless device may switch from a first BWP (e.g., BWP₁) to a second BWP (e.g., BWP₂). An active BWP of the cell may change from the first BWP to the second BWP in response to the BWP switching.

[0097] The BWP switching (e.g., BWP switching 1010, BWP switching 1012, BWP switching 1014, or BWP switching 1016 in FIG. 10) may be based on a command from the base station. The command may be a DCI comprising scheduling information for the UE in the second BWP. In case of uplink BWP switching, the first BWP and the second BWP may be uplink BWPs and the scheduling information

may be an uplink grant for uplink transmission via the second BWP. In case of downlink BWP switching, the first BWP and the second BWP may be downlink BWPs and the scheduling information may be a downlink assignment for downlink reception via the second BWP.

[0098] The BWP switching (e.g., BWP switching **1010**, BWP switching **1012**, BWP switching **1014**, or BWP switching **1016** in FIG. **10**) may be based on an expiry of a timer. The base station may configure a wireless device with a BWP inactivity timer and the wireless device may switch to a default BWP (e.g., default downlink BWP) based on the expiry of the BWP inactivity timer. The expiry of the BWP inactivity timer may be an indication of low activity on the current active downlink BWP. The base station may configure the wireless device with the default downlink BWP. If the base station does not configure the wireless device with the default BWP, the default BWP may be an initial downlink BWP. The initial active BWP may be the BWP that the wireless device receives scheduling information for remaining system information upon transition to an RRC_CONNECTED state.

[0099] A wireless device may monitor a downlink control channel of a downlink BWP. For example, the UE may monitor a set of PDCCH candidates in configured monitoring occasions in one or more configured COnTrol REsource SETs (CORESETs) according to the corresponding search space configurations. A search space configuration may define how/where to search for PDCCH candidates. For example, the search space configuration parameters may comprise a monitoring periodicity and offset parameter indicating the slots for monitoring the PDCCH candidates. The search space configuration parameters may further comprise a parameter indicating a first symbol with a slot within the slots determined for monitoring PDCCH candidates. A search space may be associated with one or more CORESETs and the search space configuration may indicate one or more identifiers of the one or more CORESETs. The search space configuration parameters may further indicate that whether the search space is a common search space or a UE-specific search space. A common search space may be monitored by a plurality of wireless devices and a UE-specific search space may be dedicated to a specific UE.

[0100] FIG. **11A** shows example arrangements of carriers in carrier aggregation in accordance with several of various embodiments of the present disclosure. With carrier aggregation, multiple NR component carriers (CCs) may be aggregated. Downlink transmissions to a wireless device may take place simultaneously on the aggregated downlink CCs resulting in higher downlink data rates. Uplink transmissions from a wireless device may take place simultaneously on the aggregated uplink CCs resulting in higher uplink data rates. The component carriers in carrier aggregation may be on the same frequency band (e.g., intra-band carrier aggregation) or on different frequency bands (e.g., inter-band carrier aggregation). The component carriers may also be contiguous or non-contiguous. This results in three possible carrier aggregation scenarios, intra-band contiguous CA **1102**, intra-band non-contiguous CA **1104** and inter-band CA **1106** as shown in FIG. **11A**. Depending on the UE capability for carrier aggregation, a UE may transmit and/or receive on multiple carriers or for a UE that is not capable of carrier aggregation, the UE may transmit and/or receive on one component carrier at a time. In this disclo-

sure, the carrier aggregation is described using the term cell and a carrier aggregation capable UE may transmit and/or receive via multiple cells.

[0101] In carrier aggregation, a UE may be configured with multiple cells. A cell of the multiple cells configured for the UE may be referred to as a Primary Cell (PCell). The PCell may be the first cell that the UE is initially connected to. One or more other cells configured for the UE may be referred to as Secondary Cells (SCells). The base station may configure a UE with multiple SCells. The configured SCells may be deactivated upon configuration and the base station may dynamically activate or deactivate one or more of the configured SCells based on traffic and/or channel conditions. The base station may activate or deactivate configured SCells using a SCell Activation/Deactivation MAC CE. The SCell Activation/Deactivation MAC CE may comprise a bitmap, wherein each bit in the bitmap may correspond to a SCell and the value of the bit indicates an activation status or deactivation status of the SCell.

[0102] An SCell may also be deactivated in response to expiry of a SCell deactivation timer of the SCell. The expiry of an SCell deactivation timer of an SCell may be an indication of low activity (e.g., low transmission or reception activity) on the SCell. The base station may configure the SCell with an SCell deactivation timer. The base station may not configure an SCell deactivation timer for an SCell that is configured with PUCCH (also referred to as a PUCCH SCell). The configuration of the SCell deactivation timer may be per configured SCell and different SCells may be configured with different SCell deactivation timer values. The SCell deactivation timer may be restarted based on one or more criteria including reception of downlink control information on the SCell indicating uplink grant or downlink assignment for the SCell or reception of downlink control information on a scheduling cell indicating uplink grant or downlink assignment for the SCell or transmission of a MAC PDU based on a configured uplink grant or reception of a configured downlink assignment.

[0103] A PCell for a UE may be an SCell for another UE and a SCell for a UE may be PCell for another UE. The configuration of PCell may be UE-specific. One or more SCells of the multiple SCells configured for a UE may be configured as downlink-only SCells, e.g., may only be used for downlink reception and may not be used for uplink transmission. In case of self-scheduling, the base station may transmit signaling for uplink grants and/or downlink assignments on the same cell that the corresponding uplink or downlink transmission takes place. In case of cross-carrier scheduling, the base station may transmit signaling for uplink grants and/or downlink assignments on a cell different from the cell that the corresponding uplink or downlink transmission takes place.

[0104] FIG. **11B** shows examples of uplink control channel groups in accordance with several of various embodiments of the present disclosure. A base station may configure a UE with multiple PUCCH groups wherein a PUCCH group comprises one or more cells. For example, as shown in FIG. **11B**, the base station may configure a UE with a primary PUCCH group **1114** and a secondary PUCCH group **1116**. The primary PUCCH group may comprise the PCell **1110** and one or more first SCells. First UCI corresponding to the PCell and the one or more first SCells of the primary PUCCH group may be transmitted by the PUCCH of the PCell. The first UCI may be, for example, HARQ feedback

for downlink transmissions via downlink CCs of the PCell and the one or more first SCells. The secondary PUCCH group may comprise a PUCCH SCell and one or more second SCells. Second UCI corresponding to the PUCCH SCell and the one or more second SCells of the secondary PUCCH group may be transmitted by the PUCCH of the PUCCH SCell. The second UCI may be, for example, HARQ feedback for downlink transmissions via downlink CCs of the PUCCH SCell and the one or more second SCells.

[0105] FIG. 12A, FIG. 12B and FIG. 12C show example random access processes in accordance with several of various embodiments of the present disclosure. FIG. 12A shows an example of four step contention-based random access (CBRA) procedure. The four-step CBRA procedure includes exchanging four messages between a UE and a base station. Msg1 may be for transmission (or retransmission) of a random access preamble by the wireless device to the base station. Msg2 may be the random access response (RAR) by the base station to the wireless device. Msg3 is the scheduled transmission based on an uplink grant indicated in Msg2 and Msg4 may be for contention resolution.

[0106] The base station may transmit one or more RRC messages comprising configuration parameters of the random access parameters. The random access parameters may indicate radio resources (e.g., time-frequency resources) for transmission of the random access preamble (e.g., Msg1), configuration index, one or more parameters for determining the power of the random access preamble (e.g., a power ramping parameter, a preamble received target power, etc.), a parameter indicating maximum number of preamble transmission, RAR window for monitoring RAR, cell-specific random access parameters and UE specific random access parameters. The UE-specific random access parameters may indicate one or more PRACH occasions for random access preamble (e.g., Msg1) transmissions. The random access parameters may indicate association between the PRACH occasions and one or more reference signals (e.g., SSB or CSI-RS). The random access parameters may further indicate association between the random access preambles and one or more reference signals (e.g., SSB or CSI-RS). The UE may use one or more reference signals (e.g., SSB(s) or CSI-RS(s)) and may determine a random access preamble to use for Msg1 transmission based on the association between the random access preambles and the one or more reference signals. The UE may use one or more reference signals (e.g., SSB(s) or CSI-RS(s)) and may determine the PRACH occasion to use for Msg1 transmission based on the association between the PRACH occasions and the reference signals. The UE may perform a retransmission of the random access preamble if no response is received with the RAR window following the transmission of the preamble. UE may use a higher transmission power for retransmission of the preamble. UE may determine the higher transmission power of the preamble based on the power ramping parameter.

[0107] Msg2 is for transmission of RAR by the base station. Msg2 may comprise a plurality of RARs corresponding to a plurality of random access preambles transmitted by a plurality of UEs. Msg2 may be associated with a random access temporary radio identifier (RA-RNTI) and may be received in a common search space of the UE. The RA-RNTI may be based on the PRACH occasion (e.g., time and frequency resources of a PRACH) in which a random

access preamble is transmitted. RAR may comprise a timing advance command for uplink timing adjustment at the UE, an uplink grant for transmission of Msg3 and a temporary C-RNTI. In response to the successful reception of Msg2, the UE may transmit the Msg3. Msg3 and Msg4 may enable contention resolution in case of CBRA. In a CBRA, a plurality of UEs may transmit the same random access preamble and may consider the same RAR as being corresponding to them. UE may include a device identifier in Msg3 (e.g., a C-RNTI, temporary C-RNTI or other UE identity). Base station may transmit the Msg4 with a PDSCH and UE may assume that the contention resolution is successful in response to the PDSCH used for transmission of Msg3 being associated with the UE identifier included in Msg3.

[0108] FIG. 12B shows an example of a contention-free random access (CFRA) process. Msg 1 (random access preamble) and Msg 2 (random access response) in FIG. 12B for CFRA may be analogous to Msg 1 and Msg 2 in FIG. 12A for CBRA. In an example, the CFRA procedure may be initiated in response to a PDCCH order from a base station. The PDCCH order for initiating the CFRA procedure by the wireless device may be based on a DCI having a first format (e.g., format 1_0). The DCI for the PDCCH order may comprise a random access preamble index, an UL/SUL indicator indicating an uplink carrier of a cell (e.g., normal uplink carrier or supplementary uplink carrier) for transmission of the random access preamble, a SS/PBCH index indicating the SS/PBCH that may be used to determine a RACH occasion for PRACH transmission, a PRACH mask index indicating the RACH occasion associated with the SS/PBCH indicated by the SS/PBCH index for PRACH transmission, etc. In an example, the CFRA process may be started in response to a beam failure recovery process. The wireless device may start the CFRA for the beam failure recovery without a command (e.g., PDCCH order) from the base station and by using the wireless device dedicated resources.

[0109] FIG. 12C shows an example of a two-step random access process comprising two messages exchanged between a wireless device and a base station. Msg A may be transmitted by the wireless device to the base station and may comprise one or more transmissions of a preamble and/or one or more transmissions of a transport block. The transport block in Msg A and Msg 3 in FIG. 12A may have similar and/or equivalent contents. The transport block of Msg A may comprise data and control information (e.g., SR, HARQ feedback, etc.). In response to the transmission of Msg A, the wireless device may receive Msg B from the base station. Msg B in FIG. 12C and Msg 2 (e.g., RAR) illustrated in FIGS. 12A and 12B may have similar and/or equivalent content.

[0110] The base station may periodically transmit synchronization signals (SSs), e.g., primary SS (PSS) and secondary SS (SSS) along with PBCH on each NR cell. The PSS/SSS together with PBCH is jointly referred to as a SS/PBCH block. The SS/PBCH block enables a wireless device to find a cell when entering to the mobile communications network or find new cells when moving within the network. The SS/PBCH block spans four OFDM symbols in time domain. The PSS is transmitted in the first symbol and occupies 127 subcarriers in frequency domain. The SSS is transmitted in the third OFDM symbol and occupies the same 127 subcarriers as the PSS. The are eight and nine

empty subcarriers on each side of the SSS. The PBCH is transmitted on the second OFDM symbol occupying 240 subcarriers, the third OFDM symbol occupying 48 subcarriers on each side of the SSS, and on the fourth OFDM symbol occupying 240 subcarriers. Some of the PBCH resources indicated above may be used for transmission of the demodulation reference signal (DMRS) for coherent demodulation of the PBCH. The SS/PBCH block is transmitted periodically with a period ranging from 5 ms to 160 ms. For initial cell search or for cell search during inactive/idle state, a wireless device may assume that the SS/PBCH block is repeated at least every 20 ms.

[0111] In NR, transmissions using of antenna arrays, with many antenna elements, and beamforming plays an important role specially in higher frequency bands. Beamforming enables higher capacity by increasing the signal strength (e.g., by focusing the signal energy in a specific direction) and by lowering the amount interference received at the wireless devices. The beamforming techniques may generally be divided to analog beamforming and digital beamforming techniques. With digital beamforming, signal processing for beamforming is carried out in the digital domain before digital-to-analog conversion and detailed control of both amplitude and phase of different antenna elements may be possible. With analog beamforming, the signal processing for beamforming is carried out in the analog domain and after the digital to analog conversion. The beamformed transmissions may be in one direction at a time. For example, the wireless devices that are in different directions relative to the base station may receive their downlink transmissions at different times. For analog receiver-side beamforming, the receiver may focus its receiver beam in one direction at a time.

[0112] In NR, the base station may use beam sweeping for transmission of SS/PBCH blocks. The SS/PBCH blocks may be transmitted in different beams using time multiplexing. The set of SS/PBCH blocks that are transmitted in one beam sweep may be referred to as a SS/PBCH block set. The period of PBCH/SSB block transmission may be a time duration between a SS/PBCH block transmission in a beam and the next SS/PBCH block transmission in the same beam. The period of SS/PBCH block is, therefore, also the period of the SS/PBCH block set.

[0113] FIG. 13A shows example time and frequency structure of SS/PBCH blocks and their associations with beams in accordance with several of various embodiments of the present disclosure. In this example, a SS/PBCH block (also referred to as SSB) set comprise L SSBs wherein an SSB in the SSB set is associated with (e.g., transmitted in) one of L beams of a cell. The transmission of SSBs of an SSB set may be confined within a 5 ms interval, either in a first half-frame or a second half-frame of a 10 ms frame. The number of SSBs in an SSB set may depend on the frequency band of operation. For example, the number of SSBs in a SSB set may be up to four SSBs in frequency bands below 3 GHz enabling beam sweeping of up to four beams, up to eight SSBs in frequency bands between 3 GHz and 6 GHz enabling beam sweeping of up to eight beams, and up to sixty four SSBs in higher frequency bands enabling beam sweeping of up to sixty four beams. The SSBs of an SSB may depend on a physical cell identity (PCI) of the cell and may be independent of which beam of the cell is used for transmission of the SSB. The PBCH of an SSB may indicate a time index parameter and the wireless device may deter-

mine the relative position of the SSB within the SSB set using the time index parameter. The wireless device may use the relative position of the SSB within an SSB set for determining the frame timing and/or determining RACH occasions for a random access process.

[0114] A wireless device entering the mobile communications network may first search for the PSS. After detecting the PSS, the wireless device may determine the synchronization up to the periodicity of the PSS. By detecting the PSS, the wireless device may determine the transmission timing of the SSS. The wireless device may determine the PCI of the cell after detecting the SSS. The PBCH of a SS/PBCH block is a downlink physical channel that carries the MIB. The MIB may be used by the wireless device to obtain remaining system information (RMSI) that is broadcast by the network. The RMSI may include System Information Block 1 (SIB1) that contains information required for the wireless device to access the cell.

[0115] As discussed earlier, the wireless device may determine a time index parameter from the SSB. The PBCH comprises a half-frame parameter indicating whether the SSB is in the first 5 ms half or the second 5 ms half of a 10 ms frame. The wireless device may determine the frame boundary using the time index parameter and the half-frame parameter. In addition, the PBCH may comprise a parameter indicating the system frame number (SFN) of the cell.

[0116] The base station may transmit CSI-RS and a UE may measure the CSI-RS to obtain channel state information (CSI). The base station may configure the CSI-RS in a UE-specific manner. In some scenarios, same set of CSI-RS resources may be configured for multiple UEs and one or more resource elements of a CSI-RS resource may be shared among multiple UEs. A CSI-RS resource may be configured such that it does not collide with a CORESET configured for the wireless device and/or with a DMRS of a PDSCH scheduled for the wireless device and/or transmitted SSBs. The UE may measure one or more CSI-RSs configured for the UE and may generate a CSI report based on the CSI-RS measurements and may transmit the CSI report to the base station for scheduling, link adaptation and/or other purposes.

[0117] NR supports flexible CSI-RS configurations. A CSI-RS resource may be configured with single or multiple antenna ports and with configurable density. Based on the number of configured antenna ports, a CSI-RS resource may span different number of OFDM symbols (e.g., 1, 2, and 4 symbols). The CSI-RS may be configured for a downlink BWP and may use the numerology of the downlink BWP. The CSI-RS may be configured to cover the full bandwidth of the downlink BWP or a portion of the downlink BWP. In some case, the CSI-RS may be repeated in every resource block of the CSI-RS bandwidth, referred to as CSI-RS with density equal to one. In some cases, the CSI-RS may be configured to be repeated in every other resource block of the CSI-RS bandwidth. CSI-RS may be non-zero power (NZP) CSI-RS or zero-power (ZP) CSI-RS.

[0118] The base station may configure a wireless device with one or more sets of NZP CSI-RS resources. The base station may configure the wireless device with a NZP CSI-RS resource set using an RRC information element (IE) NZP-CSI-RS-ResourceSet indicating a NZP CSI-RS resource set identifier (ID) and parameters specific to the NZP CSI-RS resource set. An NZP CSI-RS resource set may

comprise one or more CSI-RS resources. An NZP CSI-RS resource set may be configured as part of the CSI measurement configuration.

[0119] The CSI-RS may be configured for periodic, semi-persistent or aperiodic transmission. In case of the periodic and semi-persistent CSI-RS configurations, the wireless device may be configured with a CSI resource periodicity and offset parameter that indicate a periodicity and corresponding offset in terms of number of slots. The wireless device may determine the slots that the CSI-RSs are transmitted. For semi-persistent CSI-RS, the CSI-RS resources for CSI-RS transmissions may be activated and deactivated by using a semi-persistent (SP) CSI-CSI Resource Set Activation/Deactivation MAC CE. In response to receiving a MAC CE indicating activation of semi-persistent CSI-RS resources, the wireless device may assume that the CSI-RS transmissions will continue until the CSI-RS resources for CSI-RS transmissions are activated.

[0120] As discussed before, CSI-RS may be configured for a wireless device as NZP CSI-RS or ZP CSI-RS. The configuration of the ZP CSI-RS may be similar to the NZP CSI-RS with the difference that the wireless device may not carry out measurements for the ZP CSI-RS. By configuring ZP CSI-RS, the wireless device may assume that a scheduled PDSCH that includes resource elements from the ZP CSI-RS is rate matched around those ZP CSI-RS resources. For example, a ZP CSI-RS resource configured for a wireless device may be an NZP CSI-RS resource for another wireless device. For example, by configuring ZP CSI-RS resources for the wireless device, the base station may indicate to the wireless device that the PDSCH scheduled for the wireless device is rate matched around the ZP CSI-RS resources.

[0121] A base station may configure a wireless device with channel state information interference measurement (CSI-IM) resources. Similar to the CSI-RS configuration, configuration of locations and density of CSI-IM resources may be flexible. The CSI-IM resources may be periodic (configured with a periodicity), semi-persistent (configured with a periodicity and activated and deactivated by MAC CE) or aperiodic (triggered by a DCI).

[0122] Tracking reference signals (TRSs) may be configured for a wireless device as a set of sparse reference signals to assist the wireless in time and frequency tracking and compensating time and frequency variations in its local oscillator. The wireless device may further use the TRSs for estimating channel characteristics such as delay spread or doppler frequency. The base station may use a CSI-RS configuration for configuring TRS for the wireless device. The TRS may be configured as a resource set comprising multiple periodic NZP CSI-RS resources.

[0123] A base station may configure a UE and the UE may transmit sounding reference signals (SRSs) to enable uplink channel sounding/estimation at the base station. The SRS may support up to four antenna ports and may be designed with low cubic metric enabling efficient operation of the wireless device amplifier. The SRS may span one or more (e.g., one, two or four) consecutive OFDM symbols in time domain and may be located within the last n (e.g., six) symbols of a slot. In the frequency domain, the SRS may have a structure that is referred to as a comb structure and may be transmitted on every N th subcarrier. Different SRS transmissions from different wireless devices may have different comb structures and may be multiplexed in frequency domain.

[0124] A base station may configure a wireless device with one or more SRS resource sets and an SRS resource set may comprise one or more SRS resources. The SRS resources in an SRS resources set may be configured for periodic, semi-persistent or aperiodic transmission. The periodic SRS and the semi-persistent SRS resources may be configured with periodicity and offset parameters. The Semi-persistent SRS resources of a configured semi-persistent SRS resource set may be activated or deactivated by a semi-persistent (SP) SRS Activation/Deactivation MAC CE. The set of SRS resources included in an aperiodic SRS resource set may be activated by a DCI. A value of a field (e.g., an SRS request field) in the DCI may indicate activation of resources in an aperiodic SRS resource set from a plurality of SRS resource sets configured for the wireless device.

[0125] An antenna port may be associated with one or more reference signals. The receiver may assume that the one or more reference signals, associated with the antenna port, may be used for estimating channel corresponding to the antenna port. The reference signals may be used to derive channel state information related to the antenna port. Two antenna ports may be referred to as quasi co-located if characteristics (e.g., large-scale properties) of the channel over which a symbol is conveyed on one antenna port may be inferred from the channel over which a symbol is conveyed from another antenna port. For example, a UE may assume that radio channels corresponding to two different antenna ports have the same large-scale properties if the antenna ports are specified as quasi co-located. In some cases, the UE may assume that two antenna ports are quasi co-located based on signaling received from the base station. Spatial quasi-colocation (QCL) between two signals may be, for example, due to the two signals being transmitted from the same location and in the same beam. If a receive beam is good for a signal in a group of signals that are spatially quasi co-located, it may be assumed also be good for the other signals in the group of signals.

[0126] The CSI-RS in the downlink and the SRS in uplink may serve as quasi-location (QCL) reference for other physical downlink channels and physical uplink channels, respectively. For example, a downlink physical channel (e.g., PDSCH or PDCCCH) may be spatially quasi co-located with a downlink reference signal (e.g., CSI-RS or SSB). The wireless device may determine a receive beam based on measurement on the downlink reference signal and may assume that the determined received beam is also good for reception of the physical channels (e.g., PDSCH or PDCCCH) that are spatially quasi co-located with the downlink reference signal. Similarly, an uplink physical channel (e.g., PUSCH or PUCCH) may be spatially quasi co-located with an uplink reference signal (e.g., SRS). The base station may determine a receive beam based on measurement on the uplink reference signal and may assume that the determined received beam is also good for reception of the physical channels (e.g., PUSCH or PUCCH) that are spatially quasi co-located with the uplink reference signal.

[0127] The Demodulation Reference Signals (DM-RSs) enables channel estimation for coherent demodulation of downlink physical channels (e.g., PDSCH, PDCCCH and PBH) and uplink physical channels (e.g., PUSCH and PUCCH). The DM-RS may be located early in the transmission (e.g., front-loaded DM-RS) and may enable the receiver to obtain the channel estimate early and reduce the latency. The time-domain structure of the DM-RS (e.g.,

symbols wherein the DM-RS are located in a slot) may be based on different mapping types.

[0128] The Phase Tracking Reference Signals (PT-RSs) enables tracking and compensation of phase variations across the transmission duration. The phase variations may be, for example, due to oscillator phase noise. The oscillator phase noise may become more severe in higher frequencies (e.g., mmWave frequency bands). The base station may configure the PT-RS for uplink and/or downlink. The PT-RS configuration parameters may indicate frequency and time density of PT-RS, maximum number of ports (e.g., uplink ports), resource element offset, configuration of uplink PT-RS without transform precoder (e.g., CP-OFDM) or with transform precoder (e.g., DFT-s-OFDM), etc. The subcarrier number and/or resource blocks used for PT-RS transmission may be based on the C-RNTI of the wireless device to reduce risk of collisions between PT-RSs of wireless devices scheduled on overlapping frequency domain resources.

[0129] FIG. 13B shows example time and frequency structure of CSI-RSs and their association with beams in accordance with several of various embodiments of the present disclosure. A beam of the L beams shown in FIG. 13B may be associated with a corresponding CSI-RS resource. The base station may transmit the CSI-RSs using the configured CSI-RS resources and a UE may measure the CSI-RSs (e.g., received signal received power (RSRP) of the CSI-RSs) and report the CSI-RS measurements to the base station based on a reporting configuration. For example, the base station may determine one or more transmission configuration indication (TCI) states and may indicate the one or more TCI states to the UE (e.g., using RRC signaling, a MAC CE and/or a DCI). Based on the one or more TCI states indicated to the UE, the UE may determine a downlink receive beam and receive downlink transmissions using the receive beam. In case of a beam correspondence, the UE may determine a spatial domain filter of a transmit beam based on spatial domain filter of a corresponding receive beam. Otherwise, the UE may perform an uplink beam selection procedure to determine the spatial domain filter of the transmit beam. The UE may transmit one or more SRSs using the SRS resources configured for the UE and the base station may measure the SRSs and determine/select the transmit beam for the UE based the SRS measurements. The base station may indicate the selected beam to the UE. The CSI-RS resources shown in FIG. 13B may be for one UE. The base station may configure different CSI-RS resources associated with a given beam for different UEs by using frequency division multiplexing.

[0130] A base station and a wireless device may perform beam management procedures to establish beam pairs (e.g., transmit and receive beams) that jointly provide good connectivity. For example, in the downlink direction, the UE may perform measurements for a beam pair and estimate channel quality for a transmit beam by the base station (or a transmission reception point (TRP) more generally) and the receive beam by the UE. The UE may transmit a report indicating beam pair quality parameters. The report may comprise one or more parameters indicating one or more beams (e.g., a beam index, an identifier of reference signal associated with a beam, etc.), one or more measurement parameters (e.g., RSRP), a precoding matrix indicator (PMI), a channel quality indicator (CQI), and/or a rank indicator (RI).

[0131] FIG. 14A, FIG. 14B and FIG. 14C show example beam management processes (referred to as P1, P2 and P3, respectively) in accordance with several of various embodiments of the present disclosure. The P1 process shown in FIG. 14A may enable, based on UE measurements, selection of a base station (or TRP more generally) transmit beam and/or a wireless device receive beam. The TRP may perform a beam sweeping procedure where the TRP may sequentially transmit reference signals (e.g., SSB and/or CSI-RS) on a set of beams and the UE may select a beam from the set of beams and may report the selected beam to the TRP. The P2 procedure as shown in FIG. 14B may be a beam refinement procedure. The selection of the TRP transmit beam and the UE receive beam may be regularly reevaluated due to movements and/or rotations of the UE or movement of other objects. In an example, the base station may perform the beam sweeping procedure over a smaller set of beams and the UE may select the best beam over the smaller set. In an example, the beam shape may be narrower compared to beam selected based on the P1 procedure. Using the P3 procedure as shown in FIG. 14C, the TRP may fix its transmit beam and the UE may refine its receive beam.

[0132] A wireless device may receive one or more messages from a base station. The one or more messages may comprise one or more RRC messages. The one or more messages may comprise configuration parameters of a plurality of cells for the wireless device. The plurality of cells may comprise a primary cell and one or more secondary cells. For example, the plurality of cells may be provided by a base station and the wireless device may communicate with the base station using the plurality of cells. For example, the plurality of cells may be provided by multiple base station (e.g., in case of dual and/or multi-connectivity). The wireless device may communicate with a first base station, of the multiple base stations, using one or more first cells of the plurality of cells. The wireless device may communicate with a second base station of the multiple base stations using one or more second cells of the plurality of cells.

[0133] The one or more messages may comprise configuration parameters used for processes in physical, MAC, RLC, PCDP, SDAP, and/or RRC layers of the wireless device. For example, the configuration parameters may include values of timers used in physical, MAC, RLC, PCDP, SDAP, and/or RRC layers. For example, the configuration parameters may include parameters for configuring different channels (e.g., physical layer channel, logical channels, RLC channels, etc.) and/or signals (e.g., CSI-RS, SRS, etc.).

[0134] Upon starting a timer, the timer may start running until the timer is stopped or until the timer expires. A timer may be restarted if it is running. A timer may be started if it is not running (e.g., after the timer is stopped or after the timer expires). A timer may be configured with or may be associated with a value (e.g., an initial value). The timer may be started or restarted with the value of the timer. The value of the timer may indicate a time duration that the timer may be running upon being started or restarted and until the timer expires. The duration of a timer may not be updated until the timer is stopped or expires (e.g., due to BWP switching). This specification may disclose a process that includes one or more timers. The one or more timers may be implemented in multiple ways. For example, a timer may be used by the wireless device and/or base station to determine a time

window [t1, t2], wherein the timer is started at time t1 and expires at time t2 and the wireless device and/or the base station may be interested in and/or monitor the time window [t1, t2], for example to receive a specific signaling. Other examples of implementation of a timer may be provided.

[0135] FIG. 15 shows example components of a wireless device and a base station that are in communication via an air interface in accordance with several of various embodiments of the present disclosure. The wireless device 1502 may communicate with the base station 1542 over the air interface 1532. The wireless device 1502 may include a plurality of antennas. The base station 1542 may include a plurality of antennas. The plurality of antennas at the wireless device 1502 and/or the base station 1542 enables different types of multiple antenna techniques such as beam-forming, single-user and/or multi-user MIMO, etc.

[0136] The wireless device 1502 and the base station 1542 may have one or more of a plurality of modules/blocks, for example RF front end (e.g., RF front end 1530 at the wireless device 1502 and RF front end 1570 at the base station 1542), Data Processing System (e.g., Data Processing System 1524 at the wireless device 1502 and Data Processing System 1564 at the base station 1542), Memory (e.g., Memory 1512 at the wireless device 1502 and Memory 1542 at the base station 1542). Additionally, the wireless device 1502 and the base station 1542 may have other modules/blocks such as GPS (e.g., GPS 1514 at the wireless device 1502 and GPS 1554 at the base station 1542).

[0137] An RF front end module/block may include circuitry between antennas and a Data Processing System for proper conversion of signals between these two modules/blocks. An RF front end may include one or more filters (e.g., Filter(s) 1526 at RF front end 1530 or Filter(s) 1566 at the RF front end 1570), one or more amplifiers (e.g., Amplifier(s) 1528 at the RF front end 1530 and Amplifier(s) 1568 at the RF front end 1570). The Amplifier(s) may comprise power amplifier(s) for transmission and low-noise amplifier(s) (LNA(s)) for reception.

[0138] The Data Processing System 1524 and the Data Processing System 1564 may process the data to be transmitted or the received signals by implementing functions at different layers of the protocol stack such as PHY, MAC, RLC, etc. Example PHY layer functions that may be implemented by the Data Processing System 1524 and/or 1564 may include forward error correction, interleaving, rate matching, modulation, precoding, resource mapping, MIMO processing, etc. Similarly, one or more functions of the MAC layer, RLC layer and/or other layers may be implemented by the Data Processing System 1524 and/or the Data Processing System 1564. One or more processes described in the present disclosure may be implemented by the Data Processing System 1524 and/or the Data Processing System 1564. A Data Processing System may include an RF module (RF module 1516 at the Data Processing System 1524 and RF module 1556 at the Data Processing System 1564) and/or a TX/RX processor (e.g., TX/RX processor 1518 at the Data Processing System 1524 and TX/RX processor 1558 at the Data Processing System 1564) and/or a central processing unit (CPU) (e.g., CPU 1520 at the Data Processing System 1524 and CPU 1560 at the Data Processing System 1564) and/or a graphical processing unit (GPU) (e.g., GPU 1522 at the Data Processing System 1524 and GPU 1562 at the Data Processing System 1564).

[0139] The Memory 1512 may have interfaces with the Data Processing System 1524 and the Memory 1552 may have interfaces with Data Processing System 1564, respectively. The Memory 1512 or the Memory 1552 may include non-transitory computer readable mediums (e.g., Storage Medium 1510 at the Memory 1512 and Storage Medium 1550 at the Memory 1552) that may store software code or instructions that may be executed by the Data Processing System 1524 and Data Processing System 1564, respectively, to implement the processes described in the present disclosure. The Memory 1512 or the Memory 1552 may include random-access memory (RAM) (e.g., RAM 1506 at the Memory 1512 or RAM 1546 at the Memory 1552) or read-only memory (ROM) (e.g., ROM 1508 at the Memory 1512 or ROM 1548 at the Memory 1552) to store data and/or software codes.

[0140] The Data Processing System 1524 and/or the Data Processing System 1564 may be connected to other components such as a GPS module 1514 and a GPS module 1554, respectively, wherein the GPS module 1514 and a GPS module 1554 may enable delivery of location information of the wireless device 1502 to the Data Processing System 1524 and location information of the base station 1542 to the Data Processing System 1564. One or more other peripheral components (e.g., Peripheral Component(s) 1504 or Peripheral Component(s) 1544) may be configured and connected to the data Processing System 1524 and data Processing System 1564, respectively.

[0141] In example embodiments, a wireless device may be configured with parameters and/or configuration arrangements. For example, the configuration of the wireless device with parameters and/or configuration arrangements may be based on one or more control messages that may be used to configure the wireless device to implement processes and/or actions. The wireless device may be configured with the parameters and/or the configuration arrangements regardless of the wireless device being in operation or not in operation. For example, software, firmware, memory, hardware and/or a combination thereof and/or alike may be configured in a wireless device regardless of the wireless device being in operation or not operation. The configured parameters and/or settings may influence the actions and/or processes performed by the wireless device when in operation.

[0142] In example embodiments, a wireless device may receive one or more message comprising configuration parameters. For example, the one or more messages may comprise radio resource control (RRC) messages. A parameter of the configuration parameters may be in at least one of the one or more messages. The one or more messages may comprise information element (IEs). An information element may be a structural element that includes single or multiple fields. The fields in an IE may be individual contents of the IE. The terms configuration parameter, IE and field may be used equally in this disclosure. The IEs may be implemented using a nested structure, wherein an IE may include one or more other IEs and an IE of the one or more other IEs may include one or more additional IEs. With this structure, a parent IE contains all the offspring IEs as well. For example, a first IE containing a second IE, the second IE containing a third IE, and the third IE containing a fourth IE may imply that the first IE contains the third IE and the fourth IE.

[0143] In an example, a wireless device may receive one or more RRC messages comprising configuration param-

eters of downlink semi-Persistent Scheduling (SPS). The DL SPS configuration may be per Serving Cell and/or per BWP. In some examples, the base station may activate and/or deactivate/release the DL SPS and the activation or the deactivation/release of DL SPS may be independent among the Serving Cells. For the DL SPS, a wireless device may receive a DL assignment by PDCCH, which may be stored or may be cleared based on L1 signaling indicating SPS activation or deactivation.

[0144] In an example, the wireless device may receive RRC configuration parameters comprising: cs-RNTI: CS-RNTI for activation, deactivation, and retransmission; nrofHARQ-Processes: the number of configured HARQ processes for DL SPS; and periodicity: periodicity of configured downlink assignment for SPS. When SPS is released by upper layers, the corresponding configurations may be released.

[0145] In an example, after a downlink assignment is configured for SPS, the MAC entity may consider sequentially that the Nth downlink assignment occurs in the slot for which:

$$(\text{numberOfSlotsPerFrame} \times \text{SFN} + \text{slot number in the frame}) = [(\text{numberOfSlotsPerFrame} \times \text{SFN}(\text{start time}) + \text{slot}(\text{start time})) + N \times \text{periodicity} \times \text{numberOfSlotsPerFrame} / 10] \text{ modulo } (1024 \times \text{numberOfSlotsPerFrame})$$

where SFN(start time) and slot(start time) may be the SFN and slot, respectively, of the first transmission of PDSCH where the configured downlink assignment was (re-)initialized.

[0146] In an example, different types of transmission without dynamic grant may be configured for a wireless device in uplink. For a first type configured grant (e.g., configured grant Type 1), an uplink grant may be provided by RRC and stored as configured uplink grant. For a second type configured grant (e.g., configured grant Type 2), an uplink grant may be indicated by PDCCH, and stored or cleared as configured uplink grant based on L1 signaling indicating configured uplink grant activation or deactivation.

[0147] The configured grants Type 1 and Type 2 may be configured by RRC per Serving Cell and per BWP. In an example, multiple configurations may be active simultaneously. In an example, activation and deactivation of configured grant Type 2 may be independent among the Serving Cells.

[0148] A wireless device may receive may receive RRC configuration parameters for configured grant Type 1 comprising: cs-RNTI: indicating CS-RNTI for retransmission; periodicity: indicating periodicity of the configured grant Type 1; timeDomainOffset: indicating an offset of a resource with respect to SFN=0 in time domain; timeDomainAllocation: indicating allocation of configured uplink grant in time domain which contains startSymbolAndLength (e.g., SLIV); nrofHARQ-Processes: indicating the number of HARQ processes for configured grant.

[0149] In an example, a wireless device may receive RRC configuration parameters for configured grant Type 2 comprising: cs-RNTI: indicating CS-RNTI for activation, deactivation, and retransmission; periodicity: indicating periodicity of the configured grant Type 2; nrofHARQ-Processes: indicating the number of HARQ processes for configured grant.

[0150] In an example, upon configuration of a configured grant Type 1 for a Serving Cell by upper layers, the MAC

entity may: store the uplink grant provided by upper layers as a configured uplink grant for the indicated Serving Cell; initialize or re-initialize the configured uplink grant to start in the symbol according to timeDomainOffset and S (derived from SLIV), and to reoccur with periodicity.

[0151] In an example, after an uplink grant is configured for a configured grant Type 1, the MAC entity may consider that the uplink grant recurs associated with each symbol for which:

$$[(\text{SFN} \times \text{numberOfSlotsPerFrame} \times \text{numberOfSymbolsPerSlot}) + (\text{slot number in the frame} \times \text{numberOfSymbolsPerSlot}) + \text{symbol number in the slot}] = (\text{timeDomainOffset} \times \text{numberOfSymbolsPerSlot} + S + N \times \text{periodicity}) \text{ modulo } (1024 \times \text{numberOfSlotsPerFrame} \times \text{numberOfSymbolsPerSlot}), \text{ for all } N \geq 0.$$

[0152] In an example, after an uplink grant is configured for a configured grant Type 2, the MAC entity may consider that the uplink grant recurs associated with each symbol for which:

$$[(\text{SFN} \times \text{numberOfSlotsPerFrame} \times \text{numberOfSymbolsPerSlot}) + (\text{slot number in the frame} \times \text{numberOfSymbolsPerSlot}) + \text{symbol number in the slot}] = [(\text{SFN}(\text{start time}) \times \text{numberOfSlotsPerFrame} \times \text{numberOfSymbolsPerSlot} + \text{slot}(\text{start time}) \times \text{numberOfSymbolsPerSlot} + \text{symbol}(\text{start time})) + N \times \text{periodicity}] \text{ modulo } (1024 \times \text{numberOfSlotsPerFrame} \times \text{numberOfSymbolsPerSlot}), \text{ for all } N \geq 0.$$

where SFN(start time), slot(start time), and symbol(start time) may be the SFN, slot, and symbol, respectively, of the first transmission opportunity of PUSCH where the configured uplink grant was (re-)initialized.

[0153] When a configured uplink grant is released by upper layers, all the corresponding configurations may be released, and all corresponding uplink grants may be cleared.

[0154] In an example, if the configured uplink grant confirmation has been triggered and not cancelled; and if the MAC entity has UL resources allocated for new transmission: the MAC entity may instruct the Multiplexing and Assembly procedure to generate a Configured Grant Confirmation MAC CE; and cancel the triggered configured uplink grant confirmation.

[0155] In an example, for a configured grant Type 2, the MAC entity may clear the configured uplink grant after first transmission of Configured Grant Confirmation MAC CE triggered by the configured uplink grant deactivation. In an example, retransmissions except for repetition of configured uplink grants use uplink grants addressed to CS-RNTI.

[0156] In an example, the IE ConfiguredGrantConfig may be used to configure uplink transmission without dynamic grant according to two possible schemes. The actual uplink grant may be configured via RRC (type1) or provided via the PDCCH (addressed to CS-RNTI) (type2). The ConfiguredGrantConfig IE may comprise a subset of the following parameters: an antennaPort parameter indicating the antenna port(s) to be used for this configuration; a cg-DMRS-Configuration parameter indicating the DMRS configuration; a configuredGrantTimer parameter indicating an initial value of a configured grant timer; a dmrs-SeqInitialization parameter if transformPrecoder is disabled; a frequencyDomainAllocation parameter indicating the frequency domain resource allocation; a frequencyHopping parameter wherein a value intraSlot may enable 'Intra-slot frequency hopping'

and a value `interSlot` may enable ‘Inter-slot frequency hopping’; a `frequencyHoppingOffset` parameter indicating a frequency hopping offset used when frequency hopping is enabled; an `mcs-Table` parameter indicating the `mcs-Table` the UE may use for PUSCH without transform precoding; an `mcs-TableTransformPrecoder` parameter indicating the MCS table the UE may use for PUSCH with transform precoding; an `mcsAndTBS` parameter indicating a modulation order, target code rate and TB size; a `nrofHARQ-Processes` parameter indicating a number of HARQ processes configured; a `periodicity` parameter indicating a periodicity for UL transmission without UL grant for type 1 and type 2; a `powerControlLoopToUse` parameter indicating a closed control loop to apply; an `rbg-Size` parameter indicating a selection between configuration 1 and configuration 2 for RBG size for PUSCH; a `repK-RV` parameter indicating a redundancy version (RV) sequence to use; a `repK` parameter indicating a number of repetitions of K; a `resourceAllocation` parameter indicating configuration of resource allocation type 0 and resource allocation type 1; a `rrc-ConfiguredUplinkGrant` indicating configuration for “configured grant” transmission with fully RRC-configured UL grant (Type1) and if this field is absent the UE may use UL grant configured by DCI addressed to CS-RNTI (Type2); an `srsResourceIndicator` parameter indicating an SRS resource to be used; a `timeDomainAllocation` parameter indicating a combination of start symbol and length and PUSCH mapping type; a `timeDomainOffset` parameter indicating an offset with respect to SFN=0; a `transformPrecoder` parameter enabling or disabling transform precoding; and a `uci-OnPUSCH` parameter indicating a selection between and configuration of dynamic and semi-static beta-offset, wherein for Type 1 UL data transmission without grant, `uci-OnPUSCH` may be set to `semiStatic`.

[0157] In an example, the IE SPS-Config may be used to configure downlink semi-persistent transmission. Downlink SPS may be configured on the SpCell as well as on SCells. The IE SPS-Config may comprise one or more of a `mcs-Table` parameter indicating a MCS table the UE may use for DL SPS; an `n1PUCCH-AN` parameter indicating a HARQ resource for PUCCH for DL SPS; a `nrofHARQ-Processes` parameter indicating a number of configured HARQ processes for SPS DL; and a `periodicity` parameter indicating a periodicity for DL SPS.

[0158] In an example, a MAC entity may be configured with CS-RNTI. The MAC entity may receive an uplink grant for a PDCCH occasion for a Serving Cell belonging to a TAG that has a running `timeAlignmentTimer`. The uplink grant for this PDCCH occasion may have been received for the Serving Cell on the PDCCH for the MAC entity’s CS-RNTI.

[0159] In an example, if the NDI in the received HARQ information is 1: the MAC entity may consider the NDI for the corresponding HARQ process not to have been toggled; the MAC entity may start or restart a `configuredGrantTimer` for the corresponding HARQ process, if configured; and the MAC entity may deliver the uplink grant and the associated HARQ information to the HARQ entity.

[0160] In an example, if the NDI in the received HARQ information is 0 and if PDCCH contents indicate configured grant Type 2 deactivation, the MAC entity may trigger configured uplink grant confirmation.

[0161] In an example, if the NDI in the received HARQ information is 0 and if PDCCH contents indicate configured

grant Type 2 activation, the MAC entity may trigger configured uplink grant confirmation; the MAC entity may store the uplink grant for this Serving Cell and the associated HARQ information as configured uplink grant; the MAC entity may initialize or re-initialize the configured uplink grant for this Serving Cell to start in the associated PUSCH duration and to recur according to rules; and the MAC entity may stop the `configuredGrantTimer` for the corresponding HARQ process, if running.

[0162] In an example, higher layer configured parameters `repK` and `repK-RV` may define the K repetitions to be applied to the transmitted transport block, and the redundancy version pattern to be applied to the repetitions. If the parameter `repK-RV` is not provided in the `configuredGrant-Config`, the redundancy version for uplink transmissions with a configured grant may be set to 0. Otherwise, for the nth transmission occasion among K repetitions, $n=1, 2, \dots, K$, it may be associated with $(\text{mod}(n-1, 4)+1)$ th value in the configured RV sequence.

[0163] In an example, the initial transmission of a transport block may start at the first transmission occasion of the K repetitions if the configured RV sequence is {0,2,3,1}. The initial transmission of a transport block may start at any of the transmission occasions of the K repetitions that are associated with RV=0 if the configured RV sequence is {0,3,0,3}. The initial transmission of a transport block may start at any of the transmission occasions of the K repetitions if the configured RV sequence is {0,0,0,0}, except the last transmission occasion when K=8.

[0164] In an example, for any RV sequence, the repetitions may be terminated after transmitting K repetitions, or at the last transmission occasion among the K repetitions within the period P, or from the starting symbol of the repetition that overlaps with a PUSCH with the same HARQ process scheduled by DCI format 0_0 or 0_1, whichever is reached first. In an example, the UE may not be expected to be configured with the time duration for the transmission of K repetitions larger than the time duration derived by the periodicity P. If the UE determines that, for a transmission occasion, the number of symbols available for the PUSCH transmission in a slot is smaller than transmission duration L, the UE may not transmit the PUSCH in the transmission occasion.

[0165] In an example, for Type 1 and Type 2 PUSCH transmissions with a configured grant, when the UE is configured with `repK>1`, the UE may repeat the TB across the `repK` consecutive slots applying the same symbol allocation in each slot. A Type 1 or Type 2 PUSCH transmission with a configured grant in a slot may be omitted according to the conditions.

[0166] In an example, a wireless device may receive configuration parameters of a plurality of configured grant configurations for a BWP of a serving cell. The wireless device may be configured with separate RRC parameters for different configured grant configurations (for type 1 and/or type 2 configured grants) for a given BWP of a serving cell. In an example, some parameters may be common among different configured grant configurations.

[0167] In an example, different configured grant Type 2 configurations for a given BWP of a serving cell may be separately activated. In an example, a DCI may jointly activate two or more configured grant Type 2 configurations. In an example, different configured grant Type 2 configurations for a given BWP of a serving cell may be separately

released/deactivated. In an example, a DCI may jointly deactivate/release two or more configured grant Type 2 configurations.

[0168] In an example, one or multiple Type 1 configured grant configuration(s) and/or one or multiple Type 2 configured grant configuration(s) may be simultaneously active for a given BWP of a serving cell. In an example, same Type of multiple configured grant configurations may be simultaneously active for a given BWP of a serving cell.

[0169] In an example, $M \leq 4$ bits indication in a Release/deactivation DCI may be used for indicating which configured grant configuration(s) is/are released. The association between each state indicated by the indication and the configured grant configuration(s) may be configurable by RRC. In an example, up to 2^M states may be configurable by RRC, where each of the states may be mapped to a single or multiple configured grant configurations to be released. In an example, in case of no RRC configured state(s), separate release may be used where the release corresponds to the configured grant configuration index indicated by the indication. In an example, for activation and release of UL configured grant configuration, same field(s) may be used in a DCI format.

[0170] In an example, $M \leq 4$ bits indication in a Release/deactivation DCI may be used for indicating which DL SPS configuration(s) is/are released. The association between each state indicated by the indication and the DL SPS configuration(s) may be configurable by RRC. In an example, up to 2^M states may be configurable by RRC, where each of the states may be mapped to a single or multiple DL SPS configurations to be released. In an example, in case of no RRC configured state(s), separate release may be used where the release corresponds to the DL SPS configuration index indicated by the indication. In an example, for activation and release of DL SPS configuration, same field(s) may be used in a DCI format.

[0171] In an example, DCI format 0_0, DCI format 0_1 and/or a new DCI format may be used for scheduling PUSCH for Type 2 configured grant activation. In an example, DCI format 0_0, DCI format 0_1 and/or a new DCI format may be used for Type 2 configured grant deactivation/release.

[0172] In an example, M ($M \leq 4$) least significant bits of HARQ Process Number (HPN) field in DCI format 0_0 with CRC scrambled by CS-RNTI may be used to indicate which configuration is to be activated and which configuration(s) is/are to be released. In an example, ($M \leq 4$) least significant bits of HPN field in DCI format 0_1 with CRC scrambled by CS-RNTI may be used to indicate which configuration is to be activated. In an example, M ($M \leq 4$) least significant bits of HPN field in DCI format 0_1 with CRC scrambled by CS-RNTI may be used to indicate which configuration(s) are to be released. In an example, the M for activation (e.g., M_a) and M for deactivation/release (e.g., M_r) may be different.

[0173] In an example, at least HPN field in a new UL DCI format may be used to indicate which configuration is to be activated and/or which configuration(s) is/are to be released. In an example, other field(s) may be used with or in place of the HPN field if the number of bits for HPN field is smaller than M .

[0174] In an example, a UE may validate, for scheduling activation or scheduling release, a DL SPS assignment PDCCH or configured UL grant Type 2 PDCCH if the CRC of a corresponding DCI format is scrambled with a CS-

RNTI provided by cs-RNTI, and the new data indicator field for the enabled transport block is set to '0'. An example flow diagram for PDCCH validation for configured grant and/or DL SPS activation and/or deactivation/release is shown in FIG. 16. The validation of the DCI format may be achieved if one or more fields for the DCI format are set according to first (e.g., pre-defined values). If validation is achieved, the UE may consider the information in the DCI format as a valid activation or valid release/deactivation of DL SPS or configured UL grant Type 2. If validation is not achieved, the UE may discard the information in the DCI format.

[0175] When a wireless device receives a DCI for activation or deactivation/release of one or more uplink configured grant configurations or one or more DL SPS configurations, a validation process may be an important process prior to the activation or deactivation/release by the wireless device. The PDCCH carrying the DCI may be validated for scheduling activation or deactivation/release. Existing validation mechanisms leads to inefficient configured grant or SPS performance when DCI carries information required for identification of configured grants or SPS. There is a need to enhance the existing validation processes to improve the SPS and configured grants performance. Example embodiments enhance the validation process for configured grant and DL SPS.

[0176] In an example embodiment as shown in FIG. 17, a wireless device may receive one or more messages comprising configuration parameters. The one or more messages may comprise one or more RRC messages. The one or more messages may comprise configuration parameters of a first configured grant configuration. The configuration parameters of the first configured grant configuration may comprise a periodicity, a number of HARQ processes for the configured grant configuration, a configuration index/identifier of the configured grant configuration or a group of configured grant configurations comprising the configured grant configuration, one or more power control parameters, one or more parameters for repetition if configured, etc. The first configured grant configuration may be for a Type 2 configured grant, wherein a plurality of resources may be activated based on the receiving the configuration parameters and receiving an activation DCI indicating activation of the configured grant configuration. A Type 2 configured grant configuration may be deactivated/released based on receiving a DCI indicating the deactivation of the configured grant configuration. The wireless device may validate an activation DCI and the wireless device may activate the first configured grant configuration based on the validation being successful (achieved). The wireless device may validate a deactivation/release DCI and the wireless device may deactivate the first configured grant configuration based on the validation being successful (achieved).

[0177] In an example, the wireless device may receive a downlink control information (DCI) via a downlink control channel (e.g., PDCCH). The DCI may indicate scheduling activation of the first configured grant configuration.

[0178] In an example, the wireless device may receive a downlink control information (DCI) via a downlink control channel (e.g., PDCCH). The DCI may indicate scheduling deactivation/release of the first configured grant configuration.

[0179] The DCI may have one of a plurality of formats (e.g., an uplink DCI format such as format 0_0, format 0_1, a new format, etc.) and may comprise a plurality of fields.

The plurality of fields may comprise a HARQ process number (HPN) field. An example is shown in FIG. 18. The HPN field may comprise first bits wherein the number of the first bits may be a first number. The first number (e.g., bitwidth of the HPN field) may be pre-defined or may be a configurable parameter (e.g., based on an RRC parameter). The first bits may comprise one or more second bits and one or more third bits. The number of the one or more second bits may be a second number and the number of the one or more third bits may be a third number. The first bits may comprise one or more second bits and one or more third bits. The one or more third bits may indicate the configuration index of the first configured grant configuration. In an example, the one or more third bits may indicate a group of one or more configured grant configurations comprising the first configured grant configuration. In an example, by determining the one or more third bits, the wireless device may determine the one or more second bits from the first bits and the one or more third bits. In an example, the third number of the one or more third bits may be configurable (e.g., the one or more messages may indicate the third number) and the wireless device may determine the one or more third bits based on the first bits and the third number and the position of the one or more third bits.

[0180] In an example as shown in FIG. 17, the wireless device may validate the downlink control channel for scheduling activation. In an example, the wireless device may validate the downlink control for scheduling deactivation/release. The wireless device may validate the DCI (for scheduling activation or for scheduling deactivation/release) based on the one or more second bits of the one or more first bits of the HPN field. In an example, the validation (for scheduling activation or for scheduling deactivation/release) may be based on comparing a value of the one or more second bits with first value. In an example, the first value may be pre-determined, for example, all zeros. In an example, the first value may be configurable (e.g., with RRC configuration). The one or more messages may comprise a parameter indicating the first value for comparison and validation.

[0181] In an example, the validation (for scheduling activation or deactivation/release) may further comprises comparing a second field of the downlink control information with a second sequence of one or more second bits. For example, the second field may be a redundancy version (RV) field. The second sequence of the one or more second bits may be predetermined. For example, the second sequence of the one or more second bits may be all zeros. In an example, the second sequence of the one or more second bits may be configurable (e.g., with RRC configuration). The one or more messages may comprise a parameter indicating the value of the one or more second bits for comparison and validation.

[0182] In an example, the validating the downlink control channel (control channel carrying the downlink control information) for activation and/or for release/deactivation may be based on (e.g., in response to) a new data indicator field of the downlink control information having a first value. The first value may be zero. The wireless device may validate the downlink control for activation and/or for release/deactivation based on the new data indicator having a value of zero, otherwise the wireless device may not validate the downlink control information (control channel carrying the downlink control information).

[0183] In an example, the validating the downlink control channel (control channel carrying the downlink control information) for activation and/or for release/deactivation may be based on (e.g., in response to) the downlink control information being associated with a first RNTI. The first RNTI may be a CS-RNTI. The wireless device may validate the downlink control for activation and/or for release/deactivation based on the radio network temporary identifier associated with the downlink control information being the first RNTI, otherwise the wireless device may not validate the downlink control information (control channel carrying the downlink control information).

[0184] In an example, the validating the downlink control channel (control channel carrying the downlink control information) for activation and/or for release/deactivation may be based on (e.g., in response to) a radio network temporary identifier of the downlink control information being associated with a first RNTI and a new data indicator field of the downlink control information being a first value. The first RNTI may be a CS-RNTI and the first value may be zero. The wireless device may validate the downlink control for activation and/or for release/deactivation based on the radio network temporary identifier associated with the downlink control information being the first RNTI and the new data indicator field of the downlink control information being zero, otherwise the wireless device may not validate the downlink control information (control channel carrying the downlink control information).

[0185] In an example, the validating the downlink control channel (control channel carrying the downlink control information) for activation and/or for release/deactivation based on the HPN field (or the one or more second bits of the HPN field) may be in response to the downlink control information having one of a plurality of formats (e.g., format 0_0 or format 0_1). In response to the downlink control information not having one of the plurality of formats, the validation may not be based on the HPN field.

[0186] In an example, based on the validation for scheduling activation being successful, the wireless device may activate the configured grant configuration. The activating the configured grant configuration may comprise activating a plurality of resources based on the configured grant configuration. The wireless device may determine the plurality of resources based on one or more configuration parameters (e.g., periodicity) of the configured grant configuration and one or more parameters (e.g., resource allocation parameters) indicated by the downlink control information.

[0187] In an example, based on the validation for scheduling deactivation/release being successful, the wireless device may deactivate/release the configured grant configuration. The deactivating the configured grant configuration may comprise deactivating a plurality of resources associated with the configured grant configuration. The deactivating the plurality of resources may release the plurality of resources and the plurality of resources may be available for scheduling for the wireless device or other wireless devices. The deactivating the plurality of resources may be a physical layer deactivation and the configured grant configuration parameters (e.g., RRC configurations) may be still available for the wireless device. The wireless device may receive a new activation command indicating activation of new radio resources based on the configured grant configuration parameters.

[0188] In an example embodiment, a wireless device may receive from a base station, one or more messages comprising first configuration parameters of a first configured grant configuration. The wireless device may receive, via a downlink control channel, a downlink control information indicating scheduling activation of the first configured grant configuration, wherein the downlink control information comprises a hybrid automatic repeat request process number field comprising first bits. The wireless device may validate, for scheduling activation, the downlink control channel based on one or more second bits of the first bits. The wireless device may activate the configured grant configuration for scheduling based on the validating being successful.

[0189] In an example, the one or more third bits may be used to identify one or more configured grant configurations. The one or more third bits may indicate a single configured grant configuration index or an identifier of a group of one or more configured grant configurations. In an example, the group of one or more configured grant configurations may be for one cell (or a BWP of one cell). In an example, the group of configured grant configurations may be for one or more cells (or one or more BWPs of one or more cells). In an example, the one or more messages may comprise a parameter indicating whether the wireless device may interpret the one or more third bits as a configuration index of a single configured grant configuration or an identifier of a group of one or more configured grant configurations. In an example, the wireless device may determine whether the wireless device may interpret the one or more third bits as a configuration index of a single configured grant configuration or an identifier of a group of one or more configured grant configurations based on a value of a second field in the DCI. A value of the second field may be an explicit indication (e.g., a flag) indicating whether the wireless device may interpret the one or more third bits as a configuration index of a single configured grant configuration or an identifier of a group of one or more configured grant configurations. In an example, a value of the second field may implicitly indicate whether the wireless device may interpret the one or more third bits as a configuration index of a single configured grant configuration or an identifier of a group of one or more configured grant configurations.

[0190] In an example, the third number may be separately configured for activation and for deactivation/release. For example, the one or more third bits may indicate a configuration index/identifier of a single configured grant configuration in case of activation and may indicate an identifier of a single or a group of one or more configured grant configurations in case of release/deactivation. The third number for activation may be smaller than the third number for deactivation/release. The wireless device may receive separate configuration parameters of the third number for activation and for deactivation/release.

[0191] In an example, the wireless device may determine the third number (at least for configured grant configuration activation) based on a number of configured grant configurations configured for the wireless device. For example, for N configured grant configurations, the third number may be logarithm of N in base 2 (e.g., $\log_2(N)$). For example, for N configured grant configurations, the third number may be floor of logarithm of N in base 2 plus one (e.g., $\lfloor \log_2(N) \rfloor + 1$). For example, for N configured grant configura-

tions, the third number may be the smallest number that can indicate N different configured grant configurations.

[0192] In an example, the wireless device may determine the third number for release/deactivation based on the third number for activation plus an offset. The offset may be configurable (e.g., as an RRC parameter or based on an RRC parameter) or may be a pre-defined value.

[0193] In an example, the third number may be based on values of one or more fields of a downlink control information (e.g., the downlink control information for scheduling activation or scheduling deactivation/release).

[0194] In an example, the wireless device may determine the one or more third bits based on the first bits of the HPN field and the third number. The one or more third bits may be third number least significant bits of the first bits. For example, if the HPN field of the activation or deactivation/release DCI (e.g., the first bits) has a value of 1110 and the third number is 2, the one or more third bits is the two least significant bits of 1110 or 10. For example, if the HPN field of the activation or deactivation/release DCI (e.g., the first bits) has a value of 0111 and the third number is 3, the one or more third bits is the three least significant bits of 0111 or 111.

[0195] The wireless device may determine the one or more second bits based on the first bits of the HPN field and the one or more third bits. For example, the one or more second bits may be remaining bits of the first bits after determining the one or more third bits. For example, if the HPN field of the activation or deactivation/release DCI (e.g., the first bits) has a value of 1110 and the one or more third bits are 10, the one or more second bits are 11. For example, if the HPN field of the activation or deactivation/release DCI (e.g., the first bits) has a value of 0111 and the one or more third bits are 111, the one or more second bits is 0.

[0196] In an example, the wireless device may determine the one or more second bits as the second number most significant bits of the first bits. The wireless device may determine the second number based on the first number and the third number. The first number may a fixed number (e.g., the HPN field may have 4 bitwidth) or may be configurable (e.g., the HPN field may have a configurable bitwidth indicated by an RRC parameter). For example, if the HPN field of the activation or deactivation/release DCI (e.g., the first bits) has a value of 1110 and the second number is 2, the one or more second bits are 11. For example, if the HPN field of the activation or deactivation/release DCI (e.g., the first bits) has a value of 0111 and the second number is 1, the one or more second bits are 0.

[0197] In an example embodiment as shown in FIG. 22, the wireless device may trigger a confirmation MAC control element based on the validating for scheduling activation or the validating for scheduling release/deactivation being successful. The wireless may create the confirmation MAC CE and multiplex the confirmation MAC CE in a transport block (e.g., multiplex with data of logical channels or other MAC CEs) and may transmit the transport block using an uplink resource. The reception of the MAC CE by the base station may indicate a confirmation that the wireless device received the DCI and validated the DCI for activation and/or release/deactivation.

[0198] In an example embodiment as shown in FIG. 19, a wireless device may receive one or more messages comprising configuration parameters of a plurality of configured

grant configurations comprising a first configured grant configuration and a second configured grant configuration.

[0199] A plurality of DCI formats may be used for activation and/or release/deactivation of a configured grant configuration. The wireless device may receive a first downlink control information via a downlink control channel, wherein the first downlink control information is of a first format and the first downlink control information indicates scheduling activation of the first configured grant configuration. The wireless device may receive a second downlink control information via the downlink control channel, wherein the second downlink control information is of a second format and the second downlink control information indicates scheduling activation of the second configured grant configuration. The first downlink control information may comprise a first HPN field and the second downlink control information may comprise a second HPN field.

[0200] In an example, the wireless device may receive a first downlink control information via a downlink control channel, wherein the first downlink control information is of a first format and the first downlink control information indicates scheduling deactivation/release of the first configured grant configuration. The wireless device may receive a second downlink control information via the downlink control channel, wherein the second downlink control information is of a second format and the second downlink control information indicates scheduling deactivation/release of the second configured grant configuration. The first downlink control information may comprise a first HPN field and the second downlink control information may comprise a second HPN field.

[0201] Based on the first downlink control information having the first format, the wireless device may validate the downlink control channel for scheduling activation of the first configured grant configuration employing a value of the first HPN field. For example, the wireless device may validate the downlink control channel for scheduling activation of the first configured grant configuration based on value of one or more bits of the first bits of the HPN field.

[0202] In an example, based on the first downlink control information having the first format, the wireless device may validate the downlink control channel for scheduling deactivation/release of the first configured grant configuration employing a value of the first HPN field. For example, the wireless device may validate the downlink control channel for scheduling deactivation/release of the first configured grant configuration based on value of one or more bits of the first bits of the HPN field.

[0203] Based on the second downlink control information having the second format, the wireless device may validate the downlink control channel for scheduling activation of the second configured grant configuration without employing a value of the second HPN field. The wireless device may employ one or more values of one or more other fields of the second downlink control information for validating the downlink control channel for scheduling activation of the second configured grant configuration.

[0204] In an example, based on the second downlink control information having the second format, the wireless device may validate the downlink control channel for scheduling deactivation/release of the second configured grant configuration without employing a value of the second HPN field. The wireless device may employ one or more values of one or more other fields of the second downlink control

information for validating the downlink control channel for scheduling deactivation/release of the second configured grant configuration.

[0205] In an example embodiment, a wireless device may receive one or more messages comprising configuration parameters of a configured grant configuration. The wireless device may receive a downlink control information via a downlink control channel, wherein the downlink control information comprises an HPN field.

[0206] Based on the downlink control information having one or more first formats, the wireless device may validate the downlink control channel for scheduling activation of first configured grant configuration employing a value of the HPN field. For example, the wireless device may validate the downlink control channel for scheduling activation of the configured grant configuration based on value of one or more bits of the first bits of the HPN field.

[0207] In an example, based on the downlink control information having one or more first formats, the wireless device may validate the downlink control channel for scheduling deactivation/release of the configured grant configuration employing a value of the HPN field. For example, the wireless device may validate the downlink control channel for scheduling deactivation/release of the configured grant configuration based on value of one or more bits of the first bits of the HPN field.

[0208] Based on the downlink control information having one or more second formats, the wireless device may validate the downlink control channel for scheduling activation of the configured grant configuration without employing a value of the HPN field. The wireless device may employ one or more values of one or more other fields of the downlink control information for validating the downlink control channel for scheduling activation of the configured grant configuration.

[0209] In an example, based on the second downlink control information having one or more second formats, the wireless device may validate the downlink control channel for scheduling deactivation/release of the configured grant configuration without employing a value of the HPN field. The wireless device may employ one or more values of one or more other fields of the downlink control information for validating the downlink control channel for scheduling deactivation/release of the configured grant configuration.

[0210] A MAC PDU may comprise of one or more MAC subPDUs. A MAC subPDU may comprise of one of the following: a MAC subheader only (including padding); a MAC subheader and a MAC SDU; a MAC subheader and a MAC CE; a MAC subheader and padding. The MAC SDUs may be of variable sizes. A MAC subheader may correspond to either a MAC SDU, a MAC CE, or padding. A MAC subheader except for fixed sized MAC CE, padding, and a MAC SDU containing UL CCCH may comprise of the four header fields R/F/LCID/L. A MAC subheader for fixed sized MAC CE, padding, and a MAC SDU containing UL CCCH may comprise of the two header fields R/LCID. FIG. 23A, FIG. 23B and FIG. 23C show example MAC subheaders. For example, FIG. 23A shows an example R/F/LCID/L MAC subheader with 8-bit L field. FIG. 23B shows an example R/F/LCID/L MAC subheader with 16-bit L field and FIG. 23C shows an example R/LCID MAC subheader. MAC CEs may be placed together. DL MAC subPDU(s) with MAC CE(s) may be placed before a MAC subPDU with MAC SDU and MAC subPDU with padding. The UL

MAC subPDU(s) with MAC CE(s) may be placed after the MAC subPDU(s) with MAC SDU and before the MAC subPDU with padding in the MAC PDU. The size of padding may be zero.

[0211] FIG. 24 shows an example uplink MAC PDU (transport block) comprising a plurality of MAC subPDUs, wherein a MAC subPDU may comprise a subheader and MAC SDU (e.g. data from one or more logicals) or a MAC subPDU may comprise a subheader and a MAC control element (MAC CE). A MAC CE may be a fixed-size MAC CE wherein the length of the MAC CE may be a fixed value or may be a variable-sized MAC CE wherein the length of the MAC CE may be variable.

[0212] The subheader of a MAC subPDU may comprise a logical channel identifier (LCID). The LCID field may identify the logical channel instance of the corresponding MAC SDU or the type of the corresponding MAC CE or padding. In an example, a MAC subheader may comprise one LCID field. The LCID field size may be 6 bits. One or more first LCIDs (e.g., LCIDs for MAC CEs included in MAC subPDUs) may have pre-defined values and one or more second LCIDs (e.g., LCIDs for MAC SDUs included in subPDUs) may be configured for a logical channel/bearer, the data of which is included in the MAC subPDU. For example, an RRC parameter logicalchannelidentity may indicate an LCID of a logical channel.

[0213] When a MAC subPDU comprises a MAC SDU or a variable-sized MAC CE, the subheader contained in the MAC subPDU may have a length field. The Length field may indicate the length of the corresponding MAC SDU or variable-sized MAC CE in bytes. There may be one L field per MAC subheader except for subheaders corresponding to fixed-sized MAC CEs, padding, and MAC SDUs containing UL CCCH. The size of the L field may be indicated by the F field.

[0214] The F field (Format field) may indicate the size of the Length field. There may be one F field per MAC subheader except for subheaders corresponding to fixed-sized MAC CEs, padding, and MAC SDUs containing UL CCCH. The size of the F field may be 1 bit. The value 0 may indicate 8 bits of the Length field. The value 1 may indicate 16 bits of the Length field.

[0215] A wireless device may trigger and transmit a confirmation MAC CE based on receiving and validating DCI for activation or release/deactivation of a configured grant configuration. The transmission of confirmation MAC CE is important so that the base station may determine that an activation or deactivation/release DCI is received correctly and validated by the wireless device. When an activation or deactivation/release DCI indicates activation or deactivation/release of one or a group of configured grant configurations, existing confirmation mechanisms lead to inefficient configured grants operation for the wireless device. There is a need to enhance the confirmation process for configured grants. Example embodiments enhance the confirmation process for configured grants.

[0216] In an example embodiment as shown in FIG. 22, the wireless device may transmit a confirmation MAC control element based on a successful validation of a downlink control channel for scheduling activation or scheduling deactivation/release of a configured grant configuration.

[0217] In an example embodiment as shown in FIG. 25, the confirmation MAC CE may be a fixed-size MAC CE. The confirmation MAC CE may have a field comprising a

plurality of bits, wherein the size of the field may be fixed and the number of the plurality of bits may be a fixed number. The plurality of bits in the confirmation MAC CE field may comprise the one or more third bits that are indicated as part of the HPN field of the activation or deactivation/release DCI. The one or more third bits of the HPN field indicate the one or more configured grant configurations that are indicated for activation or deactivation/release. The one or more third bits in the MAC CE field indicate that the wireless device received the activation or deactivation/release DCI and successfully validated the activation or the deactivation/release DCI. In an example, the confirmation MAC CE may be a fixed-size MAC CE and may be included in a MAC subPDU for multiplexing in a MAC PDU (transport block) and transmission via an uplink resource.

[0218] In an example, the one or more third bits may be included as the least significant bits of the fixed-size MAC CE field. The fixed-size MAC CE field may have a size of one octet and the third number of the one or more third bits may be 3. The one or more third bits may be included as the rightmost (least significant) three bits of the MAC CE field.

[0219] In an example, the one or more third bits may be the most significant bits of the fixed-size MAC CE field. The fixed-size MAC CE field may have a size of one octet and the third number of the one or more third bits may be 3. The one or more third bits may be included as the leftmost (most significant) three bits of the MAC CE field.

[0220] In an example embodiment as shown in FIG. 26, the confirmation MAC CE may be a variable-size MAC CE. The confirmation MAC CE may have a field comprising a plurality of bits, wherein the size of the field may be variable and the number of the plurality of bits may be the third number of the one or more third bits. The plurality of bits in the confirmation MAC CE field may be the one or more third bits that are indicated as part of the HPN field of the activation or deactivation/release DCI. The one or more third bits of the HPN field indicate the one or more configured grant configurations that are indicated for activation or deactivation/release. The one or more third bits in the MAC CE field indicate that the wireless device received the activation or deactivation/release DCI and successfully validated the activation or the deactivation/release DCI. In an example, the confirmation MAC CE may be a variable-size MAC CE and may be included in a MAC subPDU for multiplexing in a MAC PDU (transport block) and transmission via an uplink resource. A MAC subheader in the MAC subPDU may comprise a length field indicating the length of the confirmation MAC CE.

[0221] In an example embodiment as shown in FIG. 20, a wireless device may receive one or more messages comprising configuration parameters. The one or more messages may comprise one or more RRC messages. The one or more messages may comprise configuration parameters of a first downlink semi-persistent scheduling (DL SPS) configuration. The configuration parameters of the first DL SPS configuration may comprise a periodicity, a number of HARQ processes for the DL SPS configuration, a configuration index/identifier of the DL SPS configuration or a group of DL SPS configurations comprising the DL SPS configuration, one or more power control parameters, one or more parameters for repetition if configured, etc. A plurality of resources may be activated based on the receiving the configuration parameters and receiving an activation DCI

indicating activation of the first DL SPS configuration. The first DL SPS configuration may be deactivated/released based on receiving a DCI indicating the deactivation of the DL SPS configuration. The wireless device may validate an activation DCI and the wireless device may activate the first DL SPS configuration based on the validation being successful (achieved). The wireless device may validate a deactivation/release DCI and the wireless device may deactivate the first DL SPS configuration based on the validation being successful (achieved).

[0222] In an example, the wireless device may receive a downlink control information (DCI) via a downlink control channel (e.g., PDCCH). The DCI may indicate scheduling activation of the first DL SPS configuration.

[0223] In an example, the wireless device may receive a downlink control information (DCI) via a downlink control channel (e.g., PDCCH). The DCI may indicate scheduling deactivation/release of the first DL SPS configuration.

[0224] The DCI may have one of a plurality of formats (e.g., a downlink DCI format such as format 1_0, format 1_1, a new format, etc.) and may comprise a plurality of fields. The plurality of fields may comprise a HARQ process number (HPN) field. An example is shown in FIG. 21. The HPN field may comprise first bits wherein the number of the first bits may be a first number. The first number (e.g., bitwidth of the HPN field) may be pre-defined or may be a configurable parameter (e.g., based on an RRC parameter). The first bits may comprise one or more second bits and one or more third bits. The number of the one or more second bits may be a second number and the number of the one or more third bits may be a third number. The one or more third bits may indicate the configuration index of the first DL SPS configuration. In an example, the one or more third bits may indicate a group of one or more DL SPS configurations comprising the first DL SPS configuration. In an example, by determining the one or more third bits, the wireless device may determine the one or more second bits from the first bits and the one or more third bits. In an example, the third number of the one or more third bits may be configurable (e.g., the one or more messages may indicate the third number) and the wireless device may determine the one or more third bits based on the first bits and the third number and the position of the one or more third bits.

[0225] In an example as shown in FIG. 20, the wireless device may validate the downlink control channel for scheduling activation. In an example, the wireless device may validate the downlink control for scheduling deactivation/release. The wireless device may validate the DCI (for scheduling activation or for scheduling deactivation/release) based on the one or more second bits of the one or more first bits of the HPN field. In an example, the validation (for scheduling activation or for scheduling deactivation/release) may be based on comparing a value of the one or more second bits with first value. In an example, the first value may be pre-determined, for example, all zeros. In an example, the first value may be configurable (e.g., with RRC configuration). The one or more messages may comprise a parameter indicating the first value for comparison and validation.

[0226] In an example, the validation (for scheduling activation or deactivation/release) may further comprises comparing a second field of the downlink control information with a second sequence of one or more second bits. For example, the second field may be a redundancy version (RV)

field. The second sequence of the one or more second bits may be predetermined. For example, the second sequence of the one or more second bits may be all zeros. In an example, the second sequence of the one or more second bits may be configurable (e.g., with RRC configuration). The one or more messages may comprise a parameter indicating the value of the one or more second bits for comparison and validation.

[0227] In an example, the validating the downlink control channel (control channel carrying the downlink control information) for activation and/or for release/deactivation may be based on (e.g., in response to) a new data indicator field of the downlink control information having a first value. The first value may be zero. The wireless device may validate the downlink control for activation and/or for release/deactivation based on the new data indicator having a value of zero, otherwise the wireless device may not validate the downlink control information (control channel carrying the downlink control information).

[0228] In an example, the validating the downlink control channel (control channel carrying the downlink control information) for activation and/or for release/deactivation may be based on (e.g., in response to) the downlink control information being associated with a first RNTI. The first RNTI may be a CS-RNTI. The wireless device may validate the downlink control for activation and/or for release/deactivation based on the radio network temporary identifier associated with the downlink control information being the first RNTI, otherwise the wireless device may not validate the downlink control information (control channel carrying the downlink control information).

[0229] In an example, the validating the downlink control channel (control channel carrying the downlink control information) for activation and/or for release/deactivation may be based on (e.g., in response to) a radio network temporary identifier of the downlink control information being associated with a first RNTI and a new data indicator field of the downlink control information being a first value. The first RNTI may be a CS-RNTI and the first value may be zero. The wireless device may validate the downlink control for activation and/or for release/deactivation based on the radio network temporary identifier associated with the downlink control information being the first RNTI and the new data indicator field of the downlink control information being zero, otherwise the wireless device may not validate the downlink control information (control channel carrying the downlink control information).

[0230] In an example, the validating the downlink control channel (control channel carrying the downlink control information) for activation and/or for release/deactivation based on the HPN field (or the one or more second bits of the HPN field) may be in response to the downlink control information having one of a plurality of formats (e.g., format 1_0 or format 1_1). In response to the downlink control information not having one of the plurality of formats, the validation may not be based on the HPN field.

[0231] In an example, based on the validation for scheduling activation being successful, the wireless device may activate the DL SPS configuration. The activating the DL SPS configuration may comprise activating a plurality of resources based on the DL SPS configuration. The wireless device may determine the plurality of resources based on one or more configuration parameters (e.g., periodicity) of the configured grant configuration and one or more param-

eters (e.g., resource allocation parameters) indicated by the downlink control information.

[0232] In an example, based on the validation for scheduling deactivation/release being successful, the wireless device may deactivate/release the DL SPS configuration. The deactivating the DL SPS configuration may comprise deactivating a plurality of resources associated with the DL SPS configuration. The deactivating the plurality of resources may release the plurality of resources and the plurality of resources may be available for scheduling for the wireless device or other wireless devices. The deactivating the plurality of resources may be a physical layer deactivation and the DL SPS configuration parameters (e.g., RRC configurations) may be still available for the wireless device. The wireless device may receive a new activation command indicating activation of new radio resources based on the DL SPS configuration parameters.

[0233] In an example embodiment, a wireless device may receive from a base station, one or more messages comprising first configuration parameters of a first DL SPS configuration. The wireless device may receive, via a downlink control channel, a downlink control information indicating scheduling activation of the first DL SPS configuration, wherein the downlink control information comprises a hybrid automatic repeat request process number field comprising first bits. The wireless device may validate, for scheduling activation, the downlink control channel based on one or more second bits of the first bits. The wireless device may activate the DL SPS configuration for scheduling based on the validating being successful.

[0234] In an example, the one or more third bits may be used to identify one or more DL SPS configurations. The one or more third bits may indicate a single DL SPS configuration index or an identifier of a group of one or more DL SPS configurations. In an example, the group of one or more DL SPS configurations may be for one cell (or a BWP of one cell). In an example, the group of DL SPS configurations may be for one or more cells (or one or more BWPs of one or more cells). In an example, the one or more messages may comprise a parameter indicating whether the wireless device may interpret the one or more third bits as a configuration index of a single DL SPS configuration or an identifier of a group of one or more DL SPS configurations. In an example, the wireless device may determine whether the wireless device may interpret the one or more third bits as a configuration index of a single DL SPS configuration or an identifier of a group of one or more DL SPS configurations based on a value of a second field in the DCI. A value of the second field may be an explicit indication (e.g., a flag) indicating whether the wireless device may interpret the one or more third bits as a configuration index of a single DL SPS configuration or an identifier of a group of one or more DL SPS configurations. In an example, a value of the second field may implicitly indicate whether the wireless device may interpret the one or more third bits as a configuration index of a single DL SPS configuration or an identifier of a group of one or more DL SPS configurations.

[0235] In an example, the third number may be separately configured for activation and for deactivation/release. For example, the one or more third bits may indicate a configuration index/identifier of a single DL SPS configuration in case of activation and may indicate an identifier of a single or a group of one or more DL SPS configurations in case of release/deactivation. The third number for activation may be

smaller than the third number for deactivation/release. The wireless device may receive separate configuration parameters of the third number for activation and for deactivation/release.

[0236] In an example, the wireless device may determine the third number (at least for DL SPS configuration activation) based on a number of DL SPS configurations configured for the wireless device. For example, for N DL SPS configurations, the third number may be logarithm of N in base 2. For example, for N DL SPS configurations, the third number may be floor of logarithm of N in base 2 plus one. For example, for N DL SPS configurations, the third number may be the smallest number that can indicate N different DL SPS configurations.

[0237] In an example, the wireless device may determine the third number for release/deactivation based on the third number for activation plus an offset. The offset may be configurable (e.g., as an RRC parameter or based on an RRC parameter) or may be a pre-defined value.

[0238] In an example, the third number may be based on values of one or more fields of a downlink control information (e.g., the downlink control information for scheduling activation or scheduling deactivation/release).

[0239] In an example, the wireless device may determine the one or more third bits based on the first bits of the HPN field and the third number. The one or more third bits may be third number least significant bits of the first bits. For example, if the HPN field of the activation or deactivation/release DCI (e.g., the first bits) has a value of 1110 and the third number is 2, the one or more third bits is the two least significant bits of 1110 or 10. For example, if the HPN field of the activation or deactivation/release DCI (e.g., the first bits) has a value of 0111 and the third number is 3, the one or more third bits is the three least significant bits of 0111 or 111.

[0240] The wireless device may determine the one or more second bits based on the first bits of the HPN field and the one or more third bits. For example, the one or more second bits may be remaining bits of the first bits after determining the one or more third bits. For example, if the HPN field of the activation or deactivation/release DCI (e.g., the first bits) has a value of 1110 and the one or more third bits are 10, the one or more second bits are 11. For example, if the HPN field of the activation or deactivation/release DCI (e.g., the first bits) has a value of 0111 and the one or more third bits are 111, the one or more second bits is 0.

[0241] In an example, the wireless device may determine the one or more second bits as the second number most significant bits of the first bits. The wireless device may determine the second number based on the first number and the third number. The first number may a fixed number (e.g., the HPN field may have 4 bitwidth) or may be configurable (e.g., the HPN field may have a configurable bitwidth indicated by an RRC parameter). For example, if the HPN field of the activation or deactivation/release DCI (e.g., the first bits) has a value of 1110 and the second number is 2, the one or more second bits are 11. For example, if the HPN field of the activation or deactivation/release DCI (e.g., the first bits) has a value of 0111 and the second number is 1, the one or more second bits are 0.

[0242] In an example embodiment, a wireless device may receive from a base station, one or more messages comprising first configuration parameters of a first configured grant configuration. The wireless device may receive, via a down-

link control channel, a downlink control information indicating scheduling release/deactivation of the first configured grant configuration, wherein the downlink control information comprises a hybrid automatic repeat request process number field comprising first bits. The wireless device may validate, for scheduling release/deactivation, the downlink control channel based on one or more second bits of the first bits. The wireless device may release/deactivate the configured grant configuration for scheduling based on the validating being successful.

[0243] In an example embodiment, a wireless device may receive from a base station, one or more messages comprising first configuration parameters of a first downlink semi-persistent scheduling (DL SPS) configuration. The wireless device may receive, via a downlink control channel, a downlink control information indicating scheduling activation of the first DL SPS configuration, wherein the downlink control information comprises a hybrid automatic repeat request process number field comprising first bits. The wireless device may validate, for scheduling activation, the downlink control channel based on one or more second bits of the first bits. The wireless device may activate the DL SPS configuration for scheduling based on the validating being successful.

[0244] In an example embodiment, a wireless device may receive from a base station, one or more messages comprising first configuration parameters of a first DL SPS configuration. The wireless device may receive, via a downlink control channel, a downlink control information indicating scheduling release/deactivation of the first DL SPS configuration, wherein the downlink control information comprises a hybrid automatic repeat request process number field comprising first bits. The wireless device may validate, for scheduling release/deactivation, the downlink control channel based on one or more second bits of the first bits. The wireless device may release/deactivate the DL SPS configuration for scheduling based on the validating being successful.

[0245] In an example, the first bits comprise the one or more second bits and one or more third bits. In an example, the one or more messages further comprise a parameter indicating a third number of the one or more third bits. In an example, the third number may be separately configured for scheduling activation and scheduling release/deactivation. In an example, the third number for scheduling release/deactivation may be larger than the third number of scheduling activation.

[0246] In an example, the third number of the one or more third bits may be based on a fourth number of a plurality of configured grant configurations comprising the first configured grant configuration.

[0247] In an example, the third number of the one or more third bits may be based on a fourth number of a plurality of DL SPS configurations comprising the first DL SPS configuration.

[0248] In an example, the third number of the one or more third bits may be logarithm (base 2) of the fourth number. In an example, the third number of the one or more third bits may be larger than or equal to the logarithm (base 2) of the fourth number. The one or more messages may comprise configuration parameters of the plurality of configured grants. The one or more messages may comprise configuration parameters of the plurality of DL SPSs.

[0249] In an example, the third number of the one or more third bits may be a configurable parameter. In an example, the first configuration parameters may comprise a parameter indicating the third number of the one or more third bits. In an example, the first configuration parameters may comprise a first parameter indicating the third number of the one or more third bits for activation and a second parameter indicating the third number of the one or more third bits for release/deactivation. In an example, the third number of the one or more third bits may be determined based on one or more second fields in the activation DCI. In an example, the third number of the one or more third bits may be determined based on one or more second field in the release/deactivation DCI.

[0250] In an example, the wireless device may determine the one or more third bits as a third number least significant bits of the first bits. For example, if the first bits are 1110 and the third number is 2, the one or more third bits may be 10.

[0251] In an example, the wireless device may determine the one or more second bits based on the first bits and the one or more third bits. For example, if the first bits are 1110 and the one or more third bits are 10, the wireless device may determine the one or more second bits to be 11.

[0252] In an example, the wireless device may determine the one or more second bits as second number most significant bits of the first bits. In an example, the wireless device may determine the second number of the one or more second bits based on the first of the first bits and the third number of the one or more third bits. The wireless device may determine the second number as the first number of the first bits minus the third number of the one or more third bits. For example, the first number may be 4 and the third number may be 2 (e.g., the third number may be indicated to the wireless device as 2). The wireless device may determine the second number to be 2. The wireless device may determine the one or more second bits as 2 most significant bits of the first bits. For example, if the first bits are 1110, the wireless device may determine the second bits as 11.

[0253] In an example, the one or more messages may comprise configuration parameters of a plurality of configured grants. The plurality of configured grant configurations may comprise the first configured grant configuration. In an example, the plurality of configured grants may be for a first BWP (UL BWP) of a cell.

[0254] In an example, the one or more messages may comprise configuration parameters of a plurality of DL SPSs. The plurality of DL SPSs may comprise the first DL SPS. In an example, the plurality of DL SPSs may be for a first BWP (DL BWP) of a cell.

[0255] In an example, the first configuration parameters may comprise a first identifier of the first configured grant configuration; and the one or more third bits may indicate the first identifier.

[0256] In an example, the first configuration parameters may comprise a first identifier of the first DL SPS configuration; and the one or more third bits may indicate the first identifier.

[0257] In an example, the first configuration parameters comprise a first identifier of the first configured grant configuration; and the one or more third bits indicate one or more identifiers comprising the first identifier.

[0258] In an example, the first configuration parameters comprise a first identifier of the first DL SPS configuration;

and the one or more third bits indicate one or more identifiers comprising the first identifier.

[0259] In an example, the wireless device may determine the first configured grant configuration of a plurality of configured grant configurations based on the one or more third bits.

[0260] In an example, the wireless device may determine the first DL SPS configuration of a plurality of DL SPS configurations based on the one or more third bits.

[0261] In an example, the wireless device may determine one or more configured grant configurations, comprising the first configured grant configuration, of a plurality of configured grant configurations based on the one or more third bits.

[0262] In an example, the wireless device may determine one or more DL SPS configurations, comprising the first DL SPS configuration, of a plurality of DL SPS configurations based on the one or more third bits.

[0263] In an example, the validating for scheduling activation or the validating for scheduling deactivation/release may comprise comparing the second bits with a sequence of one or more bits. In an example, the sequence of one or more bits may be pre-determined. In an example, the pre-determined sequence of the one or more bits may be a sequence of all zeros. In an example, the sequence of the one or more bits may be configured by RRC. In an example, the first configuration parameters may indicate the sequence of one or more bits.

[0264] In an example, the validating may further comprise comparing a second field of the downlink control information with a sequence of one or more second bits. In an example, the sequence of one or more second bits may be pre-determined. In an example, the second field may be a redundancy version field; and the predetermined sequence of one or more second bits may be a sequence of one or more zeros.

[0265] In an example, the validating the downlink control channel may be based on (e.g., in response to) a new data indicator field of the downlink control information having a first value. In an example, the first value may be zero.

[0266] In an example, the validating the downlink control channel may be based on (e.g., in response to) a radio network temporary identifier associated with the downlink control information being a first radio network temporary identifier (e.g., cs-RNTI or a new RNTI for a new DCI format).

[0267] In an example, the validating the downlink control channel for scheduling activation may be based on one or more second bits of the first bits in response to the downlink control information having one of a plurality of formats (e.g., a format 0_1 or 0_1). In response to the DCI format not having one of the plurality of formats, the validating may not be based on the one or more second bits of the first bits or may not be based on the HARQ process number field.

[0268] In an example, the validating the downlink control channel for scheduling release/deactivation may be based on one or more second bits of the first bits in response to the downlink control information having one of a plurality of formats (e.g., format 0_1 or 0_1). In response to the DCI format not having one of the plurality of formats, the validating may not be based on the one or more second bits of the first bits or may not be based on the HARQ process number field.

[0269] In an example, the wireless device may determine a plurality of resources based on a periodicity parameter,

wherein the first configuration parameters comprise the periodicity parameter. In an example, the wireless device may activate the plurality of resources based on the determining the plurality of resources.

[0270] In an example, the wireless device may transmit a confirmation MAC control element based on the activating or releasing/deactivating being successful.

[0271] In an example embodiment, a wireless device may receive from a base station, one or more messages comprising configuration parameters of a first configured grant configuration and a second configured grant configuration. The wireless device may receive, via a downlink control channel, a first downlink control information of a first format indicating scheduling activation of the first configured grant configuration, wherein the first downlink control information comprises a first hybrid automatic repeat request process number field. The wireless device may receive, via a downlink control channel, a second downlink control information of a second format indicating scheduling activation of the second configured grant configuration, wherein the second downlink control information comprises a second hybrid automatic repeat request process number field. Based on the first downlink control information being of the first format, the wireless device may validate the downlink control channel for scheduling activation of the first configured grant configuration employing a value of the first hybrid automatic repeat request field. Based on the second downlink control information being of the second format, the wireless device may validate the downlink control channel for scheduling activation of the second configured grant configuration without employing a value of the second hybrid automatic repeat request field. The wireless device may activate the first configured grant configuration for scheduling activation of the first configured grant configuration based on the validating the downlink control channel for scheduling activation of the first configured grant configuration being successful. The wireless device may activate the second configured grant configuration for scheduling activation of the second configured grant configuration based on the validating the downlink control channel for scheduling activation of the second configured grant configuration being successful.

[0272] In an example embodiment, the wireless device may receive from a base station, one or more messages comprising configuration parameters of a configured grant configuration. The wireless device may receive, via a downlink control channel, a downlink control information scheduling activation of the configured grant configuration, wherein the downlink control information comprises a hybrid automatic repeat request process number field. Based on the downlink control information being of one or more first formats, the wireless device may validate the downlink control channel for scheduling activation of the configured grant configuration, based on the hybrid automatic repeat request process number field. Based on the downlink control information being of one or more second formats, the wireless device may validate the downlink control channel for scheduling activation of the configured grant configuration not based on the hybrid automatic repeat request process number field. The wireless device may activate the configured grant configuration for scheduling activation of the configured grant configuration based on the validating being successful.

[0273] In an example, based on the downlink control information being of the one or more first formats, the validating the downlink control channel for scheduling activation of the configured grant configuration may be based on values of one or more bits of the hybrid automatic repeat request process number field.

[0274] In an example, the MAC control element comprises a field comprising a plurality of bits. A fifth number of the plurality of bits may be a fixed number. The plurality of bits may comprise the one or more third bits.

[0275] In an example, the MAC control element is a fixed-size MAC control element; and the MAC control element is in a MAC subPDU multiplexed in a MAC PDU.

[0276] In an example, the one or more third bits may be the least significant bits of the plurality of bits.

[0277] In an example, the MAC control element comprises a field comprising a plurality of bits; a fifth number of the plurality of bits may be the third number of the one or more third bits; and the plurality of bits may be the one or more third bits.

[0278] In an example, the MAC control element is a variable-size MAC control element; the MAC control element is in a MAC subPDU multiplexed in a MAC PDU; and a MAC subheader in the MAC subPDU comprises a length field indicating a length of the MAC control element.

[0279] A wireless device may be configured with periodic resources in uplink (e.g., based on uplink configured grants) or downlink (e.g., based on downlink semi-persistent scheduling (SPS)). The wireless device may be configured with or multiple uplink configured grant (e.g., Type 2 configured grant) configurations and/or one or multiple downlink SPS configurations. An uplink configured grant (e.g., Type 2 configured grant) configuration or a downlink SPS configuration may be activated or deactivated based on physical layer signaling and using an activation downlink control information (DCI) or a deactivation DCI. The wireless device may perform a validation process for a DCI in response to the DCI being associated with configured scheduling (e.g., in response to a CRC of the DCI being scrambled with a configured scheduling RNTI) and in response to an NDI field of the DCI having a value of zero. The existing validation processes for a DCI that activates or deactivates/releases an uplink configured configuration or a downlink SPS configuration is inefficient when the wireless device is configured with multiple uplink configured grant configurations and/or multiple downlink SPS configurations and leads to inefficient DCI validation, improper scheduling of periodic resources and degraded wireless device and network performance. There is a need to enhance the validation process for an activation DCI or a deactivation/release DCI that activates or deactivates/releases an uplink configured grant configuration or a downlink SPS configuration. Example embodiments enhance the validation process for a DCI that activates or deactivates/releases an uplink configured grant configuration or a downlink SPS configuration.

[0280] In example embodiments for example as shown in FIG. 27, FIG. 28, FIG. 29 and FIG. 33, a wireless device may receive one or more messages comprising configuration parameters. The one or more messages may comprise one or more RRC messages. The one or more messages may comprise configuration parameters of one or more cells. The one or more messages may comprise configuration parameters of one or more bandwidth parts (BWPs) of a cell in the one or more cells. The one or more messages may comprise

configuration parameters of one or more uplink configured grant configurations comprising an uplink configured grant configuration. The one or more uplink configured grant configurations, including the uplink configured grant configuration, may be configured for the cell of the one or more cells. The one or more uplink configured grant configurations, including the uplink configured grant configuration, may be configured for a BWP of the one or more BWPs of the cell. The uplink configured grant configuration may be a Type 2 configured grant configuration wherein the uplink configured grant configuration may be activated in response to receiving an activation DCI and the uplink configured grant configuration may be deactivated in response to receiving a deactivation DCI. The wireless device may first validate a received DCI as a valid activation DCI or a valid deactivation DCI and may activate the uplink configured grant configuration or deactivate the uplink configured grant configuration based on the DCI and based on validating the DCI as a valid activation DCI or a valid deactivation DCI.

[0281] The configuration parameters of the one or more uplink configured grant configurations may comprise an RNTI that may be used for activation or deactivation of an uplink configured grant configuration or for retransmission of a TB transmitted via a configured grant resource associated with the uplink configured grant retransmission. A DCI that may be used for activation or deactivation of the uplink configured grant configuration or for scheduling retransmission of TB that was transmitted via a configured grant resource may be associated with the RNTI. The RNTI may be a configured scheduling RNTI (CS-RNTI).

[0282] In an example when a plurality of configured grant configurations are configured, configuration parameters of an uplink configured grant configuration may comprise an identifier (e.g., a configuration index) for the uplink configured grant configuration (e.g., a separate configuration index for each of the plurality of configured grant configurations). The configuration parameters of an uplink configured grant configuration may further include a periodicity (e.g., in terms of number of symbols) that may indicate a separation between consecutive resources associated with the uplink configured grant configuration. The configuration parameters of an uplink configured grant configuration may further include a number of HARQ processes associated with the uplink configured grant configuration, one or more parameters for determining resources associated with the uplink configured grant configurations in the time domain and/or the frequency domain, one or more power control parameters, one or more repetition parameters, etc.

[0283] The wireless device may receive a DCI associated with the RNTI (e.g., CS-RNTI) that is used for activation and/or deactivation/release of a configured grant configuration. The DCI may comprise a plurality of fields including a HARQ process number field, a new data indicator (NDI) field, a redundancy version (RV) field. The wireless device may determine, based on a validation process, whether the DCI is a valid activation DCI or a valid release DCI for the uplink configured grant configuration. The wireless device may perform the validation process for the DCI based on the DCI being associated with the CS-RNTI and based on a value of the NDI field of the DCI being zero. In an example, the wireless device may be configured with multiple configured grant configurations and a value of the HARQ process number field in a DCI may indicate an activation or a deactivation for a corresponding UL configured grant

configuration (e.g., configured grant Type 2 configuration) with a same value as provided by Configuredgrantconfig-index. In an example, the wireless device may be configured with multiple configured grant configurations, including the uplink configured grant configuration, and one or more first bits of the HARQ process number field of the DCI may indicate a configuration index of the uplink configured grant configuration. In an example, the wireless device may be configured with a single configured grant configurations, e.g., the uplink configured grant configuration, and one or more bits of the HARQ process number field of the DCI may not be used for indication of a configuration index of the uplink configured grant configuration. The DCI may further comprise a redundancy version (RV) field comprising one or more second bits.

[0284] The wireless device may determine that the DCI is a valid activation DCI or a valid release/deactivation DCI. The validation process to determine that the DCI is a valid activation DCI or a valid deactivation/release DCI may be based on whether the HARQ process number field is used for indication of the configuration index of the uplink configured grant configuration. The HARQ process number field of the DCI may comprise one or more first bits and the RV field of the DCI may comprise one or more second bits. The wireless device may determine that the DCI is a valid activation DCI or a valid deactivation/release DCI (e.g., validate the DCI) based on the one or more second bits of the RV field of the DCI, for example by comparing values of the one or more second bits with one or more second predetermined values (e.g., all zeros). In an example embodiment as shown in FIG. 27, the wireless device may determine that the DCI is a valid activation DCI or a valid deactivation/release DCI (e.g., validate the DCI) further based on the one or more first bits of the HARQ process number field of the DCI (e.g., by comparing values of the one or more first bits with one or more first predetermined values (e.g., all zeros)) in response to the one or more first bits of the HARQ process number field of the DCI not indicating (e.g., not being used for indication of) the configuration index of the uplink configured grant configuration. The wireless device may determine whether the one or more first bits of the HARQ process number field indicates (e.g., is used for indication of) the configuration index of the uplink configured grant configuration. The wireless device may determine that the DCI is a valid activation DCI or a valid deactivation/release DCI (e.g., validate the DCI) based on the one or more first bits of the HARQ process number field (e.g., by comparing values of the one or more first bits with one or more first predetermined values (e.g., all zeros)) and the one or more second bits of the RV field (e.g., by comparing values of the one or more second bits with one or more second predetermined values (e.g., all zeros)) in response to the determination that the one or more first bits of the HARQ process number field of the DCI not indicating (e.g., not being used for indication of) the configuration index of the uplink configured grant configuration. The wireless device may determine that the DCI is a valid activation DCI or a valid deactivation/release DCI (e.g., validate the DCI) based on the one or more second bits of the RV field and not based on the one or more first bits of the HARQ process number field in response to the determination that the one or more first bits of the HARQ process number field of the DCI indicates (e.g., used for indication of) the configuration index of the uplink configured grant configuration. In an example

embodiment as shown in FIG. 28, in response to the one or more first bits of the HARQ process number field of the DCI not indicating (e.g., not being used for indication) of the configuration index of the uplink configured grant, the wireless device may determine that the DCI is a valid activation DCI or a valid deactivation/release DCI (e.g., validate the DCI) based on the one or more second bits of the RV field of the DCI (e.g., by comparing values of the one or more second bits with one or more second predetermined values (e.g., all zeros)) and based on the one or more first bits of the HARQ process number field of the DCI (e.g., by comparing values of the one or more first bits with one or more first predetermined values (e.g., all zeros)). In an example embodiment as shown in FIG. 29, in response to the one or more first bits of the HARQ process number field of the DCI indicating (e.g., being used for indication) of the configuration index of the uplink configured grant, the wireless device may determine that the DCI is a valid activation DCI or a valid deactivation/release DCI (e.g., validate the DCI) based on the one or more second bits of the RV field of the DCI (e.g., by comparing values of the one or more second bits with one or more second predetermined values (e.g., all zeros)) and not based on the one or more first bits of the HARQ process number field of the DCI.

[0285] In an example embodiment as shown in FIG. 33, the wireless device may determine that the wireless device is a valid activation DCI or a valid release/deactivation DCI. The validation process for the received DCI (e.g., the determination that the DCI is a valid activation DCI or a valid release/deactivation DCI) at the wireless device may or may not be based on one or more bits of the HARQ process number field of the received DCI. The determination that the DCI is a valid activation DCI or a valid release/deactivation DCI may be based on whether one or more bits of the HARQ process field of the DCI are used for indication of a configuration index of the uplink configured grant configuration.

[0286] In an example, the DCI may be used for deactivation/release of multiple uplink configured grant configurations. The wireless device may receive a configuration parameter indicating a deactivation state list. The deactivation state list may comprise a plurality of deactivation states wherein each deactivation state, in the plurality of deactivation states, may indicate one or more configuration indexes of one or more uplink configured grant configurations. A value of the HARQ process number field of DCI may indicate a deactivation state that indicates one or more configuration indexes of one or more configured grant configurations including a configuration index of the uplink configured grant configuration. The wireless device may determine that the DCI is a valid deactivation/release DCI based on the one or more second bits of the RV field of the DCI and not based on the one or more first bits of the HARQ process number field of the DCI in response to determining that the HARQ process number field of the DCI is used for indication of the deactivation state which indicates the one or more configuration indexes of the one or more uplink configured grant configurations including the configuration index of the uplink configured grant configuration.

[0287] In an example, the wireless device may transmit a confirmation control element (e.g., a confirmation MAC CE). The wireless device may transmit the confirmation control element in response to determining that the DCI is a valid activation or valid release DCI (e.g., in response to

validating the DCI or the validation of the DCI being successful). In an example that the wireless device is configured with a plurality of configured grant configurations, the confirmation control element may indicate the uplink configured grant configuration in the plurality of uplink configured grant configurations. In an example, the confirmation control element may indicate that the DCI, indicating activation or deactivation of the uplink configured grant configuration in the plurality of uplink configured grant configurations, is received.

[0288] The wireless device may activate or deactivate a plurality of resources associated with the uplink configured grant configuration based on determining that the DCI is a valid activation DCI or a valid deactivation/release DCI (e.g., based on the validation of the DCI being successful). For example, in case of the DCI being an activation DCI, the plurality of resources may be determined based on the activation DCI and the configuration parameters of the uplink configured grant configuration. For example, in case of the DCI being a deactivation/release DCI, the plurality of resources may be determined based on a previously received activation DCI and the configuration parameters of the uplink configured grant configuration. The activation or deactivation of the plurality of resources may be based on an offset from the timing of the reception of the DCI and may be based on the DCI processing time and/or wireless device capabilities.

[0289] In example embodiments for example as shown in FIG. 30, FIG. 31, FIG. 32 and FIG. 34, a wireless device may receive one or more messages comprising configuration parameters. The one or more messages may comprise one or more RRC messages. The one or more messages may comprise configuration parameters of one or more cells. The one or more messages may comprise configuration parameters of one or more bandwidth parts (BWPs) of a cell in the one or more cells. The one or more messages may comprise configuration parameters of one or more downlink semi-persistent scheduling (SPS) configurations comprising a downlink SPS configuration. The one or more SPS configurations, including the downlink SPS configuration, may be configured for the cell of the one or more cells. The one or more downlink SPS configurations, including the downlink SPS configuration, may be configured for a BWP of the one or more BWPs of the cell. The downlink SPS configuration may be activated in response to receiving an activation DCI and the downlink SPS configuration may be deactivated in response to receiving a deactivation DCI. The wireless device may first validate a received DCI as a valid activation DCI or a valid deactivation DCI and may activate the downlink SPS configuration or deactivate the downlink SPS configuration based on the DCI and based on validating the DCI as a valid activation DCI or a valid deactivation DCI.

[0290] The configuration parameters of the one or more downlink SPS configurations may comprise an RNTI that may be used for activation or deactivation of a downlink SPS configuration or for retransmission of a TB transmitted via a downlink SPS resource associated with the downlink SPS retransmission. A DCI that may be used for activation or deactivation of the downlink SPS configuration or for scheduling retransmission of TB that was received via a downlink SPS resource may be associated with the RNTI. The RNTI may be a configured scheduling RNTI (CS-RNTI).

[0291] In an example when a plurality of downlink SPS configurations may be configured, configuration parameters of a downlink SPS configuration may comprise an identifier (e.g., a configuration index) for the downlink SPS configuration (e.g., a separate configuration index for each of the plurality of downlink SPS configurations). The configuration parameters of a downlink SPS configuration may further include a periodicity (e.g., in terms of number of symbols) that may indicate a separation between consecutive resources associated with the downlink SPS configuration. The configuration parameters of a downlink SPS configuration may further include a number of HARQ processes associated with the downlink SPS configuration, one or more parameters for determining resources associated with the downlink SPS configuration in the time domain and/or the frequency domain, etc.

[0292] The wireless device may receive a DCI associated with the RNTI (e.g., CS-RNTI) that is used for activation and/or deactivation/release of a downlink SPS configuration. The DCI may comprise a plurality of fields including a HARQ process number field, a new data indicator (NDI) field, a redundancy version (RV) field. The wireless device may determine, based on a validation process, whether the DCI is a valid activation DCI or a valid release DCI for the downlink SPS configuration. The wireless device may perform the validation process for the DCI based on the DCI being associated with the CS-RNTI and based on a value of the NDI field of the DCI being zero. In an example, the wireless device may be configured with multiple downlink SPS configurations and a value of the HARQ process number field in a DCI may indicate an activation or a deactivation for a corresponding DL SPS configuration with a same value as provided by SPSconfig-index. In an example, the wireless device may be configured with multiple downlink SPS configurations, including the downlink SPS configuration, and one or more first bits of the HARQ process number field of the DCI may indicate a configuration index of the downlink SPS configuration. In an example, the wireless device may be configured with a single downlink SPS configurations, e.g., the downlink SPS configuration, and one or more bits of the HARQ process number field of the DCI may not be used for indication of a configuration index of the downlink SPS configuration. The DCI may further comprise a redundancy version (RV) field comprising one or more second bits.

[0293] The wireless device may determine that the wireless device is a valid activation DCI or a valid release/deactivation DCI. The validation process to determine that the DCI is a valid activation DCI or a valid deactivation/release DCI may be based on whether the HARQ process number field is used for indication of the configuration index of the downlink SPS configuration. The HARQ process number field of the DCI may comprise one or more first bits and the RV field of the DCI may comprise one or more second bits. The wireless device may determine that the DCI is a valid activation DCI or a valid deactivation/release DCI (e.g., validate the DCI) based on the one or more second bits of the RV field of the DCI, for example by comparing values of the one or more second bits with one or more second predetermined values (e.g., all zeros). In an example embodiment as shown in FIG. 30, the wireless device may determine that the DCI is a valid activation DCI or a valid deactivation/release DCI (e.g., validate the DCI) further based on the one or more first bits of the HARQ process

number field of the DCI (e.g., by comparing values of the one or more first bits with one or more first predetermined values (e.g., all zeros)) in response to the one or more first bits of the HARQ process number field of the DCI not indicating (e.g., not being used for indication of) the configuration index of the downlink SPS configuration. The wireless device may determine whether the one or more first bits of the HARQ process number field indicates (e.g., is used for indication of) the configuration index of the downlink SPS configuration. The wireless device may determine that the DCI is a valid activation DCI or a valid deactivation/release DCI (e.g., validate the DCI) based on the one or more first bits of the HARQ process number field (e.g., by comparing values of the one or more first bits with one or more first predetermined values (e.g., all zeros)) and the one or more second bits of the RV field (e.g., by comparing values of the one or more second bits with one or more second predetermined values (e.g., all zeros)) in response to the determination that the one or more first bits of the HARQ process number field of the DCI does not indicate (e.g., are not used for indication of) the configuration index of the downlink SPS configuration. The wireless device may determine that the DCI is a valid activation DCI or a valid deactivation/release DCI (e.g., validate the DCI) based on the one or more second bits of the RV field and not based on the one or more first bits of the HARQ process number field in response to the determination that the one or more first bits of the HARQ process number field of the DCI indicates (e.g., used for indication of) the configuration index of the downlink SPS configuration. In an example embodiment as shown in FIG. 31, in response to the one or more first bits of the HARQ process number field of the DCI not indicating (e.g., not being used for indication of) the configuration index of the downlink SPS configuration, the wireless device may determine that the DCI is a valid activation DCI or a valid deactivation/release DCI (e.g., validate the DCI) based on the one or more second bits of the RV field of the DCI (e.g., by comparing values of the one or more second bits with one or more second predetermined values (e.g., all zeros)) and based on the one or more first bits of the HARQ process number field of the DCI (e.g., by comparing values of the one or more first bits with one or more first predetermined values (e.g., all zeros)). In an example embodiment as shown in FIG. 32, in response to the one or more first bits of the HARQ process number field of the DCI indicating (e.g., being used for indication of) the configuration index of the downlink SPS configuration, the wireless device may determine that the DCI is a valid activation DCI or a valid deactivation/release DCI (e.g., validate the DCI) based on the one or more second bits of the RV field of the DCI (e.g., by comparing values of the one or more second bits with one or more second predetermined values (e.g., all zeros)) and not based on the one or more first bits of the HARQ process number field of the DCI.

[0294] In an example embodiment as shown in FIG. 34, the wireless device may determine that the wireless device is a valid activation DCI or a valid release/deactivation DCI. The validation process for the received DCI (e.g., the determination that the DCI is a valid activation DCI or a valid release/deactivation DCI) at the wireless device may or may not be based on one or more bits of the HARQ process number field of the received DCI. The determination that the DCI is a valid activation DCI or a valid release/deactivation DCI may be based on whether one or more bits of the HARQ

process field of the DCI are used for indication of a configuration index of the downlink SPS configuration.

[0295] In an example, the DCI may be used for deactivation/release of multiple downlink SPS configurations. The wireless device may receive a configuration parameter indicating a deactivation state list. The deactivation state list may comprise a plurality of deactivation states wherein each deactivation state, in the plurality of deactivation states, may indicate one or more configuration indexes of one or more downlink SPS configurations. A value of the HARQ process number field of DCI may indicate a deactivation state that indicates one or more configuration indexes of one or more downlink SPS configurations including a configuration index of the downlink SPS configuration. The wireless device may determine that the DCI is a valid deactivation/release DCI based on the one or more second bits of the RV field of the DCI and not based on the one or more first bits of the HARQ process number field of the DCI in response to determining that the HARQ process number field of the DCI is used for indication of the deactivation state which indicates the one or more configuration indexes of the one or more downlink SPS configurations including the configuration index of the downlink SPS configuration.

[0296] The wireless device may activate or deactivate a plurality of resources associated with the downlink SPS configuration based on determining that the DCI is a valid activation DCI or a valid deactivation/release DCI (e.g., based on the validation of the DCI being successful). For example, in case of the DCI being an activation DCI, the plurality of resources may be determined based on the activation DCI and the configuration parameters of the downlink SPS configuration. For example, in case of the DCI being a deactivation/release DCI, the plurality of resources may be determined based on a previously received activation DCI and the configuration parameters of the downlink SPS configuration. The activation or deactivation of the plurality of resources may be based on an offset from the timing of the reception of the DCI and may be based on the DCI processing time and/or wireless device capabilities.

[0297] In accordance with various exemplary embodiments in the present disclosure, a device (e.g., a wireless device, a base station and/or alike) may include one or more processors and may include memory that may store instructions. The instructions, when executed by the one or more processors, cause the device to perform actions as illustrated in the accompanying drawings and described in the specification. The order of events or actions, as shown in a flow chart of this disclosure, may occur and/or may be performed in any logically coherent order. In some examples, at least two of the events or actions shown may occur or may be performed at least in part simultaneously and/or in parallel. In some examples, one or more additional events or actions may occur or may be performed prior to, after, or in between the events or actions shown the flow charts of the present disclosure.

[0298] FIG. 35 shows an example flow diagram in accordance with several of various embodiments of the present disclosure. At 3510, a wireless device may receive configuration parameters of an uplink configured grant configuration. At 3520, the wireless device may receive a downlink control information (DCI) comprising: a hybrid automatic repeat request (HARQ) process number field comprising one or more first bits; and a redundancy version field comprising one or more second bits. At 3530, the wireless device may

determine whether a first value of the one or more first bits indicates a configuration index of the uplink configured grant configuration. At **3540**, the wireless device may determine that the DCI is a valid activation DCI or a valid release DCI: based on the one or more second bits and not based on the one or more first bits in response to the first value indicating the configuration index; and based on the one or more second bits and the one or more first bits in response to the first value not indicating the configuration index. At **3550**, the wireless device may activate or deactivate a plurality of resources associated with the uplink configured grant configuration.

[0299] In example embodiments, the wireless device may receive a configuration parameter indicating one or more deactivation states. The first value of the one or more first bits of the HARQ process number field may indicate a first deactivation state of the one or more deactivation states. The first deactivation state may correspond to one or more uplink configured grant configurations comprising the uplink configured grant configurations. The wireless device may determine that the DCI is a valid release/deactivation DCI at **3540** and may deactivate the plurality of resources associated with the uplink configured grant configuration at **3550**.

[0300] In example embodiments, the wireless device may transmit a confirmation control element. The confirmation control element may comprise a field with a second value. The second value may indicate that the DCI indicating activation or deactivation/release of the uplink configured grant configuration is received. The second value may indicate that the DCI indicating activation or deactivation/release of the uplink configured grant configuration, in a plurality of uplink configured grant confirmations, is received.

[0301] In an example embodiment, the configuration parameters of the uplink configured grant configuration, received at **3510**, may comprise a first radio network temporary identifier (RNTI). The first RNTI may be a configured scheduling RNTI (CS-RNTI). The DCI received at **3520** may be associated with the first radio network temporary identifier. In an example embodiment, the DCI may further comprise a new data indicator field. The determining, at **3540**, that the DCI is a valid activation DCI or a valid release/deactivation DCI may further be based on the DCI being associated with the first radio network temporary identifier and based on a third value of the new data indicator field being zero.

[0302] In an example embodiment, the wireless device may be configured with a plurality of configured grant configurations comprising the uplink configured grant configuration. In an example embodiment, the wireless device may receive second configuration parameters of bandwidth part of a cell. The plurality of uplink configured grant configurations may be for the bandwidth part. The first value of the one or more first bits of the HARQ process number field may indicate the configuration index of the uplink configured grant configuration in the plurality of uplink configured grant configurations. In an example embodiment, the configuration parameters of the uplink configured grant configuration may comprise a first parameter indicating the configuration index. The determining, at **3540**, that the DCI is a valid activation DCI or a valid release/deactivation DCI may be based on the one or more second bits and not based on the one or more first bits.

[0303] In an example embodiment, in response to the first value indicating the configuration index, the determining, at **3540**, that the DCI is a valid activation DCI or a valid release/deactivation DCI may be based on comparing the one or more second bits with one or more second predetermined values. In an example embodiment, in response to the first value not indicating the configuration index, the determining, at **3540**, that the DCI is a valid activation DCI or a valid release/deactivation DCI may be based on comparing the one or more second bits with one or more second predetermined values and comparing the one or more first bits with one or more first predetermined values. In example embodiments, the one or more first predetermined values may be all zeros. In example embodiments, the one or more second predetermined values may be all zeros.

[0304] In an example embodiment, a wireless device may receive configuration parameters of an uplink configured grant configuration. The wireless device may receive a downlink control information (DCI) comprising: a hybrid automatic repeat request (HARQ) process number field comprising one or more first bits; and a redundancy version field comprising one or more second bits. The wireless device may determine that a first value of the one or more first bits does not indicate a configuration index of the uplink configured grant configuration. The wireless device may determine that the DCI is a valid activation DCI or a valid release/deactivation DCI based on the one or more second bits and the one or more first bits. The wireless device may activate or deactivate a plurality of resources associated with the uplink configured grant configuration.

[0305] FIG. 36 shows an example flow diagram in accordance with several of various embodiments of the present disclosure. At **3610**, a wireless device may receive configuration parameters of an uplink configured grant configuration. At **3620**, the wireless device may receive a downlink control information (DCI) comprising: a hybrid automatic repeat request (HARQ) process number field comprising one or more first bits; and a redundancy version field comprising one or more second bits. At **3630**, in response to a first value of the one or more first bits not indicating a configuration index of the uplink configured grant configuration, the wireless device may determine that the DCI is a valid activation DCI or a valid release/deactivation DCI based on the one or more second bits and the one or more first bits. At **3640**, the wireless device may activate or deactivate a plurality of resources associated with the uplink configured grant configuration.

[0306] In an example embodiment, a wireless device may receive configuration parameters of an uplink configured grant configuration. The wireless device may receive a downlink control information (DCI) comprising: a hybrid automatic repeat request (HARQ) process number field comprising one or more first bits; and a redundancy version field comprising one or more second bits. The wireless device may determine that a first value of the one or more first bits indicates a configuration index of the uplink configured grant configuration. The wireless device may determine that the DCI is a valid activation DCI or a valid release/deactivation DCI based on the one or more second bits and not based on the one or more first bits. The wireless device may activate or deactivate a plurality of resources associated with the uplink configured grant configuration.

[0307] FIG. 37 shows an example flow diagram in accordance with several of various embodiments of the present

disclosure. At **3710**, a wireless device may receive configuration parameters of an uplink configured grant configuration. At **3720**, the wireless device may receive a downlink control information (DCI) comprising: a hybrid automatic repeat request (HARQ) process number field comprising one or more first bits; and a redundancy version field comprising one or more second bits. At **3730**, in response to a first value of the one or more first bits indicating a configuration index of the uplink configured grant configuration, determining that the DCI is a valid activation DCI or a valid release/deactivation DCI based on the one or more second bits and not based on the one or more first bits. At **3740**, the wireless device may activate or deactivate a plurality of resources associated with the uplink configured grant configuration.

[0308] FIG. 38 shows an example flow diagram in accordance with several of various embodiments of the present disclosure. At **3810**, a wireless device may receive configuration parameters of a downlink semi-persistent scheduling (SPS) configuration. At **3820**, the wireless device may receive a downlink control information (DCI) comprising: a hybrid automatic repeat request (HARQ) process number field comprising one or more first bits; and a redundancy version field comprising one or more second bits. At **3830**, the wireless device may determine whether a first value of the one or more first bits indicates a configuration index of the downlink SPS configuration. At **3840**, the wireless device may determine that the DCI is a valid activation DCI or a valid release/deactivation DCI: based on the one or more second bits and not based on the one or more first bits in response to the first value indicating the configuration index; and based on the one or more second bits and the one or more first bits in response to the first value not indicating the configuration index. At **3850**, the wireless device may activate or deactivate a plurality of resources associated with the downlink SPS configuration.

[0309] In example embodiments, the wireless device may receive a configuration parameter indicating one or more deactivation states. The first value of the one or more first bits of the HARQ process number field may indicate a first deactivation state of the one or more deactivation states. The first deactivation state may correspond to one or more downlink SPS configurations comprising the downlink SPS configurations. The wireless device may determine that the DCI is a valid release/deactivation DCI at **3840** and may deactivate the plurality of resources associated with the downlink SPS configuration at **3850**.

[0310] In an example embodiment, the configuration parameters of the downlink SPS configuration, received at **3810**, may comprise a first radio network temporary identifier (RNTI). The first RNTI may be a configured scheduling RNTI (CS-RNTI). The DCI received at **3820** may be associated with the first radio network temporary identifier. In an example embodiment, the DCI may further comprise a new data indicator field. The determining, at **3840**, that the DCI is a valid activation DCI or a valid release/deactivation DCI may further be based on the DCI being associated with the first radio network temporary identifier and based on a third value of the new data indicator field being zero.

[0311] In an example embodiment, the wireless device may be configured with a plurality of downlink SPS configurations comprising the downlink SPS configuration. In an example embodiment, the wireless device may receive second configuration parameters of bandwidth part of a cell. The plurality of downlink SPS configurations may be for the

bandwidth part. The first value of the one or more first bits of the HARQ process number field may indicate the configuration index of the downlink SPS configuration in the plurality of downlink SPS configurations. In an example embodiment, the configuration parameters of the downlink SPS configuration may comprise a first parameter indicating the configuration index. The determining, at **3840**, that the DCI is a valid activation DCI or a valid release/deactivation DCI may be based on the one or more second bits and not based on the one or more first bits.

[0312] In an example embodiment, in response to the first value indicating the configuration index, the determining, at **3840**, that the DCI is a valid activation DCI or a valid release/deactivation DCI may be based on comparing the one or more second bits with one or more second predetermined values. In an example embodiment, in response to the first value not indicating the configuration index, the determining, at **3840**, that the DCI is a valid activation DCI or a valid release/deactivation DCI may be based on comparing the one or more second bits with one or more second predetermined values and comparing the one or more first bits with one or more first predetermined values. In example embodiments, the one or more first predetermined values may be all zeros. In example embodiments, the one or more second predetermined values may be all zeros.

[0313] In an example embodiment, a wireless device may receive configuration parameters of a downlink semi-persistent scheduling (SPS) configuration. The wireless device may receive a downlink control information (DCI) comprising: a hybrid automatic repeat request (HARQ) process number field comprising one or more first bits; and a redundancy version field comprising one or more second bits. The wireless device may determine that a first value of the one or more first bits does not indicate a configuration index of the downlink SPS configuration. The wireless device may determine that the DCI is a valid activation DCI or a valid release/deactivation DCI based on the one or more second bits and the one or more first bits. The wireless device may activate or deactivate a plurality of resources associated with the uplink configured grant configuration.

[0314] FIG. 39 shows an example flow diagram in accordance with several of various embodiments of the present disclosure. At **3910**, a wireless device may receive configuration parameters of a downlink semi-persistent scheduling (SPS) configuration. At **3920**, the wireless device may receive a downlink control information (DCI) comprising: a hybrid automatic repeat request (HARQ) process number field comprising one or more first bits; and a redundancy version field comprising one or more second bits. At **3930**, in response to a first value of the one or more first bits not indicating a configuration index of the downlink SPS configuration, the wireless device may determine that the DCI is a valid activation DCI or a valid release/deactivation DCI based on the one or more second bits and the one or more first bits. At **3940**, the wireless device may activate or deactivate a plurality of resources associated with the downlink SPS configuration.

[0315] In an example embodiment, a wireless device may receive configuration parameters of a downlink semi-persistent scheduling (SPS) configuration. The wireless device may receive a downlink control information (DCI) comprising: a hybrid automatic repeat request (HARQ) process number field comprising one or more first bits; and a redundancy version field comprising one or more second

bits. The wireless device may determine that that a first value of the one or more first bits indicates a configuration index of the downlink SPS configuration. The wireless device may determine that the DCI is a valid activation DCI or a valid release/deactivation DCI based on the one or more second bits and not based on the one or more first bits. The wireless device may activate or deactivate a plurality of resources associated with the uplink configured grant configuration.

[0316] FIG. 40 shows an example flow diagram in accordance with several of various embodiments of the present disclosure. At **4010**, a wireless device may receive configuration parameters of a downlink semi-persistent scheduling (SPS) configuration. At **4020**, the wireless device may receive a downlink control information (DCI) comprising: a hybrid automatic repeat request (HARQ) process number field comprising one or more first bits; and a redundancy version field comprising one or more second bits. At **4030**, in response to a first value of the one or more first bits indicating a configuration index of the downlink SPS configuration, determining that the DCI is a valid activation DCI or a valid release/deactivation DCI based on the one or more second bits and not based on the one or more first bits. At **4040**, the wireless device may activate or deactivate a plurality of resources associated with the downlink SPS configuration.

[0317] FIG. 41 shows an example flow diagram in accordance with several of various embodiments of the present disclosure. At **4110**, a wireless device may receive configuration parameters of an uplink configured grant configuration. At **4120**, the wireless device may receive a downlink control information (DCI) comprising a hybrid automatic repeat request (HARQ) process number field. At **4130**, the wireless device may determine that the DCI is a valid activation DCI or a valid release/deactivation DCI, wherein the determining is based on whether one or more bits of the HARQ process number field are used for indication of a configuration index of the uplink configured grant configuration. At **4140**, the wireless device may activate or deactivate a plurality of resources associated with the uplink configured grant configuration.

[0318] In an example embodiment, the DCI, received at **4120**, may further comprise a redundancy version field. The determining, at **4130**, may further be based on one or more second bits of the redundancy version field.

[0319] FIG. 42 shows an example flow diagram in accordance with several of various embodiments of the present disclosure. At **4210**, a wireless device may receive configuration parameters of an downlink semi-persistent scheduling (SPS) configuration. At **4220**, the wireless device may receive a downlink control information (DCI) comprising a hybrid automatic repeat request (HARQ) process number field. At **4230**, the wireless device may determine that the DCI is a valid activation DCI or a valid release/deactivation DCI, wherein the determining is based on whether one or more bits of the HARQ process number field are used for indication of a configuration index of the downlink SPS configuration. At **4240**, the wireless device may activate or deactivate a plurality of resources associated with the downlink SPS configuration.

[0320] In an example embodiment, the DCI, received at **4220**, may further comprise a redundancy version field. The determining, at **4230**, may further be based on one or more second bits of the redundancy version field.

[0321] Various exemplary embodiments of the disclosed technology are presented as example implementations and/or practices of the disclosed technology. The exemplary embodiments disclosed herein are not intended to limit the scope. Persons of ordinary skill in the art will appreciate that various changes can be made to the disclosed embodiments without departure from the scope. After studying the exemplary embodiments of the disclosed technology, alternative aspects, features and/or embodiments will become apparent to one of ordinary skill in the art. Without departing from the scope, various elements or features from the exemplary embodiments may be combined to create additional embodiments. The exemplary embodiments are described with reference to the drawings. The figures and the flowcharts that demonstrate the benefits and/or functions of various aspects of the disclosed technology are presented for illustration purposes only. The disclosed technology can be flexibly configured and/or reconfigured such that one or more elements of the disclosed embodiments may be employed in alternative ways. For example, an element may be optionally used in some embodiments or the order of actions listed in a flowchart may be changed without departure from the scope.

[0322] An example embodiment of the disclosed technology may be configured to be performed when deemed necessary, for example, based on one or more conditions in a wireless device, a base station, a radio and/or core network configuration, a combination thereof and/or alike. For example, an example embodiment may be performed when the one or more conditions are met. Example one or more conditions may be one or more configurations of the wireless device and/or base station, traffic load and/or type, service type, battery power, a combination of thereof and/or alike. In some scenarios and based on the one or more conditions, one or more features of an example embodiment may be implemented selectively.

[0323] In this disclosure, the articles “a” and “an” used before a group of one or more words are to be understood as “at least one” or “one or more” of what the group of the one or more words indicate. The use of the term “may” before a phrase is to be understood as indicating that the phrase is an example of one of a plurality of useful alternatives that may be employed in an embodiment in this disclosure.

[0324] In this disclosure, an element may be described using the terms “comprises”, “includes” or “consists of” in combination with a list of one or more components. Using the terms “comprises” or “includes” indicates that the one or more components are not an exhaustive list for the description of the element and do not exclude components other than the one or more components. Using the term “consists of” indicates that the one or more components is a complete list for description of the element. In this disclosure, the term “based on” is intended to mean “based at least in part on”. The term “based on” is not intended to mean “based only on”. In this disclosure, the term “and/or” used in a list of elements indicates any possible combination of the listed elements. For example, “X, Y, and/or Z” indicates X; Y; Z; X and Y; X and Z; Y and Z; or X, Y, and Z.

[0325] Some elements in this disclosure may be described by using the term “may” in combination with a plurality of features. For brevity and ease of description, this disclosure may not include all possible permutations of the plurality of features. By using the term “may” in combination with the plurality of features, it is to be understood that all permu-

tations of the plurality of features are being disclosed. For example, by using the term “may” for description of an element with four possible features, the element is being described for all fifteen permutations of the four possible features. The fifteen permutations include one permutation with all four possible features, four permutations with any three features of the four possible features, six permutations with any two features of the four possible features and four permutations with any one feature of the four possible features.

[0326] Although mathematically a set may be an empty set, the term set used in this disclosure is a nonempty set. Set B is a subset of set A if every element of set B is in set A. Although mathematically a set has an empty subset, a subset of a set is to be interpreted as a non-empty subset in this disclosure. For example, for set A={subcarrier1, subcarrier2}, the subsets are {subcarrier1}, {subcarrier2} and {subcarrier1, subcarrier2}.

[0327] In this disclosure, the phrase “based on” may be used equally with “based at least on” and what follows “based on” or “based at least on” indicates an example of one of plurality of useful alternatives that may be used in an embodiment in this disclosure. The phrase “in response to” may be used equally with “in response at least to” and what follows “in response to” or “in response at least to” indicates an example of one of plurality of useful alternatives that may be used in an embodiment in this disclosure. The phrase “depending on” may be used equally with “depending at least on” and what follows “depending on” or “depending at least on” indicates an example of one of plurality of useful alternatives that may be used in an embodiment in this disclosure. The phrases “employing” and “using” and “employing at least” and “using at least” may be used equally in this in this disclosure and what follows “employing” or “using” or “employing at least” or “using at least” indicates an example of one of plurality of useful alternatives that may be used in an embodiment in this disclosure.

[0328] The example embodiments disclosed in this disclosure may be implemented using a modular architecture comprising a plurality of modules. A module may be defined in terms of one or more functions and may be connected to one or more other elements and/or modules. A module may be implemented in hardware, software, firmware, one or more biological elements (e.g., an organic computing device and/or a neurocomputer) and/or a combination thereof and/or alike. Example implementations of a module may be as software code configured to be executed by hardware and/or a modeling and simulation program that may be coupled with hardware. In an example, a module may be implemented using general-purpose or special-purpose processors, digital signal processors (DSPs), microprocessors, microcontrollers, application-specific integrated circuits (ASICs), programmable logic devices (PLDs) and/or alike. The hardware may be programmed using machine language, assembly language, high-level language (e.g., Python, FORTRAN, C, C++ or the like) and/or alike. In an example, the function of a module may be achieved by using a combination of the mentioned implementation methods.

What is claimed is:

1. A non-transitory computer-readable medium storing instructions that, when executed, cause a wireless device to:
 - validate a downlink control information (DCI), wherein the validation is:
 - based on one or more second bits in a redundancy version field of the DCI; and
 - based on comparing one or more first bits in a hybrid automatic repeat request process number field of the

DCI with a first sequence of one or more bits, determined using a first configuration parameter, in response to:

- the one or more first bits not indicating a configuration index; and
- the DCI having a first format; and
- not based on comparing the one or more first bits with the first sequence in response to the DCI having a second format; and

transmit a confirmation in response to the validation.

2. The non-transitory computer-readable medium of claim 1, wherein the instructions, when executed, further cause the wireless device to receive configuration parameters of a configured grant configuration, wherein the DCI indicates activation of the configured grant configuration.

3. The non-transitory computer-readable medium of claim 1, wherein the confirmation is based on a medium access control (MAC) control element (CE).

4. The non-transitory computer-readable medium of claim 1, wherein the DCI is associated with a configured scheduling radio network temporary identifier.

5. The non-transitory computer-readable medium of claim 4, wherein:

- the DCI further comprises a new data indicator field; and
- the validation is further based on the DCI being associated with the configured scheduling radio network temporary identifier and based on a value of the new data indicator field being zero.

6. The non-transitory computer-readable medium of claim 1, wherein the first configuration parameter indicates a first number of bits in the first sequence of the one or more bits.

7. The non-transitory computer-readable medium of claim 6, wherein the first sequence of one or more bits are all zeros.

8. The non-transitory computer-readable medium of claim 1, wherein the validation is based on comparing the one or more second bits with a second sequence of one or more bits determined using a second configuration parameter.

9. The non-transitory computer-readable medium of claim 8, wherein the second configuration parameter indicates a second number of bits in the second sequence of the one or more bits.

10. The non-transitory computer-readable medium of claim 9, wherein the second sequence of one or more bits are all zeros.

11. A non-transitory computer-readable medium storing instructions that, when executed, cause a base station to:

- transmit, to a wireless device, a downlink control information (DCI) comprising:
 - a hybrid automatic repeat request process number field comprising one or more first bits; and
 - a redundancy version field comprising one or more second bits;

- receive a transport block in response to validation of the DCI by the wireless device, wherein the validation is:
 - based on the one or more second bits;
 - based on comparing the one or more first bits with a first sequence of one or more bits, determined using a first configuration parameter, in response to:
 - the one or more first bits not indicating a configuration index; and
 - the DCI having a first format; and
 - not based on comparing the one or more first bits with the first sequence in response to the DCI having a second format.

12. The non-transitory computer-readable medium of claim **11**, wherein the instructions, when executed, further cause the base station to transmit configured grant configuration parameters of an uplink configured grant configuration, wherein the DCI indicates activation of the configured grant configuration.

13. The non-transitory computer-readable medium of claim **11**, wherein a plurality of resources are activated in response to the validation.

14. The non-transitory computer-readable medium of claim **11**, wherein the DCI is associated with a configured scheduling radio network temporary identifier.

15. The non-transitory computer-readable medium of claim **14**, wherein:

the DCI further comprises a new data indicator field; and
the validation is further based on the DCI being associated with the configured scheduling radio network temporary identifier and based on a value of the new data indicator field being zero.

16. The non-transitory computer-readable medium of claim **11**, wherein the first configuration parameter indicates a first number of bits in the first sequence of the one or more bits.

17. The non-transitory computer-readable medium of claim **16**, wherein the first sequence of one or more bits are all zeros.

18. The non-transitory computer-readable medium of claim **11**, wherein the validation is based on comparing the one or more second bits with a second sequence of one or more bits determined using a second configuration parameter.

19. The non-transitory computer-readable medium of claim **18**, wherein the second configuration parameter indicates a second number of bits in the second sequence of the one or more bits.

20. The non-transitory computer-readable medium of claim **11**, wherein the second sequence of one or more bits are all zeros.

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